

DOMOLOGY

THREE BOOKS IN ONE

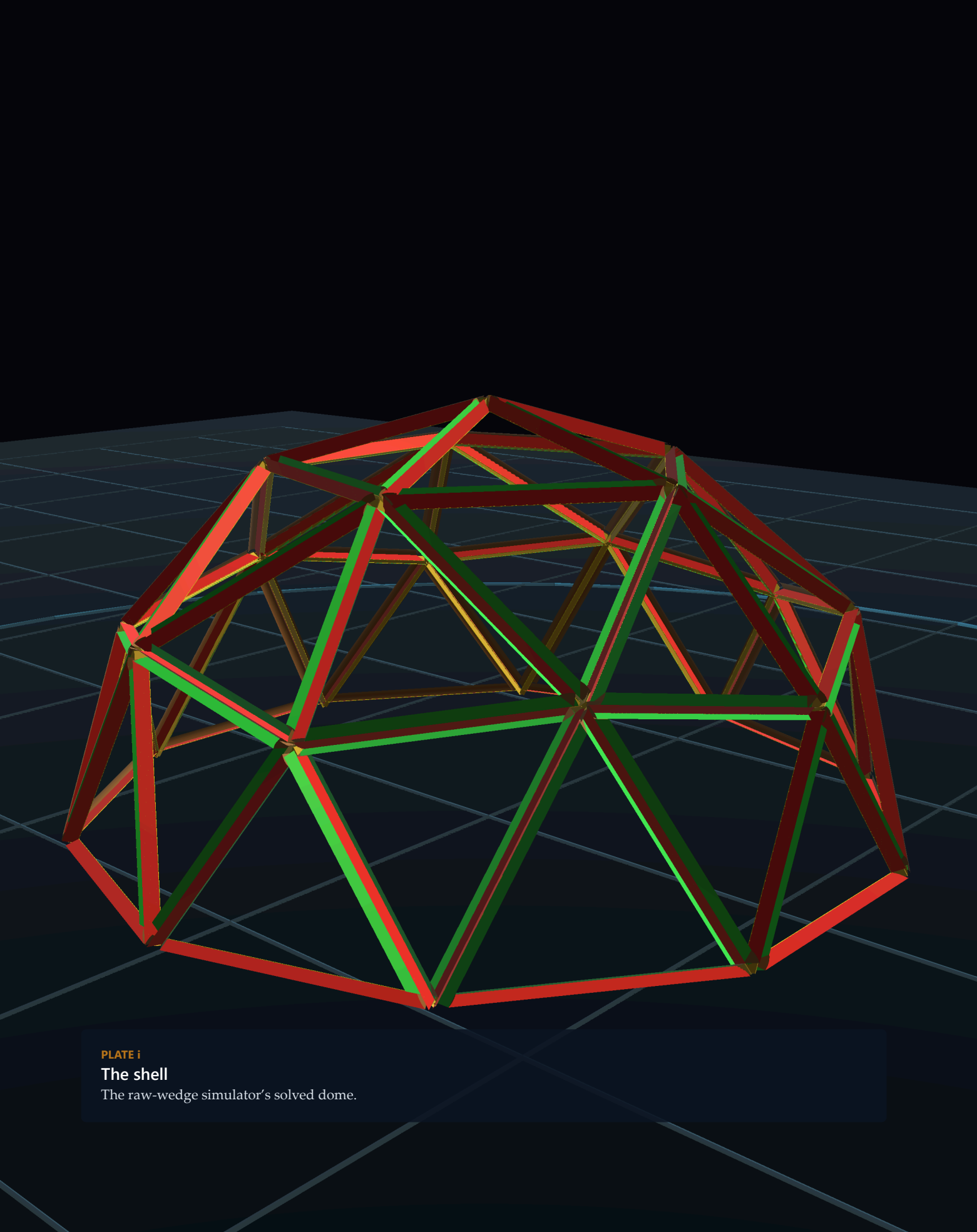


PLATE I

The shell

The raw-wedge simulator's solved dome.

Domology

The Geodesic Dome: the Science, the Journey, and the Build

THREE BOOKS IN ONE

BOOK ONE The Dome
BOOK TWO The Journey
BOOK THREE The Build

[AUTHOR NAME]

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This copy was built on September 12, 2026 from the DomeSim repository at commit ea2d9f0. Every number in it was computed by the project's code at that moment; the appendix The Numbers Behind This Book lists each one, its value, and what computes it.

The plates were rendered by the project's own tools: the DomeSim film engine, Dome Forge, the Dome Home Assembly Line and the Dome Creator.

This book describes how one person designed and built a dome. It is not engineering, legal or financial advice. Check your local building code, have any structure people will live in reviewed by a qualified engineer, and use a chainsaw only with training and protective equipment.

ISBN [ISBN]

[DEDICATION: a line of your own]

Every house is an argument about what a person is worth.

The Round House Creed

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Three Books in One

How to read this book, and what every number in it promises.

The Round House Creed

The argument this whole project was built to make.

Every house is an argument about what a person is worth.

The straight wall argues one way — stack them, price them, and let the market decide who sleeps inside. The triangle argues another. The triangle says the load should be *shared*: that no member carries the roof alone, that strength comes from how things are joined and not from how much they cost.

A 2V dome is forty triangles and two lengths of wood. Thirty short. Thirty-five long. Cut them once, cut them true, and those same two numbers raise a shelter on a mountainside, on a lot outside Tulsa, on a field three weeks after the water went down. The geometry does not check your credit. The geometry does not know your zip code. A sphere encloses the most room with the least skin whether you are a developer or a widow — and that indifference is the most democratic thing left in construction.

So we built the whole argument into code.

One floor: no staircase to age out of, no doorway that needs an apology. Seams that drink the rain instead of leaking it. A south face that earns its keep in sunlight. Forty triangles from **two jigs** — which means a shop, which means a line, which means the second house goes up faster than the first, and the hundredth faster still. Not a handmade miracle. A repeatable answer.

We will not tell you it beats everything. We will show you the numbers and let you disagree. Every figure in this repository — every strut, every dollar, every labor-hour, every tree — is **computed, never claimed**. Where the case is strong, we make it. Where it is uncertain, we say so out loud. Where the honest answer is *not yet*, we wrote that down too — because a housing argument that only survives while nobody checks it isn't an argument. It's a sales pitch.

This is for the person who has done everything right and still can't get a key.

Build one. Then build the next one faster.

BOOK ONE

The Dome

Why the geodesic dome encloses the most, with the least, from the fewest parts.

- 1 The Shape That Encloses the Most
- 2 Triangles Do Not Rack
- 3 Twelve Points and a Golden Ratio
- 4 Divide, Push, Repeat
- 5 Two Lengths, Forty Triangles
- 6 The Missing Fifteen Degrees
- 7 Hubs, Hubless, and the Seam
- 8 Wind, Water, and Weight
- 9 The Dome Family
- 10 What a Dome Costs, and What It Saves

A / LONG 72.0 in

B / SHORT 63.5 in

1

The Shape That Encloses the Most

Half a sphere is the cheapest wall there is, and the numbers say by how much.

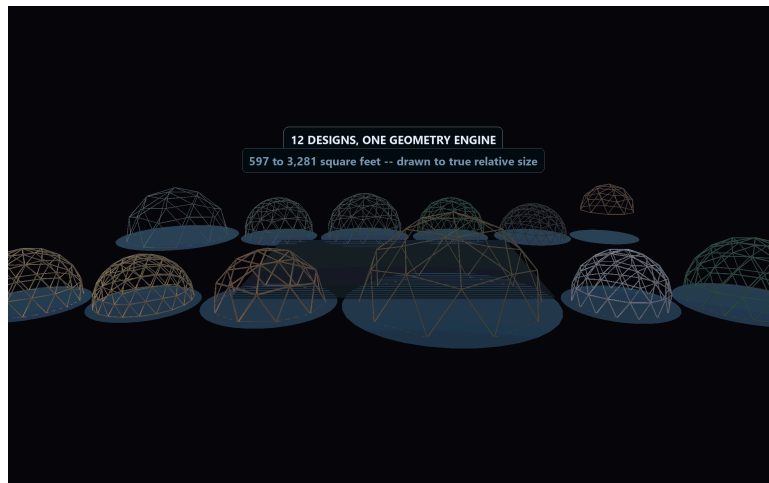


PLATE 1.1 Twelve domes, one geometry engine

All 12 designs in the Dome Creator, each rebuilt from the simulator's own model and stood side by side at true relative scale.

Rendered by the DomeSim film engine from "Every Dome In The World", chapter "world_open", 90% through. The lineup is frozen after the last design has risen, with its names kept.

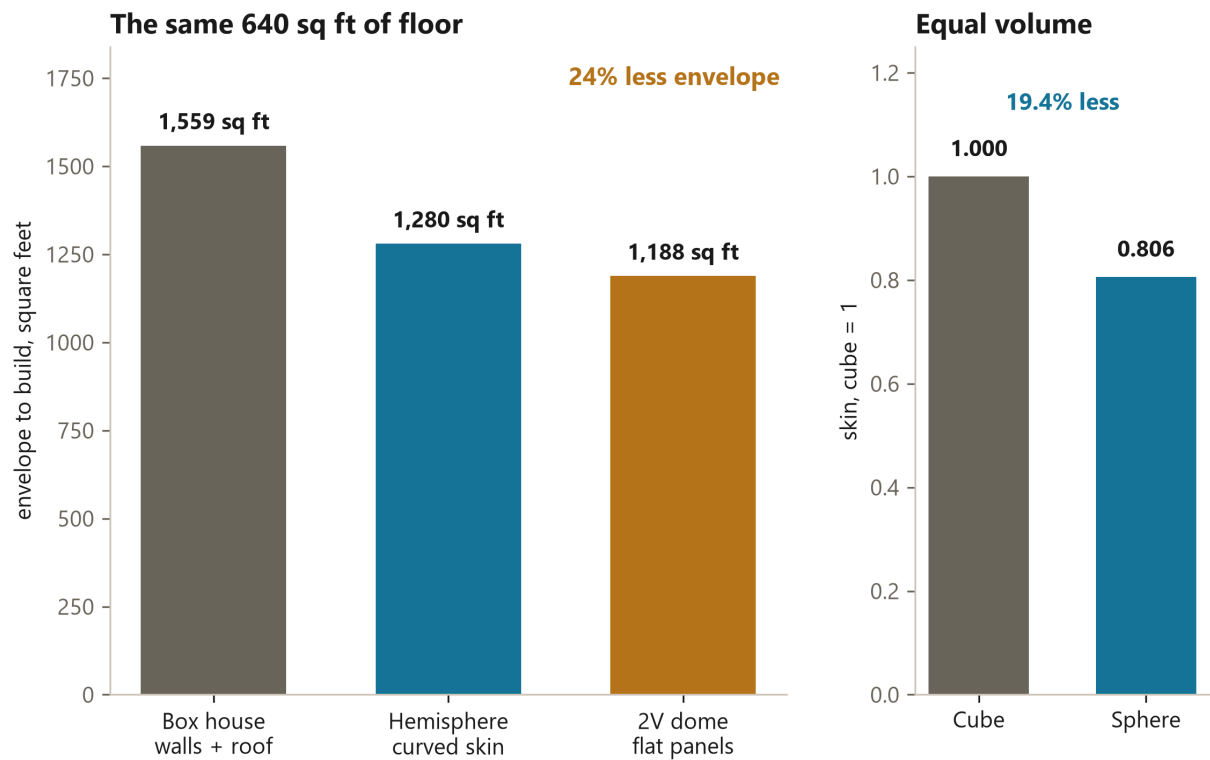
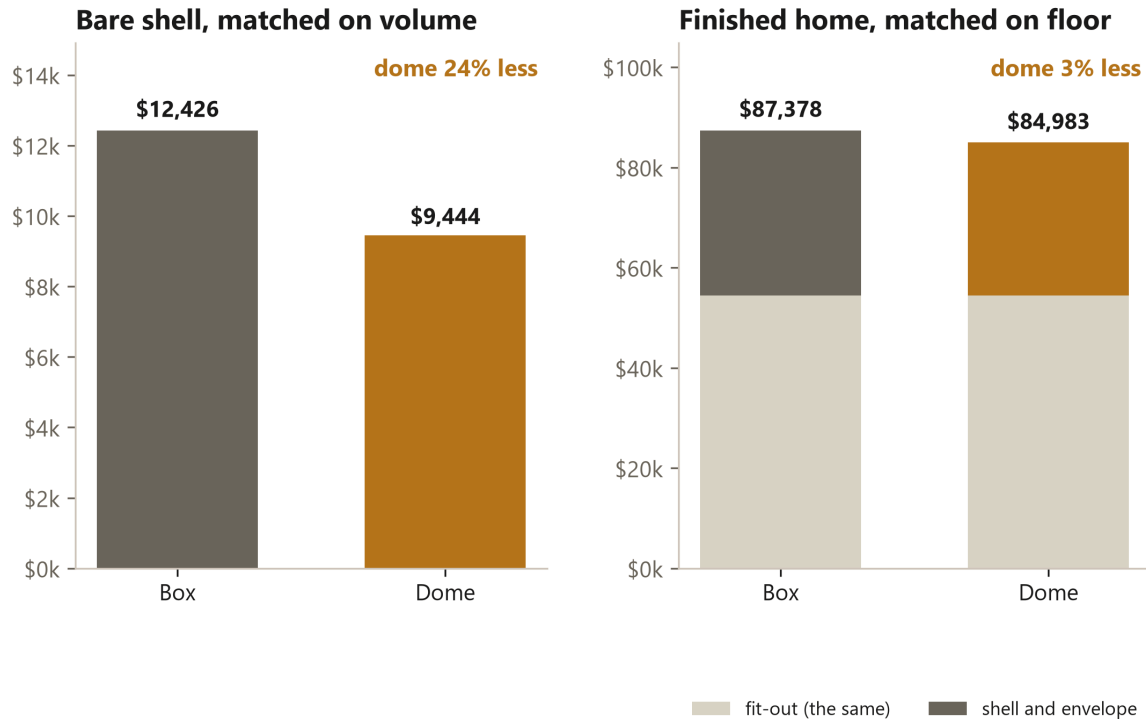


PLATE 1.2 Less skin for the same floor

On the same 640 square feet of floor, a box house needs 1,559 square feet of walls and roof and a 2V hemisphere of flat panels 1,188: 24 percent less envelope. A sphere wraps a volume in 19.4 percent less skin than a cube.

Drawn when the book is built. From domology.science, using al_build's comparison house.

**PLATE 1.3** Where the saving stops

One rate card, two shapes. As a bare shell the dome costs 24 percent less than the box; finished, only 3 percent less, because the kitchen, bath and wiring cost the same in either shape.

Drawn when the book is built. From `al_build.building_comparisons`, with the fit-out drawn as its own band.

WORKED MATH**Sphere against box**

equal volume, sphere against cube:

area ratio = $(36 \pi)^{1/3} / 6 = 0.8060$

the sphere needs 19.4% less skin

equal floor, 640 sq ft, a hemisphere against a box house:

box 22.2 x 28.8 ft, walls 9 ft, flat roof

box skin = walls + roof = 1,559 sq ft (2.44 per sq ft of floor)

hemisphere radius 14.3 ft: curved skin 1,280 sq ft (2.00)

its 2V flat panels: 1,188 sq ft (1.86)

same floor, 24% less envelope to build, insulate and seal

Computed by `domology.science.sheet_sphere_box`. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

Skin per square foot of floor

skin you must build and seal, per square foot of floor:

Split-Log Homestead	2260 / 1217 = 1.86
Whole Trunk Lodge - 20 ft	6092 / 3281 = 1.86
Continuous Steel Arc Hanga	4018 / 2164 = 1.86
Treehouse Canopy Dome	1108 / 597 = 1.86
...	
Rebar Garden Dome	2333 / 991 = 2.35
Concrete Monocoque Form	2776 / 1179 = 2.35
Woodland Hex Mirror	2333 / 991 = 2.35

best: Split-Log Homestead at 1.86
worst: Woodland Hex Mirror at 2.35
the spread is frequency and where the dome gets cut,
not the material bolted to it

1.86 against 2.35 square feet of envelope per square foot of floor -- the cheapest wall is the one the geometry never asks you to build

Computed by two_v_demo.world_facts.steps_efficiency. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

The same rate card, two shapes

one rate card, two shapes (al_build's comparison):

storage shed, matched on volume (3,840 cu ft):

BOX	framing	1,032 ft	skin	1,184 sq ft	build \$	12,426
DOME	framing	465 ft	skin	874 sq ft	build \$	9,444

home, finished, matched on floor (640 sq ft):

STICK	envelope	1,559 sq ft	build \$	87,378	(\$137/sq ft)
DOME	envelope	1,507 sq ft	build \$	84,983	(\$133/sq ft)

the dome costs 24% less as a bare shell and 3% less finished -- the fit-out is the same either way, which is why the gap narrows

Computed by al_build.building_comparisons. Every line above is printed by that code when the book is built; none of it is typed.

2

Triangles Do Not Rack

Why forty triangles hold what forty squares cannot.

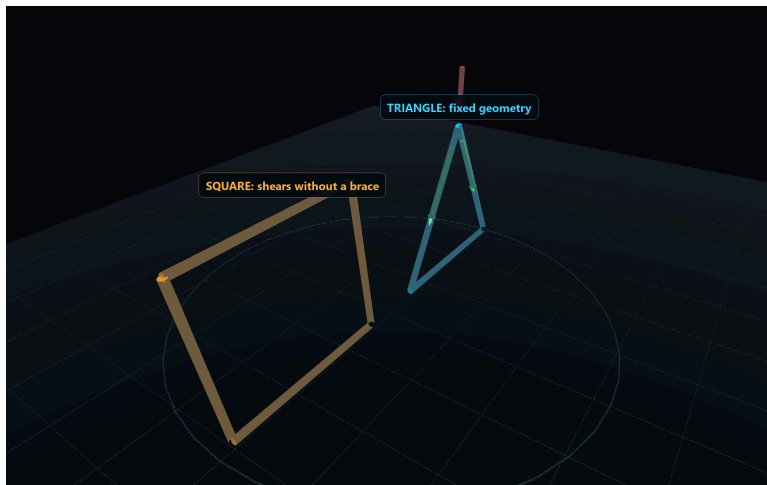


PLATE 2.1 A triangle does not lean

Three sides fix a triangle's three angles; four sides fix nothing about a square's. At the moment of greatest shear the square has leaned and the triangle has not moved.

Rendered by the DomeSim film engine from "2V Geodesic Masterclass", chapter "triangles", 25% through. Stopped at the instant the square leans furthest.

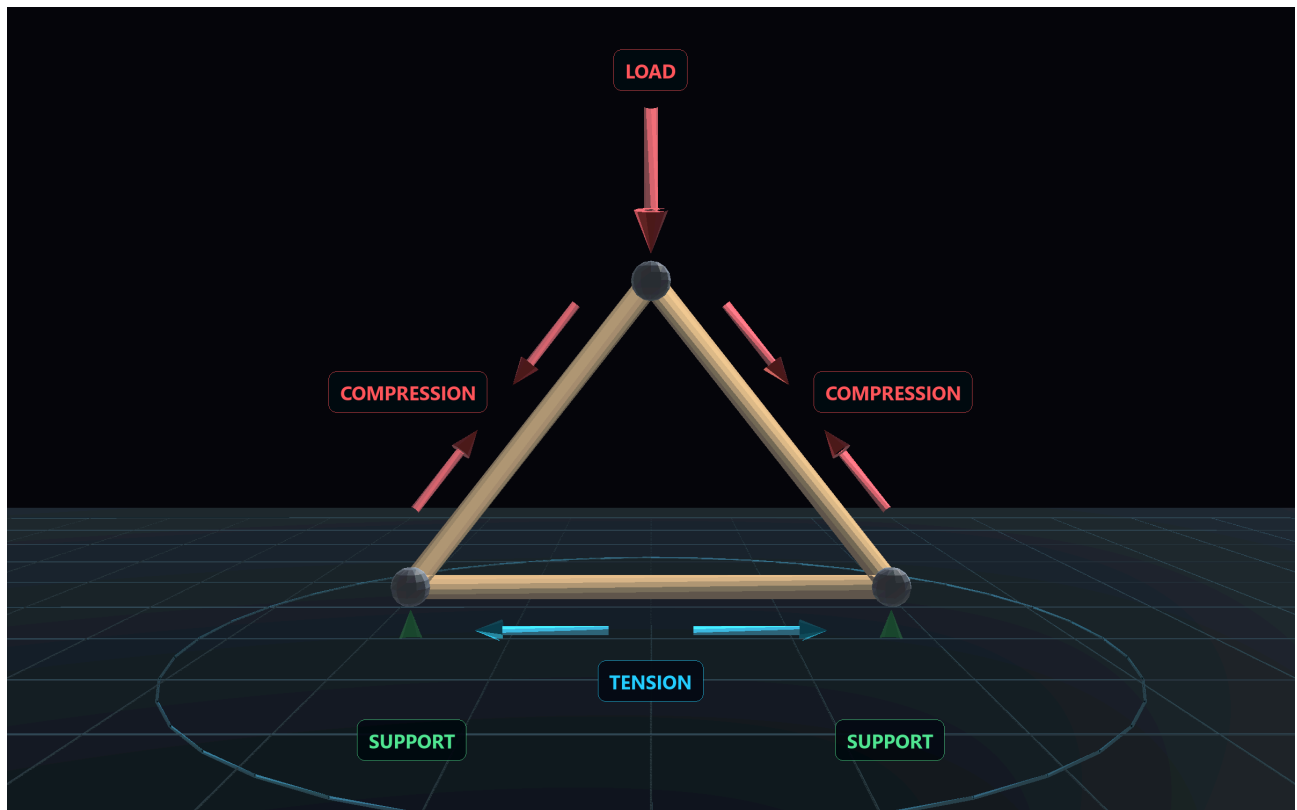


PLATE 2.2 Compression and tension in one triangle

Load a triangle at its apex and the two sloping members push back in compression while the tie across the bottom pulls in tension. Every member of a triangulated shell works along its length.

Built for this book from the film engine's visual objects. Built for the book: three members, their forces, and the supports.

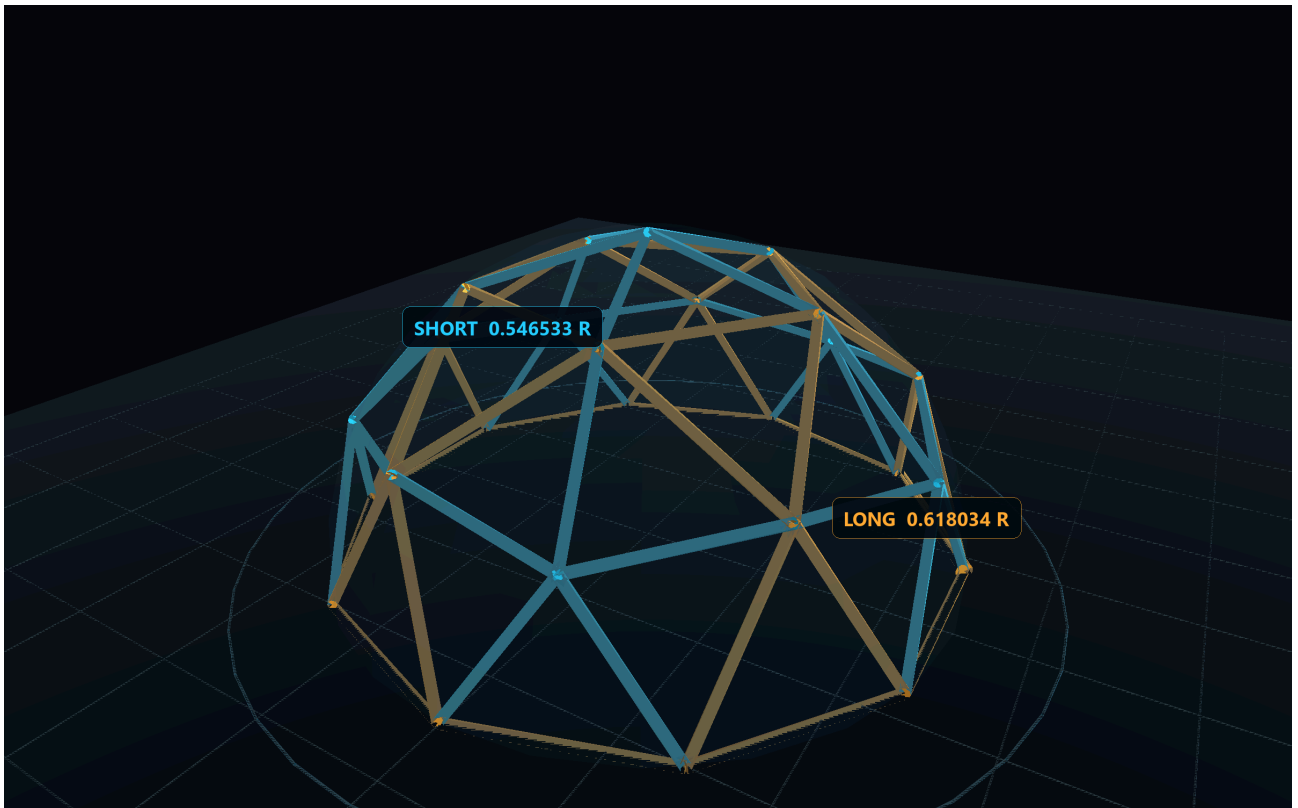


PLATE 2.3 Forty triangles, two shapes

The 2V dome's struts sorted by length: 35 long and 30 short in a hubbed hemisphere, in two triangle shapes.

Rendered by the DomeSim film engine from "2V Geodesic Masterclass", chapter "classes", 92% through. The classes scene once every strut has taken its class colour.

3

Twelve Points and a Golden Ratio

Every geodesic dome begins as the most symmetrical solid there is.

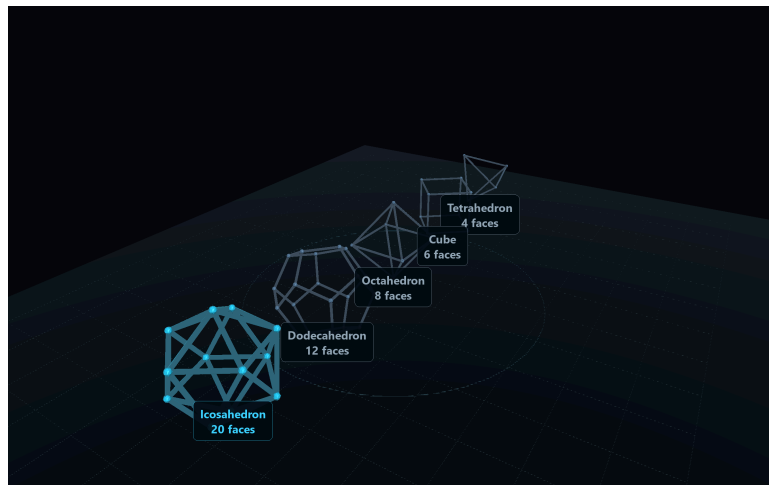
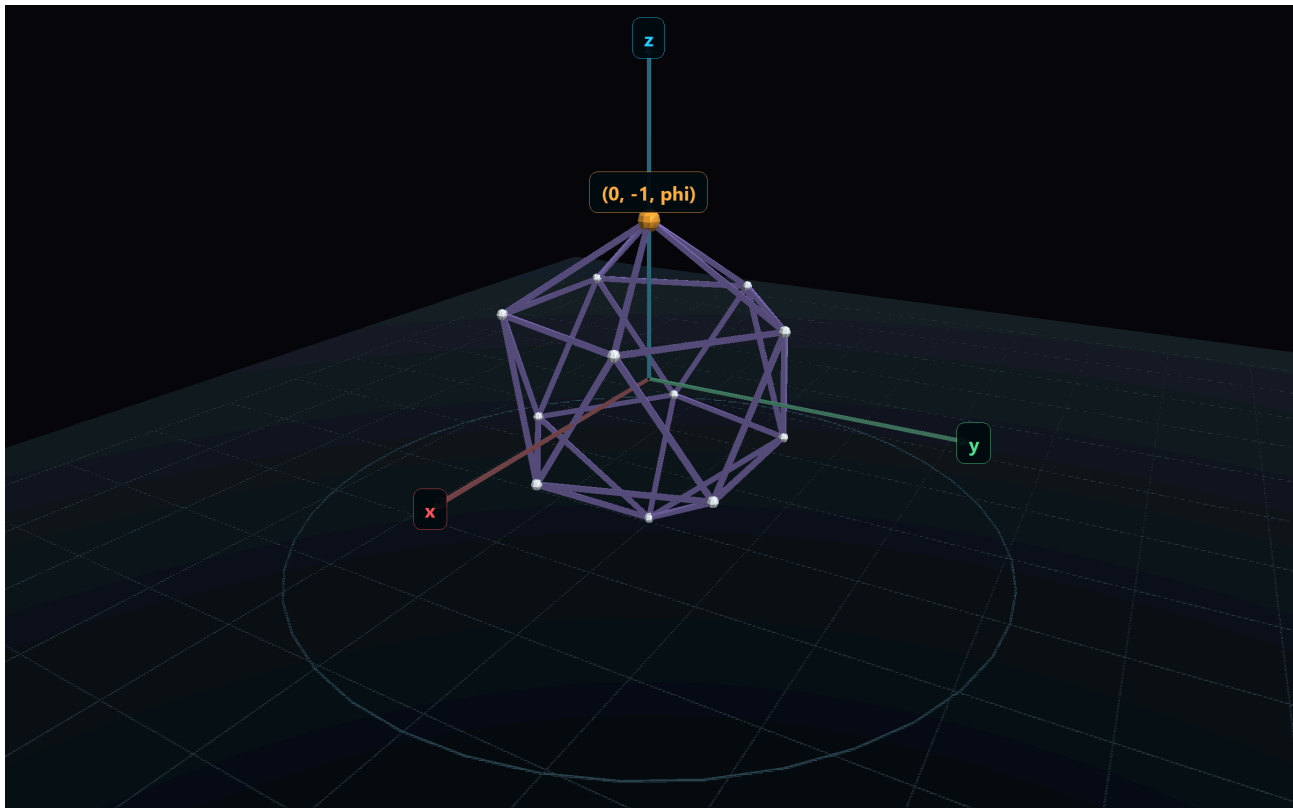


PLATE 3.1 The five Platonic solids

The five solids whose faces are all the same regular polygon. The icosahedron, with 20 equilateral triangles, is the most sphere-like of them.

Rendered by the DomeSim film engine from "2V Geodesic Masterclass", chapter "platonic", 80% through. Stopped once every edge has drawn, with the icosahedron highlighted.

**PLATE 3.2** Twelve corners from one number

Every corner of the icosahedron sits at a combination of 0, ± 1 and $\pm\varphi$, where $\varphi = 1.618034$. One vertex is marked.

Rendered by the DomeSim film engine from "2V Geodesic Masterclass", chapter "phi", 92% through. The coordinates scene with its axes and one vertex labelled.

WORKED MATH**Twelve points from one ratio**

```
start with one number and nothing else:
phi = (1 + sqrt 5) / 2 = 1.618033989
  phi is the number whose square is itself plus one
build 12 points by putting 0, +/-1 and +/-phi in every order:
  (0, +/-1, +/-phi), (+/-1, +/-phi, 0), (+/-phi, 0, +/-1)
  3 families x 4 sign choices = 12 points
measure the closest pair of those points:
  nearest-neighbour distance = 2.000000000
measure each point's distance from the centre:
  |v| = sqrt(1 + phi^2) = 1.902113033 (all 12 the same)
  |v| means length: sqrt(x^2 + y^2 + z^2)
every point is 2.000 from its neighbours and 1.902113 from the middle
```

One irrational number places twelve points in perfectly even space -- that is an icosahedron, and it is the entire starting stock of the dome.

Computed by two_v_demo.scratch_facts.steps_phi. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**Onto the sphere**

the 12 raw points sit on a sphere of an awkward radius:
 $|v| = 1.902113033$ (not 1, not anything useful)
 divide every point by its own length -- 'normalize':
 $v_hat = v / |v|$ same direction, length set to 1
 check: every point is now at radius 1.000000000 to 1.000000000
 the edge that measured 2.000000 becomes:
 $2 / \sqrt{1 + \phi^2} = 1.051462224$
 that number is a CHORD FACTOR: a multiplier, not a length
 length in inches = chord factor x radius in inches
 so the model is unit-free -- pick the size later, once, and
 the same numbers serve a playhouse and a hangar

Normalizing turns one specific solid into a reusable recipe: every distance becomes a multiple of a radius you have not chosen yet.

Computed by two_v_demo.scratch_facts.steps_normalize. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**Euler's count**

the icosahedron, counted off the model on the left:
 $V = \text{corners} = 12$
 $F = \text{triangle faces} = 20$
 every face has 3 sides and every side is shared by 2 faces:
 $E = F \times 3 / 2 = 20 \times 3 / 2 = 30$
 counted directly off the model: $E = 30$
 Euler's rule, true of any closed surface without holes:
 $V - E + F = 12 - 30 + 20 = 2$
 a closed surface always gives 2 -- anything else is a bug:
 a missing face, a doubled corner, a hole in the mesh
 run it again after subdividing, on 42 corners and 80 faces:
 $42 - 120 + 80 = 2$

$V - E + F = 2$ is the cheapest proof in geometry that the shape in memory really closed up.

Computed by two_v_demo.scratch_facts.steps_euler. Every line above is printed by that code when the book is built; none of it is typed.

4

Divide, Push, Repeat

How a flat solid becomes a sphere: split every edge and push the new points out.

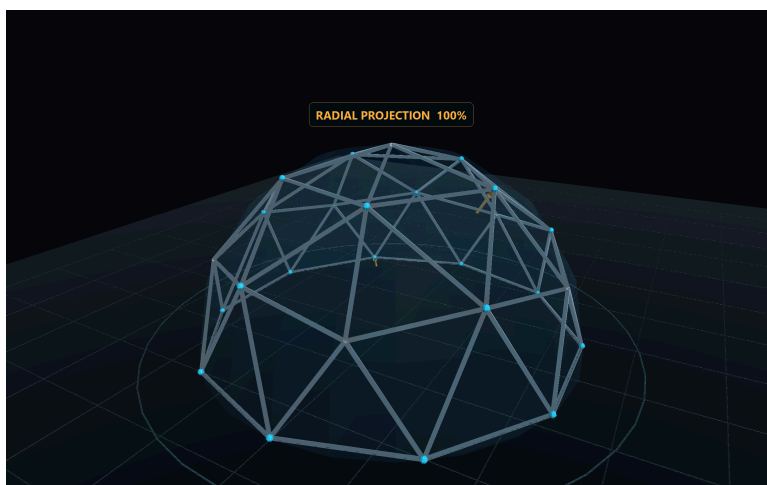


PLATE 4.1 Push the new points out

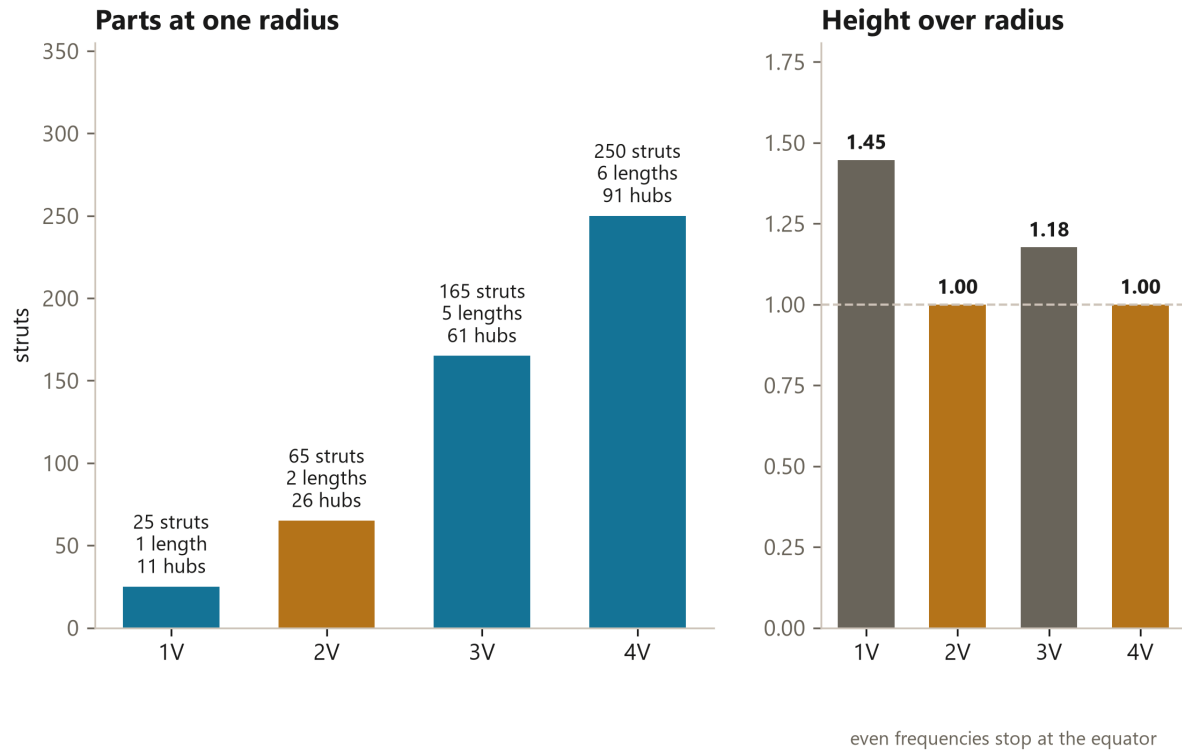
Move each new point out to the sphere. The flat faces bend into a dome and the edges fall into exactly two lengths.

Rendered by the DomeSim film engine from "2V Geodesic Masterclass", chapter "project", 90% through. Near the end of the projection, with the points on the sphere.

**PLATE 4.2** Split every edge

Mark the middle of every edge and each triangle becomes four: the icosahedron's 20 faces become 80.

Rendered by the DomeSim film engine from "2V Geodesic Masterclass", chapter "subdivide", 85% through. Stopped after the midpoints appear, before they are pushed out.

**PLATE 4.3** What frequency costs

Built at one radius: 25, 65, 165 and 250 struts for 1V to 4V, in 1, 2, 5 and 6 lengths. Even frequencies cut exactly at the equator.

Drawn when the book is built. From world_facts.frequency_ladder, which builds each dome with the simulator.

WORKED MATH**Split every edge**

take one edge of the icosahedron, between corners a and b

the halfway point is the plain average of the two:

```
m = (a + b) / 2    add the coordinates, halve them
|a| = 1.000000    |b| = 1.000000    (both on the sphere)
|m| = 0.850650808    (NOT on the sphere)
```

a straight line between two points on a ball cuts inside it:

```
shortfall = 1 - 0.850651 = 0.149349192 of the radius
on this lesson's 9.7 ft radius dome, that is 17.38 in
```

leave the 30 midpoints where they landed and you get a
faceted icosahedron with more triangles -- not a dome

Halving an edge is the easy half of subdivision: those points sit 14.93% of the radius too deep, and have to be pushed out.

Computed by two_v_demo.scratch_facts.steps_midpoint. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

Push the midpoints out

push each midpoint straight out from the centre until it reaches the sphere -- and that is the same divide as before:
 $p = m / |m|$ m chose the direction, the divide fixes the distance
 $|m| = 0.850650808 \rightarrow |p| = 1.000000000$
 distance travelled = $1 - |m| = 0.149349192$ radii
 the direction never changes, only the distance from centre:
 each point slides along its own ray, so the face keeps its shape and simply bulges outward
 this is the single line that separates a subdivided icosahedron from a geodesic sphere
 and it is what breaks the equal edges: the four small triangles in each face are no longer the same size

One division by a length -- $p = m / |m|$ -- is the entire difference between a faceted ball and a geodesic dome.

Computed by two_v_demo.scratch_facts.steps_project. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

Two lengths fall out

measure all 120 edges of the finished sphere. They fall into exactly two lengths -- no more, no fewer:
 $SHORT = 0.546533058 \ R \ x60$
 $LONG = 0.618033989 \ R \ x60$
 R is the radius; these are multipliers, not inches
 four independent routes to that same SHORT number:
 1. subtract coordinates: $|u - v| = 0.546533058$
 2. central angle: $\theta = \arccos(u \cdot v) = 31.717474 \text{ deg}$
 $\text{chord} = 2 \sin(\theta / 2) = 0.546533058$
 3. law of cosines on the same triangle: identical
 4. measure it in CAD: identical
 routes 1 and 2 differ by $1.11e-16$
 the ratio nobody guesses: $LONG / SHORT = 1.130826361$
 and $LONG = 0.618033989 = 1 / \phi$, exactly

Two lengths, 30 of one and 35 of the other in the finished dome, in a ratio of 1.130826 -- which is not the golden ratio, and that surprise is the whole point.

Computed by two_v_demo.scratch_facts.steps_chords. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**What frequency costs**

same 5 m radius, four frequencies, all built and measured:

1V: 25 struts in 1 length class , 15 panels, 11 hubs

2V: 65 struts in 2 length classes, 40 panels, 26 hubs

3V: 165 struts in 5 length classes, 105 panels, 61 hubs

4V: 250 struts in 6 length classes, 160 panels, 91 hubs

now the part nobody mentions -- where the dome gets cut:

even frequencies (2V, 4V) have a ring of hubs exactly on the equator,

so height = radius exactly: 5.00 m, 5.00 m

odd frequencies (1V, 3V) have no such ring; the cut lands above it,

so they stand taller than a hemisphere: 7.24 m, 5.88 m

frequency buys smoothness and costs part variety -- and an odd frequency quietly buys you extra height you may not have wanted

Computed by two_v_demo.world_facts.steps_frequency. Every line above is printed by that code when the book is built; none of it is typed.

5

Two Lengths, Forty Triangles

The 2V hemisphere: sixty-five edges, two lengths, two shapes, two jigs.

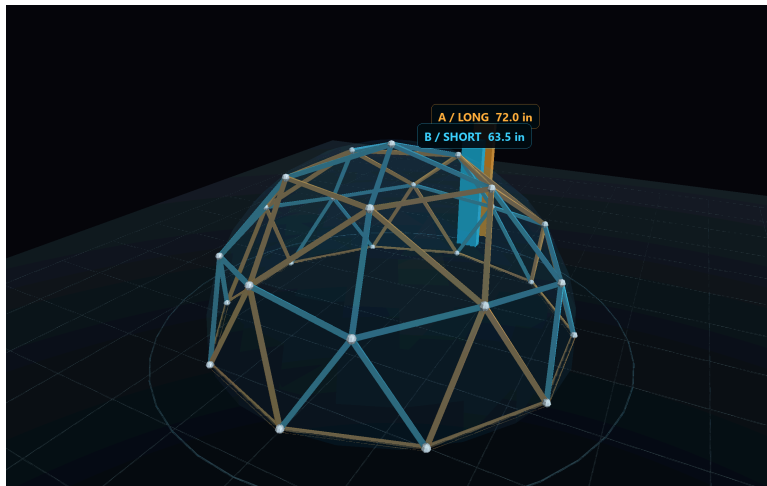


PLATE 5.1 Two lengths of board

The 2V hemisphere and the two boards it is cut from. Whatever the size, the long strut is 1.1308 times the short one.

Rendered by the DomeSim film engine from "2V Geodesic Masterclass", chapter "welcome", 92% through. The masterclass opening, once the dome has drawn, beside its two boards.

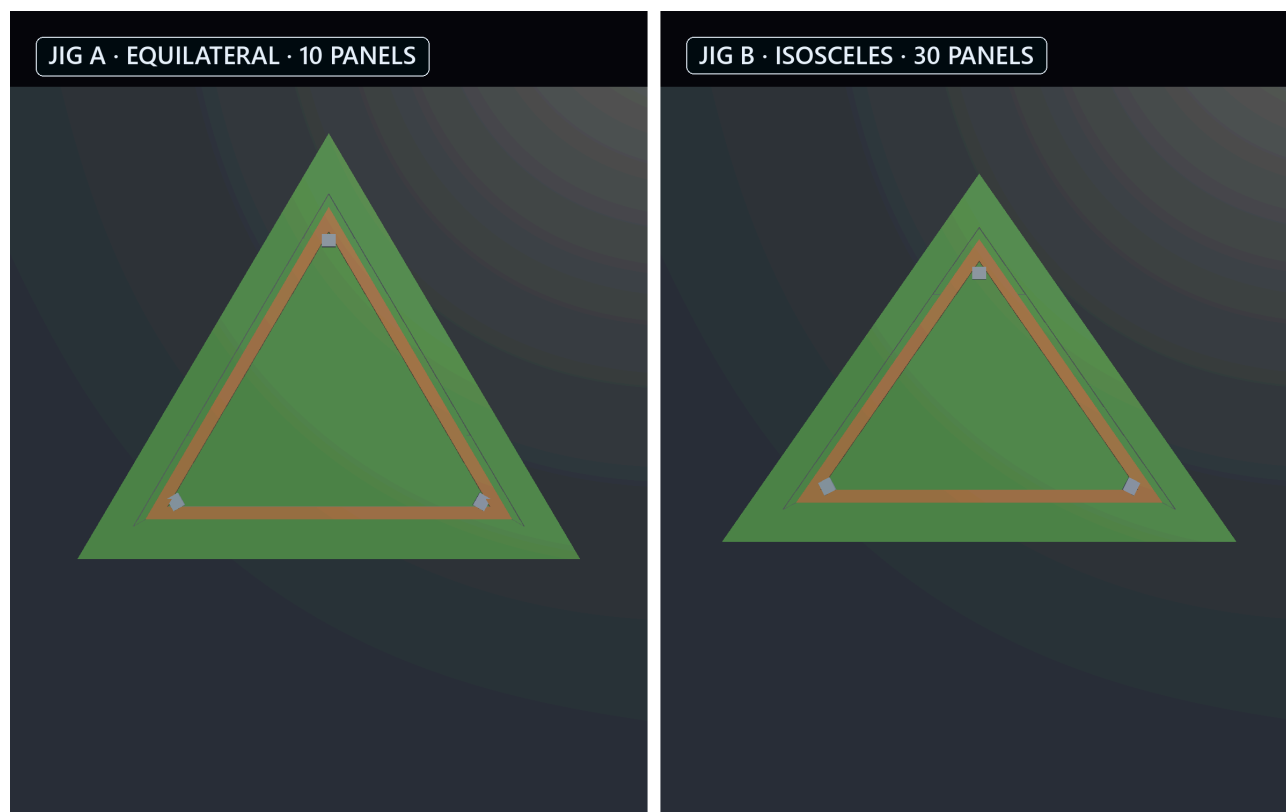


PLATE 5.2 Two jigs make every triangle

The two jigs that make all 40 triangles: 10 equilateral on the left, 30 isosceles on the right, each loaded with its three boards.

Two renders side by side. Dome Forge's jig shop, both jigs loaded and seen from directly above so their true shapes compare, each named.

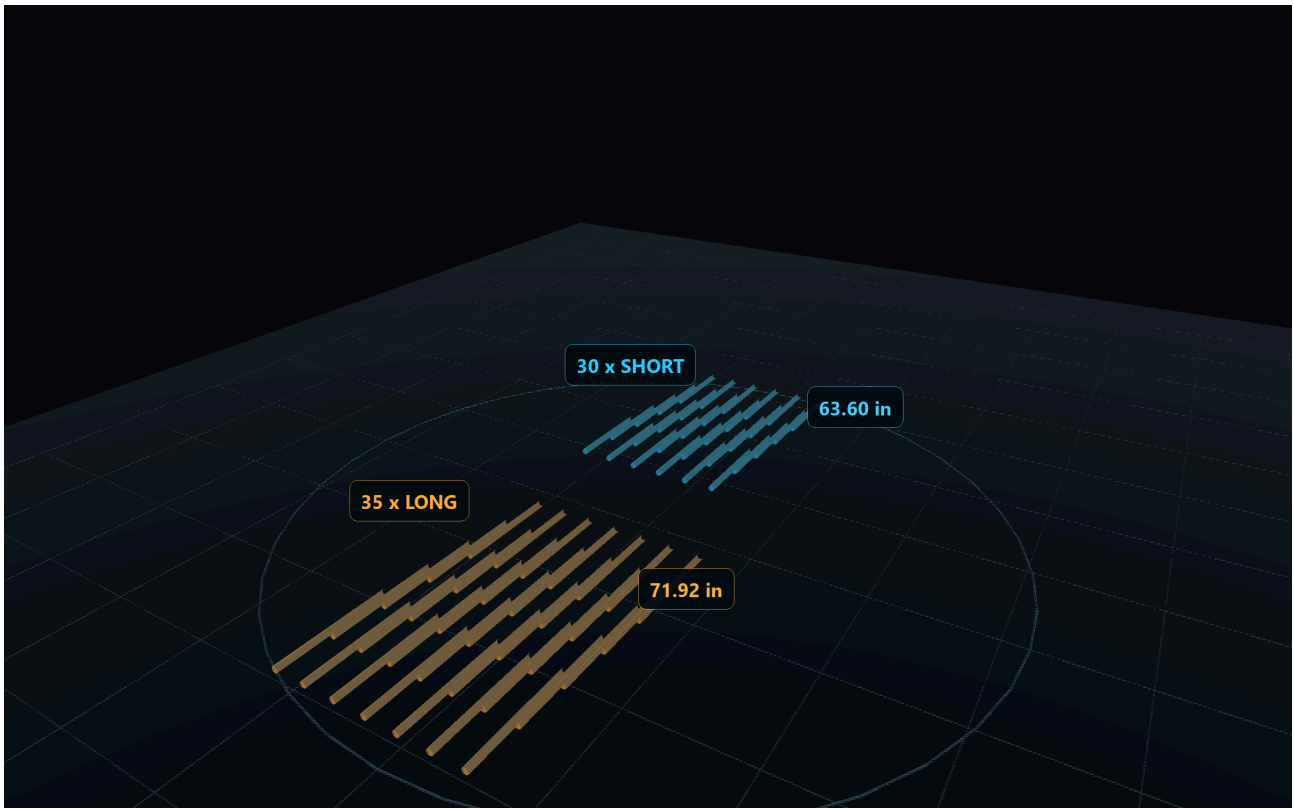


PLATE 5.3 The whole cut list

Every member of the hemisphere is one of two lengths, cut 35 and 30 times.

Rendered by the DomeSim film engine from "2V Geodesic Masterclass", chapter "cut_list", 92% through. The masterclass cut-list scene at full reveal.

WORKED MATH

Counting the half sphere

a dome is the top half of the sphere. Keep every face whose corners all sit at $z \geq 0$, and count what survives:

triangular panels	= 40
SHORT struts	= 30
LONG struts	= 35
struts total	= 65
hubs (corners where struts meet)	= 26
base ring corners	= 10

cross-check: 40 panels x 3 sides = 120 side slots, but interior sides are shared by two panels, which is why 120 slots close into only 65 real struts

not one of these was typed in -- they are counted, now, off the same model being drawn on the left of this screen

40 panels, 65 struts, 26 hubs: a complete shelter, described by two lengths and a count.

Computed by two_v_demo.scratch_facts.steps_counts. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**The reference dome**

declared: long strut A = 72 in (the tools' reference board)
 radius R = A / 0.618034 = 116.50 in = 9.71 ft
 short strut B = 0.546533 x R = 63.67 in
 diameter = 2R = 19.42 ft; height = R
 floor = πR^2 = 296 sq ft
 curved skin = 2 πR^2 = 592 sq ft; flat panels 550 sq ft
 volume = $(2/3) \pi R^3$ = 1,916 cu ft

two boards, 72 in and 63.67 in, enclose 296 sq ft

Computed by domology.science.sheet_reference. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**Two jigs build the dome**

40 triangles, two shapes:
 jig A, equilateral: 10 needed, corners 60 deg, saw miter 60.000 deg
 jig B, isosceles: 30 needed, corners 55.569 / 55.569 / 68.862 deg
 saw miters 62.215 and 55.569 deg
 edge bevel: 9.0146 deg on LONG, 11.2295 deg on SHORT
 every angle re-measured off the assembled 3-D faces before the shop drawing is trusted

two jigs make all 40 triangles

Computed by dome_forge.jigs.jig_specs. Every line above is printed by that code when the book is built; none of it is typed.

6

The Missing Fifteen Degrees

Why flat triangles curve, and where the curve is hiding.

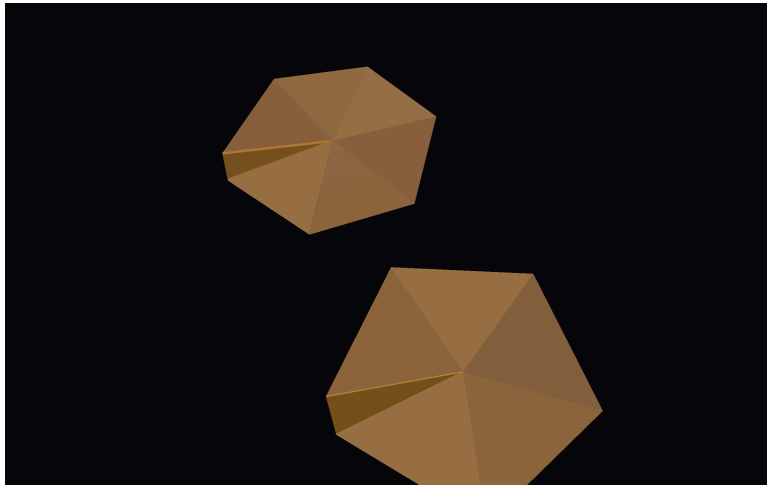


PLATE 6.1 Why a flat fan rises

At a five-way hub the flat corners fall 15.69 degrees short of a full turn, so closing the fan lifts its centre into the curve.

Rendered by Dome Forge with its interface hidden. The jig shop's last stage, which assembles a fan around one hub.

Pentagon cover, long strut 72 in, seam allowance 2 in

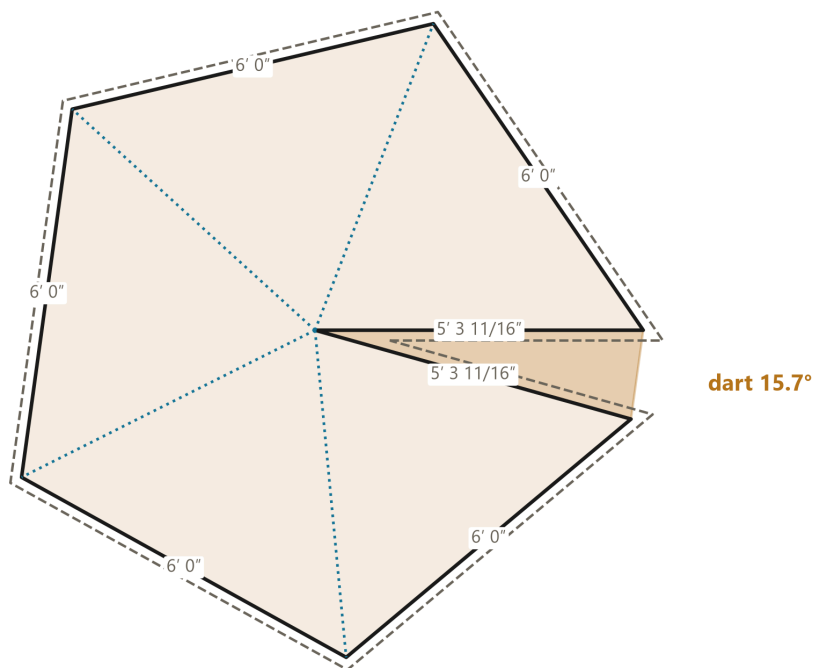


PLATE 6.2 A pentagon, laid flat

Five triangles around a point, developed flat at the reference dome's size, with the 15.69-degree dart that closes to make the curve. The dashed line is the cut with seam allowance; dotted lines are folds.

Drawn when the book is built. From Dome Forge's cover-pattern geometry, dome_forge.patterns.

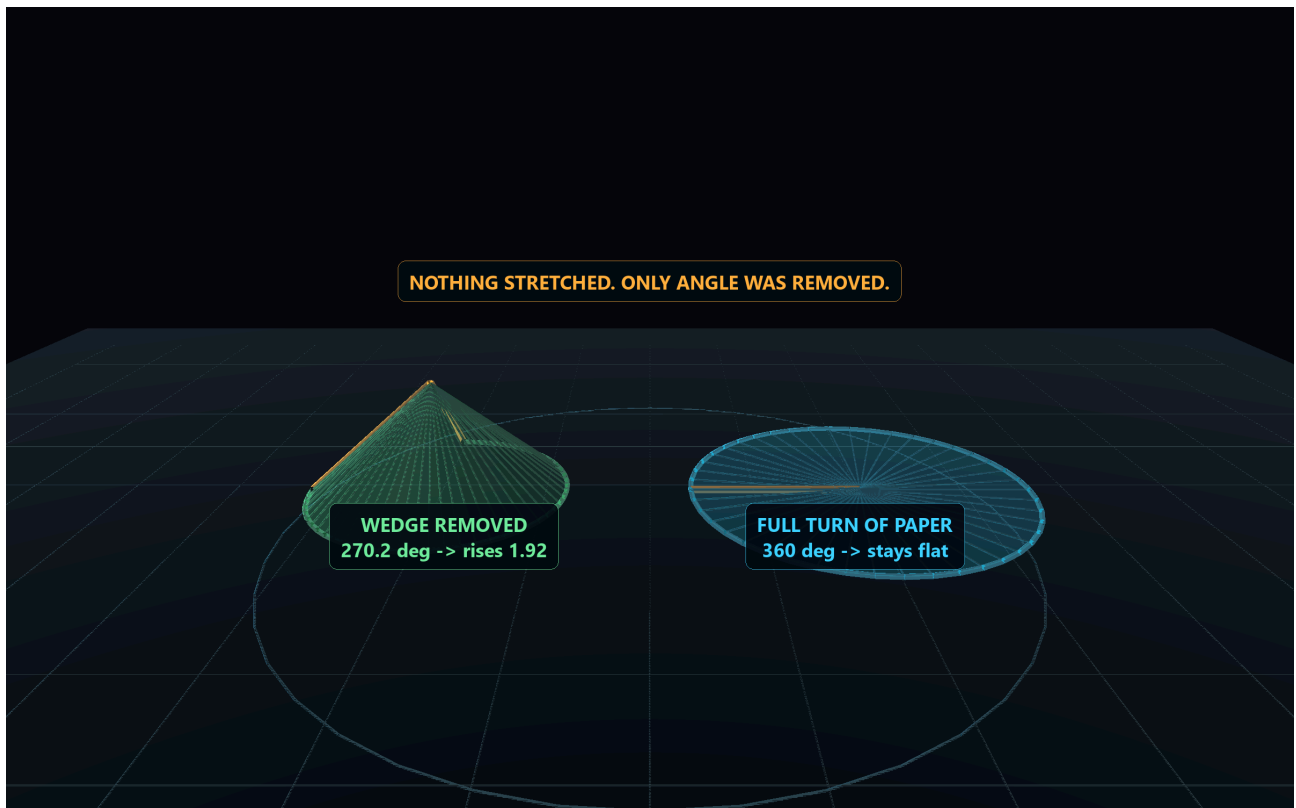


PLATE 6.3 Curvature is bought with missing angle

A sheet of hexagons stays flat, because three hexagon corners fill a full turn. A closed cage of polygons is always short by 720 degrees in total.

Rendered by the DomeSim film engine from "Hexagonal Dome Masterclass", chapter "buy_curve", 80% through. The hex masterclass's deficit scene.

WORKED MATH

The missing angle

a flat equilateral corner is 60 deg; six fill 360 deg exactly (flat)
 at a 5-way hub the flat corners sum to 344.31 deg: short by 15.69
 at a 6-way hub they sum to 342.28 deg: short by 17.72
 this hemisphere: 6 five-way hubs, 10 six-way
 Descartes: a closed convex cage is short exactly 720 deg in total
 the icosahedron's 12 corners $\times$ 60 deg = 720 deg

the missing angle is not a flaw to hide -- it is the curve

Computed by dome_forge.jigs.vertex_report. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**Always twelve pentagons**

H = 20 hexagons, P pentagons, every corner 3-way

$F = P + H$ $E = (5P + 6H)/2$ $V = (5P + 6H)/3$

$V - E + F = 2 \rightarrow P/6 = 2 \rightarrow P = 12$

check: $V \ 60 - E \ 90 + F \ 32 = 2$

try any number of hexagons: 12, 12, 12

a closed cage of pentagons and hexagons always has exactly twelve pentagons

Computed by two_v_demo.hex_geometry.euler_pentagon_proof. Every line above is printed by that code when the book is built; none of it is typed.

7

Hubs, Hubless, and the Seam

Where a dome actually fails, and three honest ways to join one.

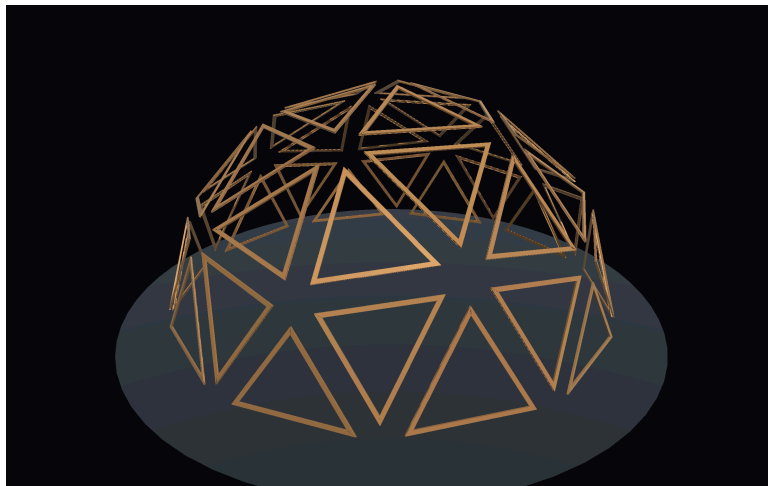
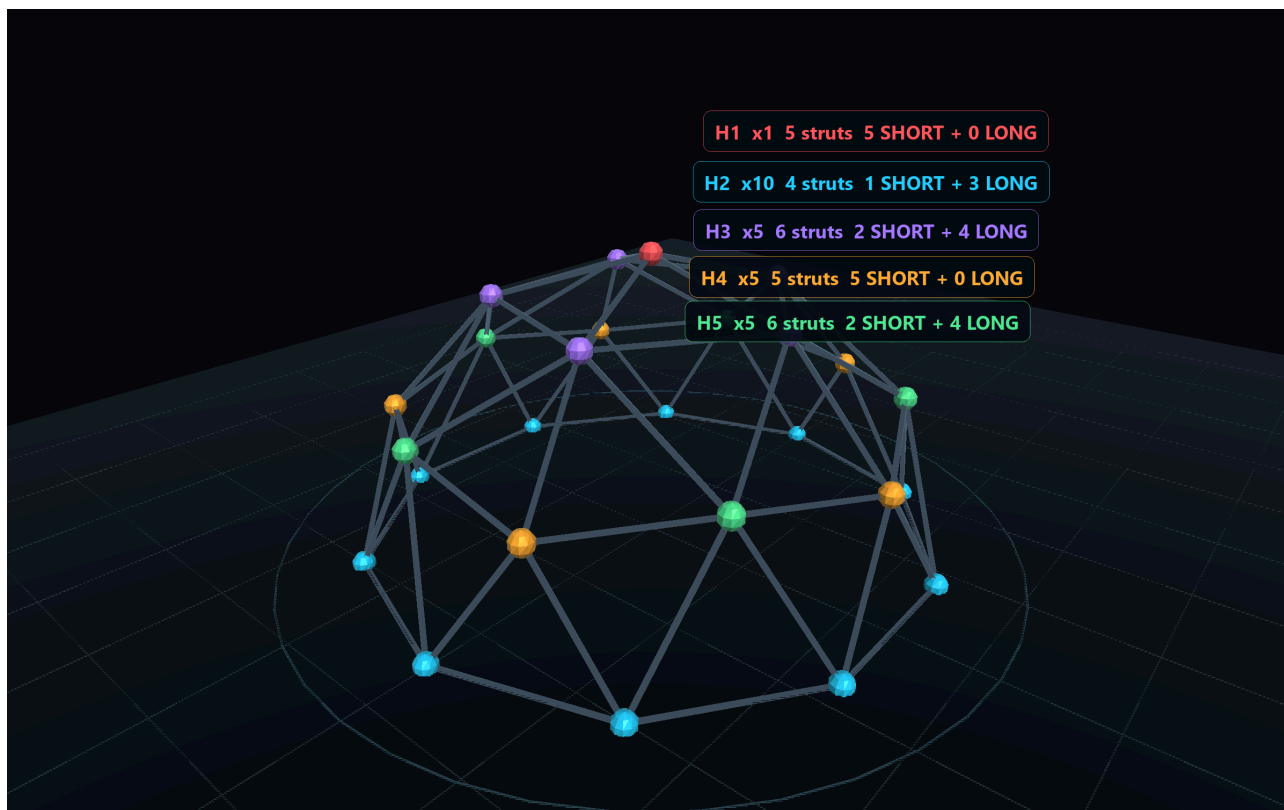


PLATE 7.1 Forty separate triangles

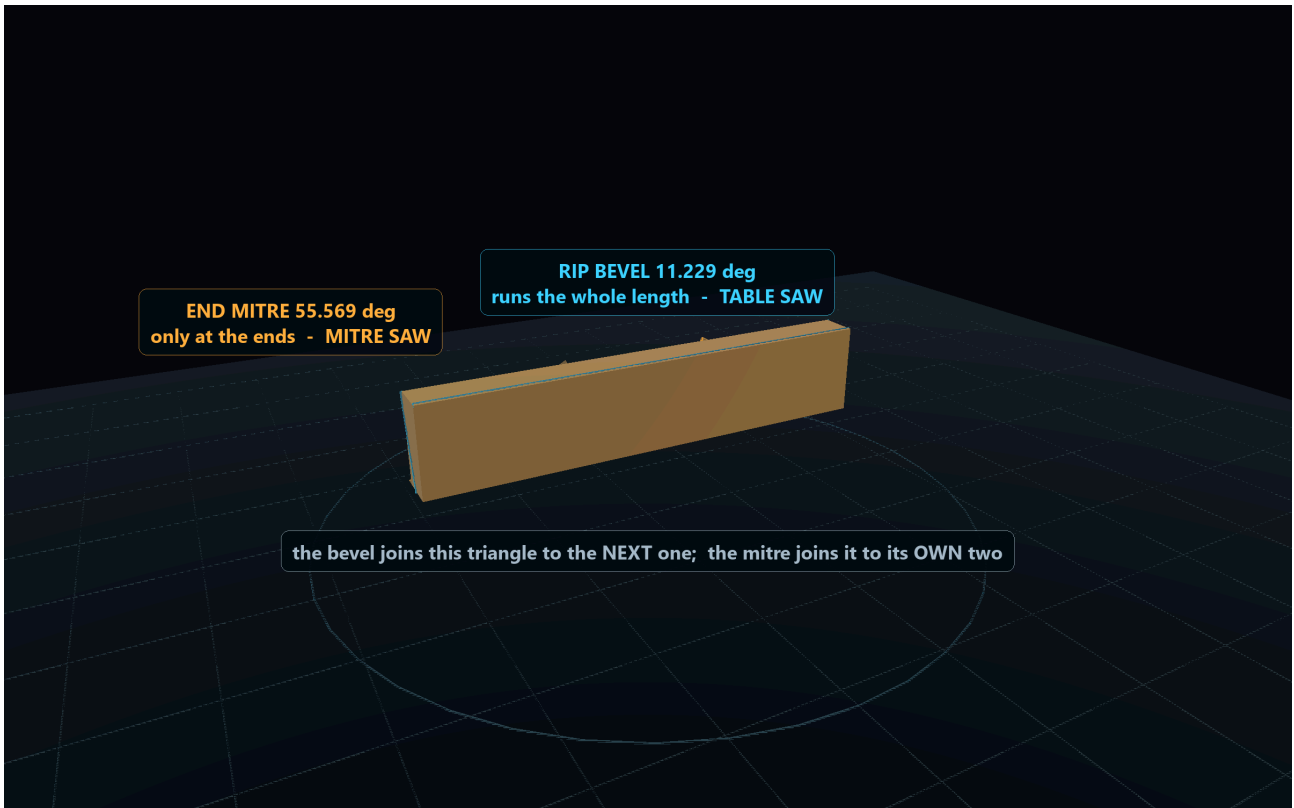
The hubless frame: 40 triangles, each bringing its own three members, bolted to their neighbours. Here every triangle is lifted straight out from the centre so each can be seen whole.

Rendered by Dome Forge with its interface hidden. Only the panel assemblies shown, fills off; every face selected and popped out.

**PLATE 7.2 A dome on hubs**

65 struts meet at 26 hubs; each hub gathers up to six members at compound angles.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "hubs", 85% through. The construction masterclass's hub scene near its end.

**PLATE 7.3** Two angles at once

A hubless strut's end needs a mitre and a bevel together: 6 distinct setups in all, 6 past a typical saw's stop.

Rendered by the DomeSim film engine from "The Compound Cut, Both Machines", chapter "two_angles", 70% through. The compound-cut film's opening scene, moved in so its labels separate.

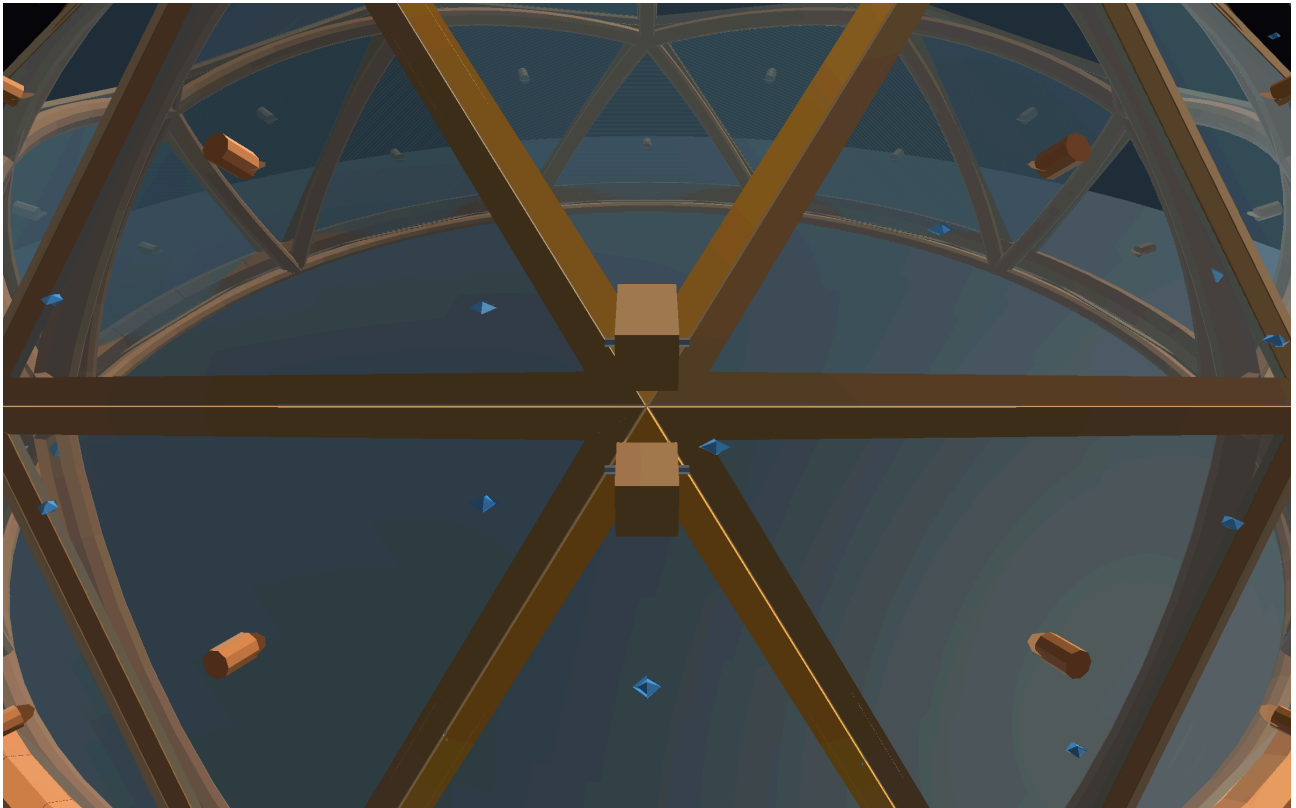


PLATE 7.4 An hourglass and its waist

Two equilateral triangles meeting at a single point, joined by wooden braces that carry the load into their sides rather than their end-grain tips.

Rendered by Dome Forge with its interface hidden. The group editor on the first hourglass, moved in close on its braced waist.

WORKED MATH

Hubs or no hubs

both are 2V domes from this world:

Whole Trunk Lodge - 20 ft -- Hub & Strut

struts 65, hubs 26, bolts 0

Split-Log Homestead -- Hubless Doubled

struts 120, hubs 0, bolts 130

the hubless frame carries 1.8x the sticks

and exactly 0 hub connectors

every triangle brings its own three boards, so each shared

seam ends up two boards thick and bolts to its neighbour

and because its struts are split logs, the model reports 30 trees to harvest at 4 wedges a log

you trade wood for hardware: more sticks, zero hubs, and nothing on the critical path you cannot cut yourself

Computed by two_v_demo.world_facts.steps_hub_vs_hubless. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**A hubless frame, counted**

40 triangles, each bringing its own three members:

40 x 3 = 120 members

65 edges: 55 shared seams (two members each) + 10 rim edges (one)

check: 55 x 2 + 10 = 120

6 distinct compound saw setups; 6 past a typical saw's stop

no hubs to buy: the joint is the seam between two panels

Computed by two_v_demo.hubless_geometry.hubless_summary. Every line above is printed by that code when the book is built; none of it is typed.

8

Wind, Water, and Weight

What the shape does for you when the weather turns, and what it does not.

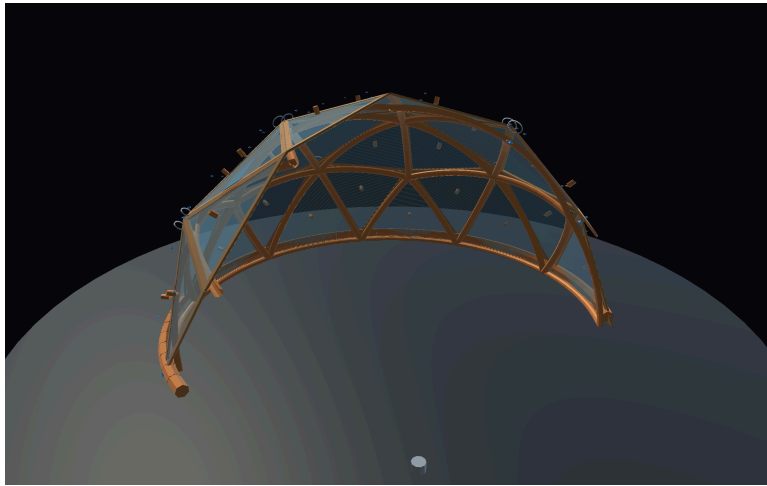


PLATE 8.1 The water dome in section

Dished panels shed rain to micro-drains; seam veins carry anything that gets past a seam down to a collector ring, a downpipe and a cistern.

Rendered by Dome Forge with its interface hidden. The default water stack with the cutaway open and the flow mid-run.

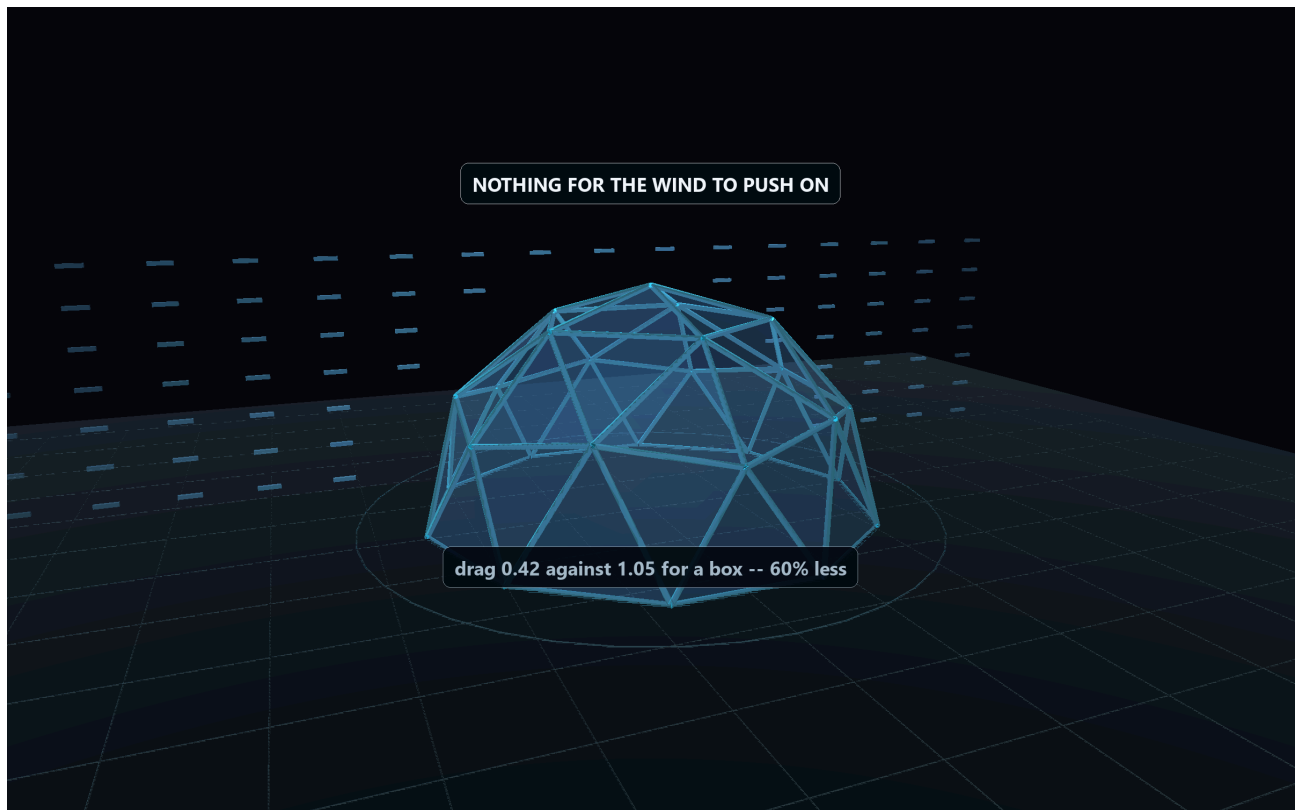


PLATE 8.2 Wind around a round body

Wind parts around a curved shell instead of piling against a flat wall.

Rendered by the DomeSim film engine from "Build A Dome: The Campaign, Version Two", chapter "wind", 70% through. The campaign film's wind scene.

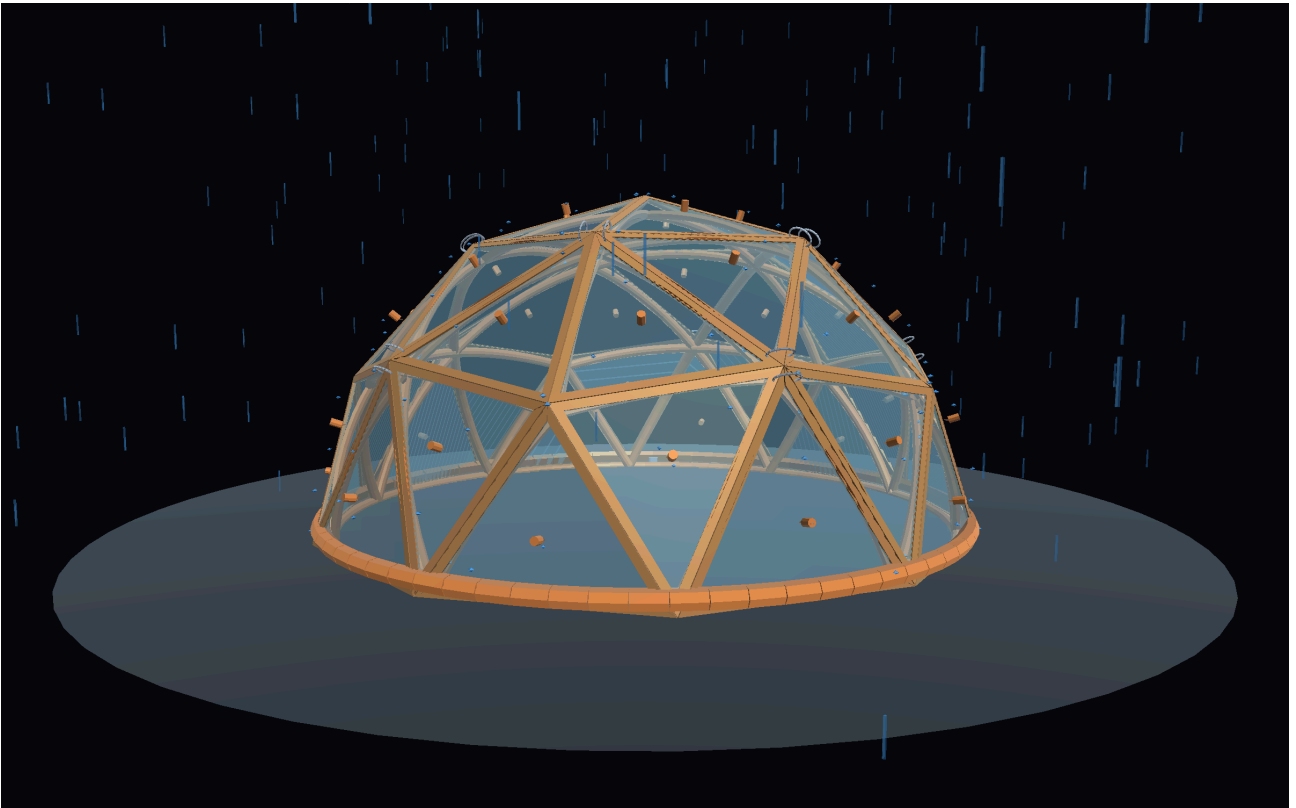


PLATE 8.3 In the rain

The same dome from outside, each panel dished to its own low point.

Rendered by Dome Forge with its interface hidden. Rain and panel runoff on, from low on the ground.

9

The Dome Family

Hex, zome, and higher frequencies: when a different dome is the better answer.



PLATE 9.1 A soccer-ball dome

The truncated icosahedron as a dome: pentagons and hexagons in place of triangles, and always 12 pentagons.

Rendered by the DomeSim film engine from "Hexagonal Dome Masterclass", chapter "one_hexagon", 80% through. The hex masterclass's soccer-ball scene.

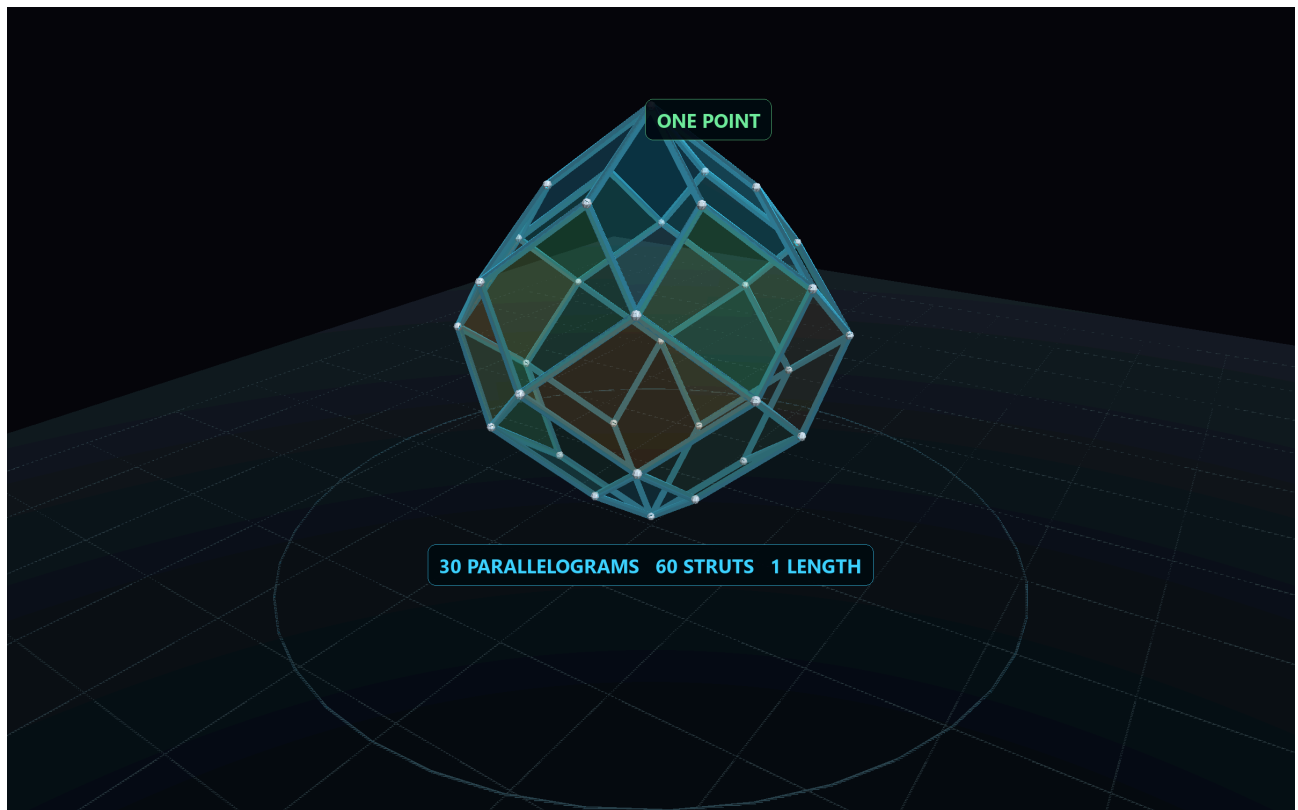


PLATE 9.2 A zome

A zome: 30 flat four-sided panels spiralling up to a single point.

Rendered by the DomeSim film engine from "Zome Construction Masterclass", chapter "what", 85% through. The zome masterclass's opening zome, fully drawn.

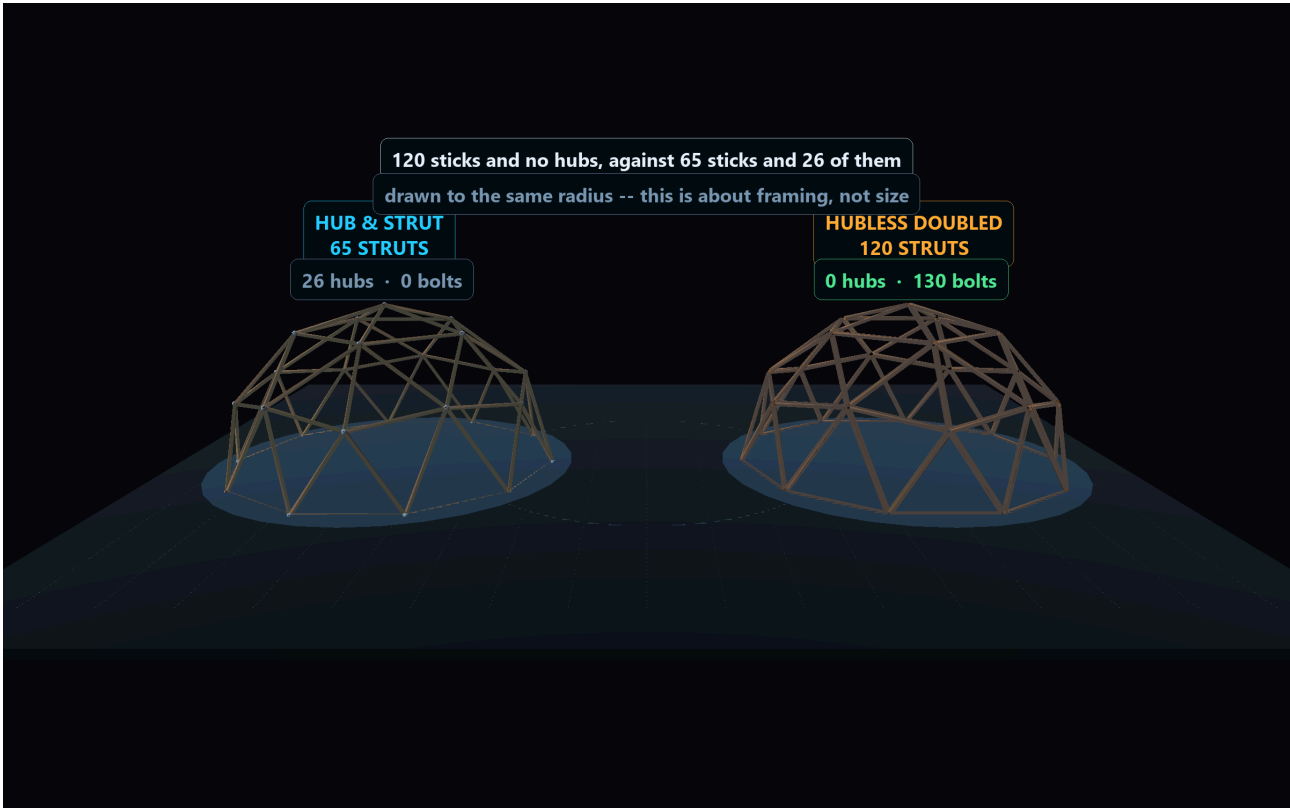


PLATE 9.3 Three ways to frame
 Hub and strut, hubless doubled triangles, and continuous arcs, compared on real designs from the catalogue.
 Rendered by the DomeSim film engine from "Every Dome In The World", chapter "framing", 80% through. The world film's framing scene.

WORKED MATH
Twelve designs, counted

12 shipped designs, each rebuilt live:

Timber Workshop	3V	165 struts	105 panels	819 sq ft
Glass Studio Loft	4V	250 struts	160 panels	845 sq ft
Split-Log Homestead	2V	120 struts	40 panels	1217 sq ft
Whole Trunk Lodge - 20 ft	2V	65 struts	40 panels	3281 sq ft
Grow Dome	3V	165 struts	105 panels	819 sq ft
Hex Cell Pavilion	3V	165 struts	105 panels	1179 sq ft
Continuous Steel Arc Hanga	2V	65 struts	40 panels	2164 sq ft
Rebar Garden Dome	3V	165 struts	105 panels	991 sq ft
Concrete Monocoque Form	3V	165 struts	105 panels	1179 sq ft
Woodland Hex Mirror	3V	165 struts	105 panels	991 sq ft
Woodland Square Mirror	4V	250 struts	160 panels	1023 sq ft
Treehouse Canopy Dome	2V	65 struts	40 panels	597 sq ft

one geometry engine, 12 buildings, 597 to 3281 square feet -- and not one of these numbers was typed in

Computed by two_v_demo.world_facts.steps_catalogue. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**Size is a number you type**

part counts depend on frequency alone, never on size:

Timber Workshop $r = 5.0 \text{ m} \rightarrow 165 \text{ struts, } 105 \text{ panels}$

Grow Dome $r = 5.0 \text{ m} \rightarrow 165 \text{ struts, } 105 \text{ panels}$

Hex Cell Pavilion $r = 6.0 \text{ m} \rightarrow 165 \text{ struts, } 105 \text{ panels}$

Rebar Garden Dome $r = 5.5 \text{ m} \rightarrow 165 \text{ struts, } 105 \text{ panels}$

same three numbers, different buildings

smallest in the catalogue: Treehouse Canopy Dome, 28 ft across, 597 sq ft

largest: Whole Trunk Lodge - 20 ft, 65 ft across, 3281 sq ft

that is 5.5x the floor from the same list of operations

5.5 times the house for the same parts list -- size is a number you type, not a harder build

Computed by two_v_demo.world_facts.steps_scale. Every line above is printed by that code when the book is built; none of it is typed.

10

What a Dome Costs, and What It Saves

The honest arithmetic: where the shape saves money, and where it stops.

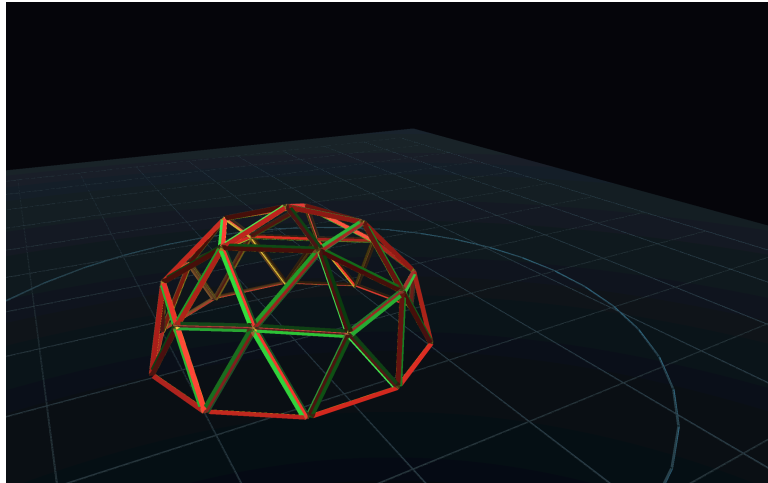


PLATE 10.1 The shell being priced

The simulator's solved wedge dome: the shell whose framing the cost chapters price.

Rendered by the DomeSim film engine from "Why Build This Way", chapter "question", 60% through. Why Build This Way's dome scene.

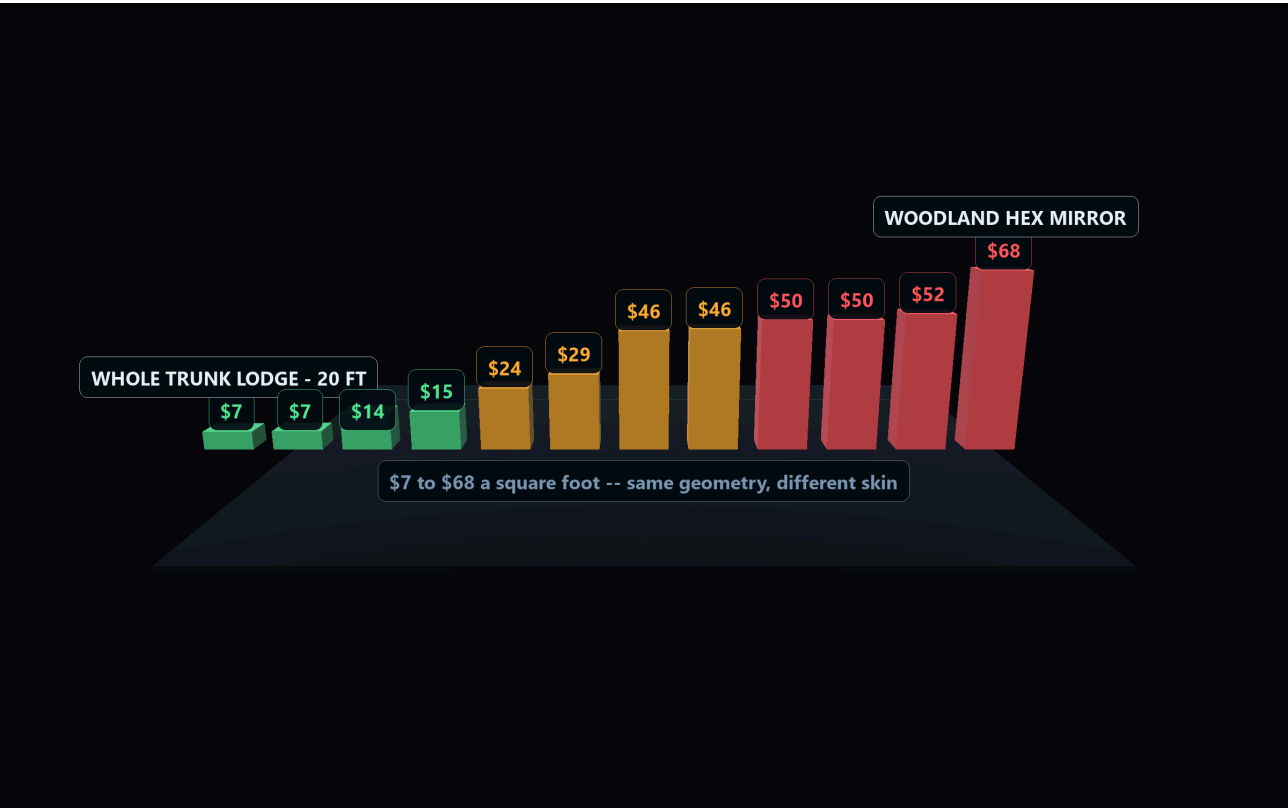


PLATE 10.2 The same shape, clad twelve ways
Across the catalogue a square foot costs from 7 to 68 dollars: the cladding decides the price, not the shape.
Rendered by the DomeSim film engine from “Every Dome In The World”, chapter “cladding”, 80% through. The world film's cladding scene.

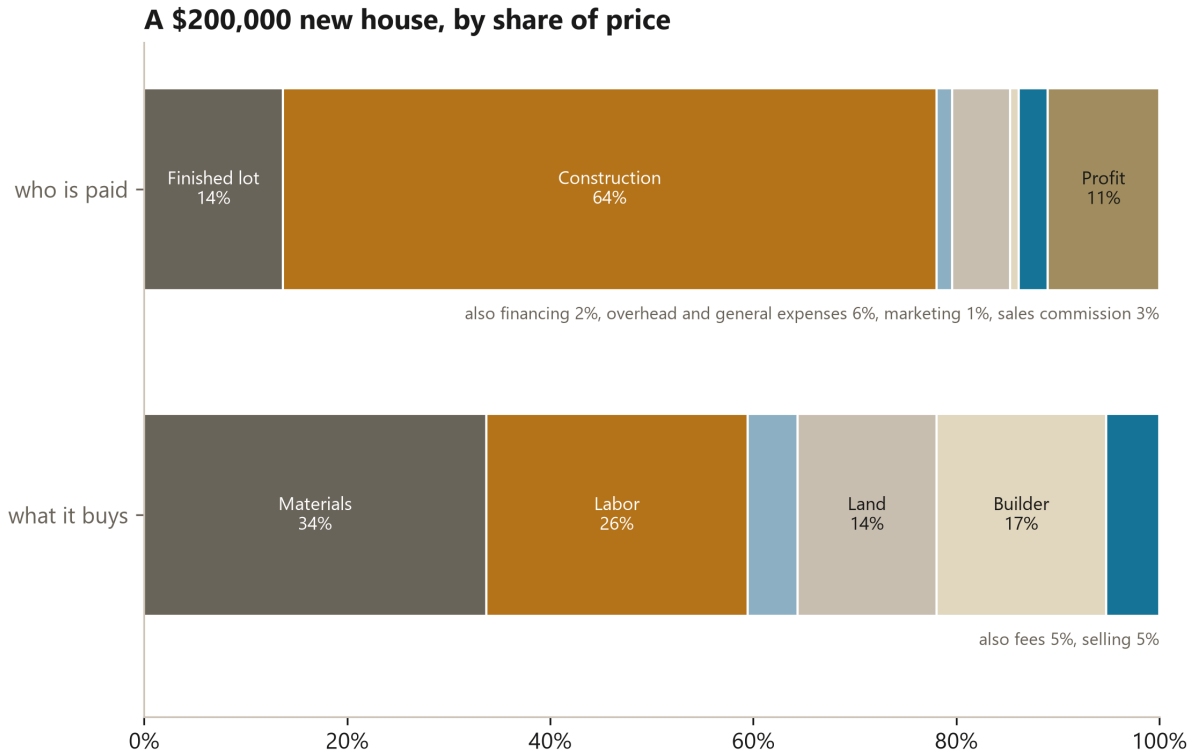


PLATE 10.3 Where a new house's price goes

A new house's price from the builders' survey, first by who is paid, then by what the money buys.

Drawn when the book is built. From house_economics, NAHB's 2024 cost survey.

WORKED MATH

Twelve designs, priced

every design priced off its own bill of materials

(frame + hubs + panels + layers + foundation + fit-out):

Whole Trunk Lodge - 20 ft	\$ 22,747	\$ 7/sq ft	Canvas
Rebar Garden Dome	\$ 7,002	\$ 7/sq ft	Open
Grow Dome	\$ 11,489	\$ 14/sq ft	Polycarb Twinw
Split-Log Homestead	\$ 18,067	\$ 15/sq ft	Plywood
Timber Workshop	\$ 19,514	\$ 24/sq ft	Plywood
Glass Studio Loft	\$ 24,670	\$ 29/sq ft	Glass Window
Treehouse Canopy Dome	\$ 27,243	\$ 46/sq ft	Hex Composite
Continuous Steel Arc Hanga	\$ 100,454	\$ 46/sq ft	Open
Concrete Monocoque Form	\$ 58,661	\$ 50/sq ft	Concrete Form
Woodland Square Mirror	\$ 50,968	\$ 50/sq ft	Square Mirror
Hex Cell Pavilion	\$ 61,355	\$ 52/sq ft	Hex Composite
Woodland Hex Mirror	\$ 67,346	\$ 68/sq ft	Hex Mirror

\$7 to \$68 a square foot across the same geometry -- what you clad it in decides the price, not the shape

Computed by two_v_demo.world_facts.steps_economics. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

What the house figures rest on

PUBLISHED NAHB 2024 cost survey: who gets a house's price, and 37 line items of building it (41 builders; not representative)
PUBLISHED Census: 7.6 months, start to finish; BLS wages, May 2025
PUBLISHED Lumber peak: \$1,500 per 1,000 bd ft; +\$35,872 on an average home
PUBLISHED Manufactured homes: \$86.62 a sq ft; factory split from 2002
THE AUTHOR'S \$200,000 house, \$100,000 trailer, 40% labor, 80 hours of shell work
DECIDED the 12% federal bracket; state tax left out
ESTIMATED dealer 25%, labor by item, run share 35%, curve +25%

So every estimate below is a guess that can be changed, and says so

Computed by two_v_demo.house_economics.source_lines. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

One shell on one sheet

300 sq ft of shell in 80 hours: 3.75 sq ft an hour
\$4,800 of commercial framing displaced
\$60 an hour production value; 2.5x the framing labor alone
\$10,922 of 30-year payments avoided: \$136.53 an hour, not a wage
80 hours of life kept, against earning \$25 an hour to buy it
Worth building while your take-home is under \$50 an hour
One shell a year for 30 years: \$144,000 of framing, 9,000 sq ft

One hour: \$60 of shelter today, \$137 of payments never made

Computed by two_v_demo.why_build_economics.score_lines. Every line above is printed by that code when the book is built; none of it is typed.

12 DESIGNS, ONE GEOMETRY ENGINE

597 to 3,281 square feet -- drawn to true relative size

BOOK TWO

The Journey

How one person built a dome in software before building one in wood, and what each tool taught.

- 11 A Dome in Software First
- 12 The Factory Line
- 13 Teaching the Geometry to Myself
- 14 Dome Forge and the Water Dome
- 15 The Frankendome
- 16 The Cut That Changed It
- 17 Two Trees and a Fortnight
- 18 What I Got Wrong, and What Is Still Unproven

11

A Dome in Software First

A walkable world where a dome could be built, priced and argued with before a single cut.

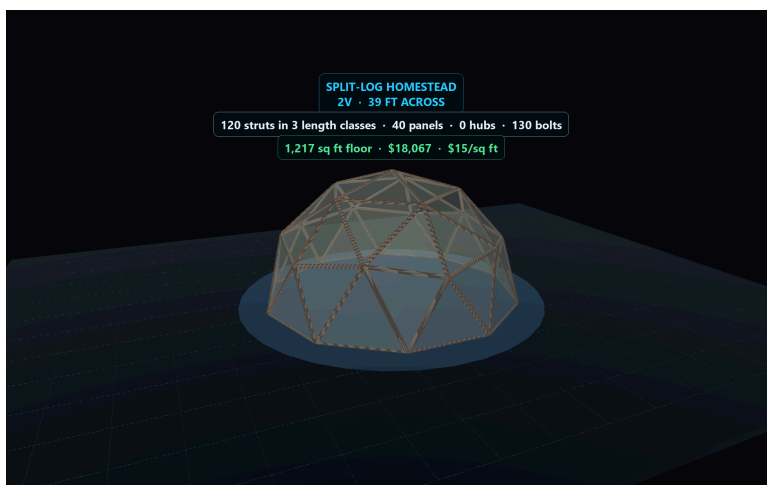


PLATE 11.1 The Split-Log Homestead

Half-round logs on the long seams, quarter-round on the short: a preset built from split wood, rebuilt by the simulator.

Rendered by the DomeSim film engine from "Every Dome In The World", chapter "show_02", 80% through. The world film's Split-Log Homestead chapter.

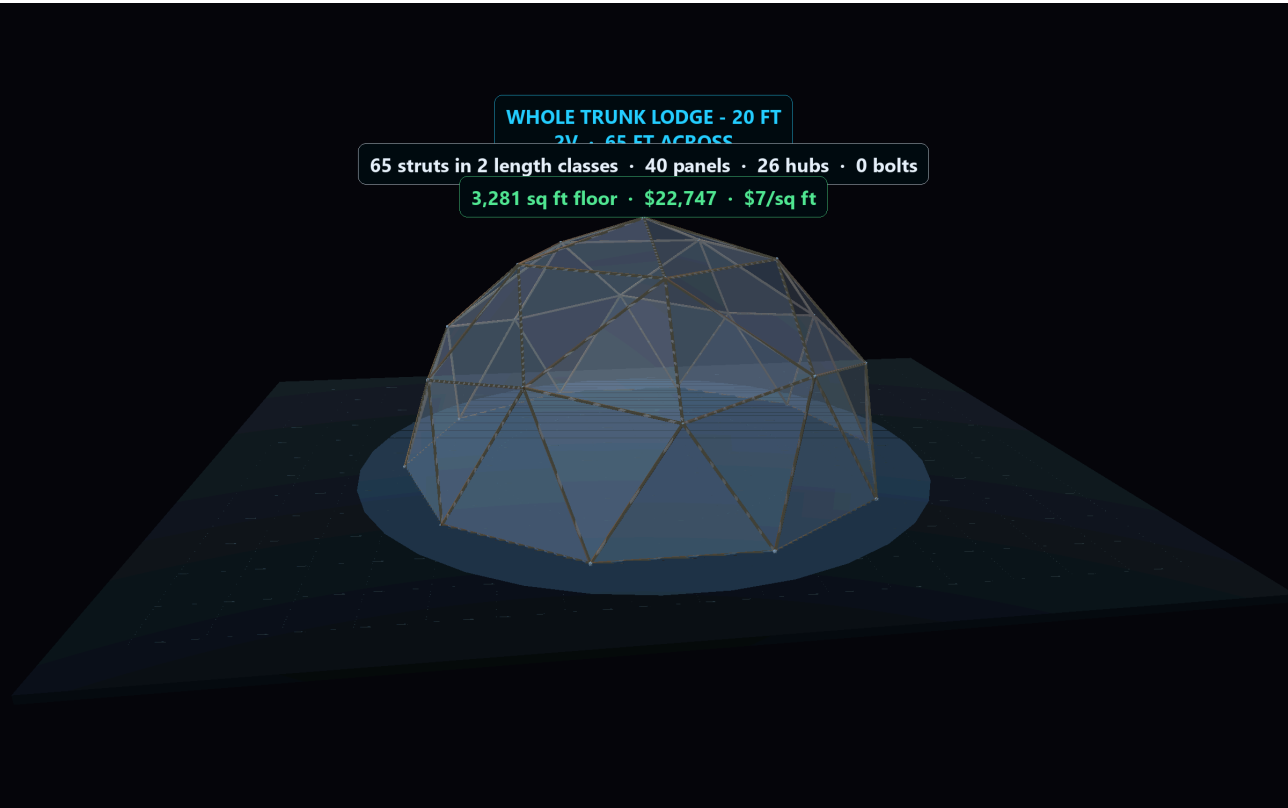


PLATE 11.2 The Whole Trunk Lodge

Whole trunks as struts: a dome sized to the trees that frame it.

Rendered by the DomeSim film engine from “Every Dome In The World”, chapter “show_03”, 80% through. The world film’s Whole Trunk Lodge chapter.

WORKED MATH

Twelve designs, counted

12 shipped designs, each rebuilt live:

Timber Workshop	3V	165 struts	105 panels	819 sq ft
Glass Studio Loft	4V	250 struts	160 panels	845 sq ft
Split-Log Homestead	2V	120 struts	40 panels	1217 sq ft
Whole Trunk Lodge - 20 ft	2V	65 struts	40 panels	3281 sq ft
Grow Dome	3V	165 struts	105 panels	819 sq ft
Hex Cell Pavilion	3V	165 struts	105 panels	1179 sq ft
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Rebar Garden Dome	3V	165 struts	105 panels	991 sq ft
Concrete Monocoque Form	3V	165 struts	105 panels	1179 sq ft
Woodland Hex Mirror	3V	165 struts	105 panels	991 sq ft
Woodland Square Mirror	4V	250 struts	160 panels	1023 sq ft
Treehouse Canopy Dome	2V	65 struts	40 panels	597 sq ft

one geometry engine, 12 buildings, 597 to 3281 square feet -- and not one of these numbers was typed in

Computed by two_v_demo.world_facts.steps_catalogue. Every line above is printed by that code when the book is built; none of it is typed.

12

The Factory Line

Fifteen stations, four products, and the question an investor asks.

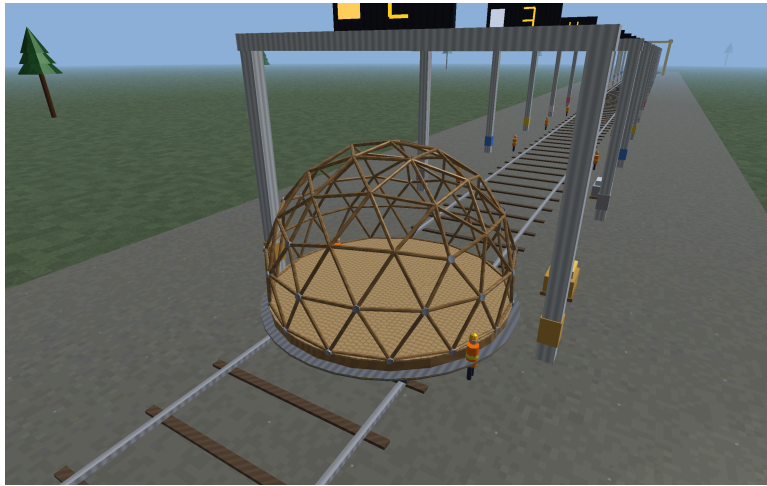


PLATE 12.1 The shell on the line

The frame raised on the transfer carriage at the framing station, before anything hides it.

Rendered by the Dome Home Assembly Line. The line stopped as the framing station finishes.

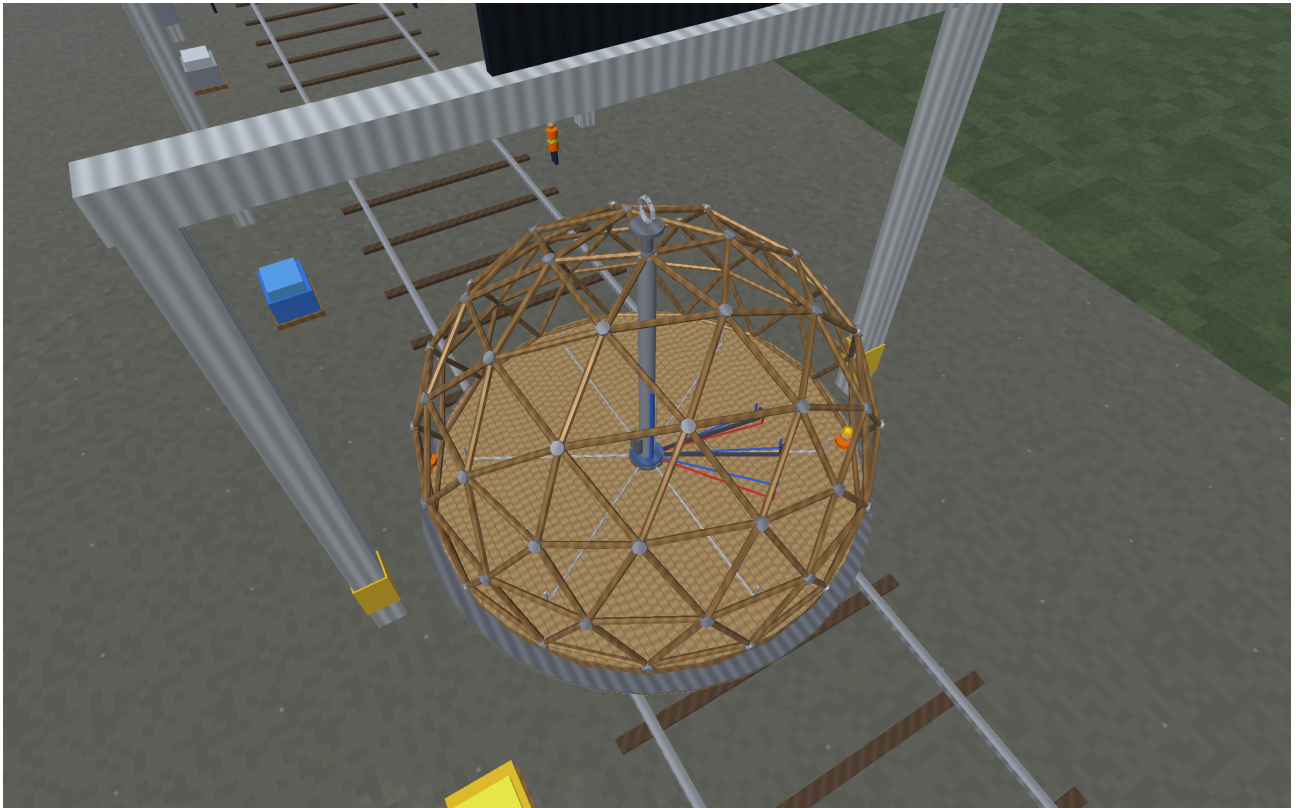


PLATE 12.2 Services before skin

Water and power routed through the floor to one central column while the shell is still open.

Rendered by the Dome Home Assembly Line. The line at the power station, seen from above.



PLATE 12.3 Fit-out through a cutaway

Kitchen, bath and bedroom fitted at the interior station, seen through a roof cutaway.

Rendered by the Dome Home Assembly Line. The interior station with the cutaway forced open.

13

Teaching the Geometry to Myself

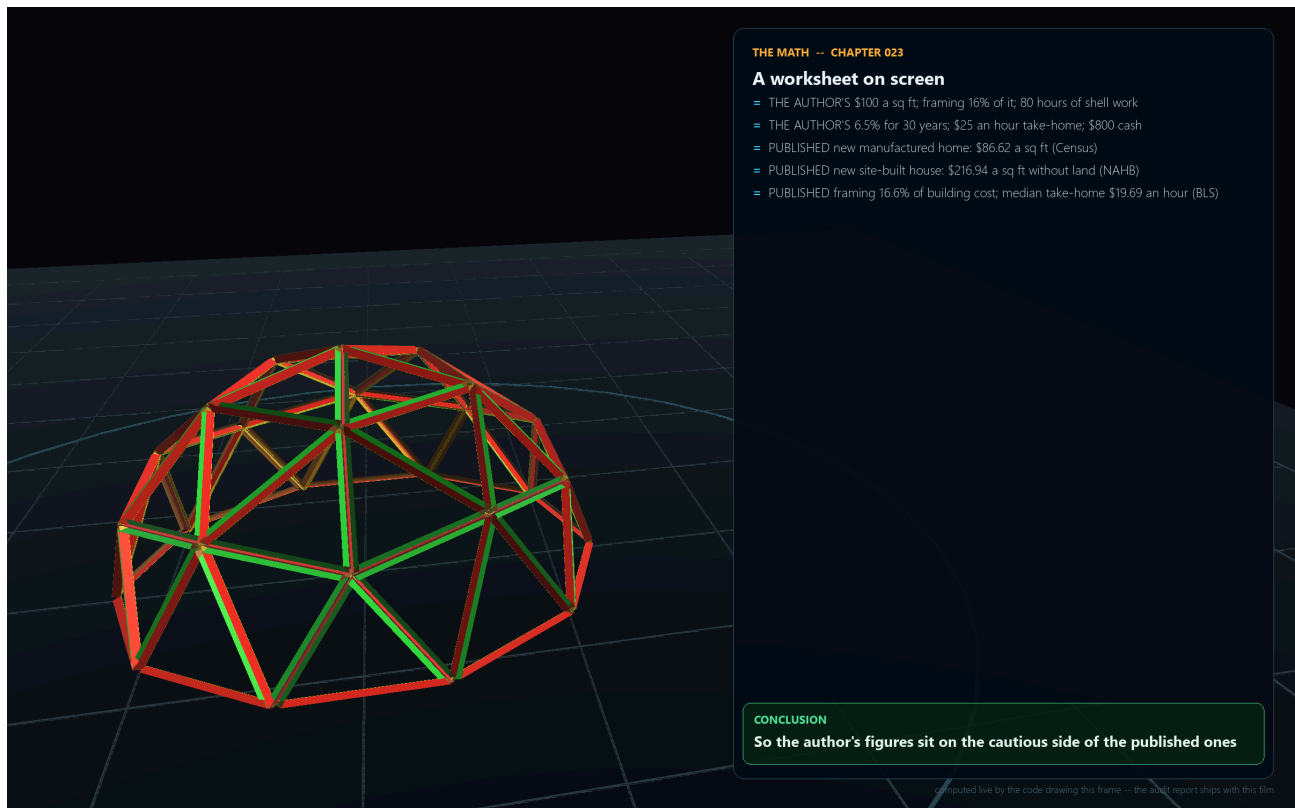
A film engine, a lesson from scratch, and the rule that every number on screen is computed.



PLATE 13.1 From geometry to a pixel

The whole chain the from-scratch film derives on camera: points, edges, triangles, the camera, the screen.

Rendered by the DomeSim film engine from "From Scratch: Every Calculation Behind a Dome on Screen", chapter "pipeline", 90% through. The from-scratch film's pipeline scene, moved in so each stage reads.

**PLATE 13.2 A worksheet on screen**

A film's math screen: the picture stays live while every figure is derived beside it. This book prints the same worksheets.

Rendered by the DomeSim film engine from "Why Build This Way", chapter "sources", 97% through. Why Build This Way's sources chapter with its worksheet kept.

WORKED MATH**From factors to inches**

the model is still unit-free. Two measured boards fix it:

measured LONG = 72.0 in

measured SHORT = 63.5 in

each board on its own implies a radius:

R from LONG = $72.0 / 0.618034 = 116.498$ in

R from SHORT = $63.5 / 0.546533 = 116.187$ in

they disagree, because tapes and saws are not exact

least squares picks the radius that misses both by least:

best-fit R = 116.362 in = 9.70 ft

residuals = +0.084 in and -0.096 in

multiply the two chord factors by that radius:

30 SHORT @ 63.596 in

35 LONG @ 71.916 in

hub-centre to hub-centre; the physical cut is that minus

whatever the connector eats at each end

floor 295.4 sq ft, height 9.70 ft, span 19.39 ft

One multiplication turns the unit-free model into a cut list for a 19.4 foot dome.

Computed by two_v_demo.scratch_facts.steps_scale. Every line above is printed by that code when the book is built; none of it is typed.

14

Dome Forge and the Water Dome

Layers, hubless triangles, a jig shop, and seams that drink the rain.

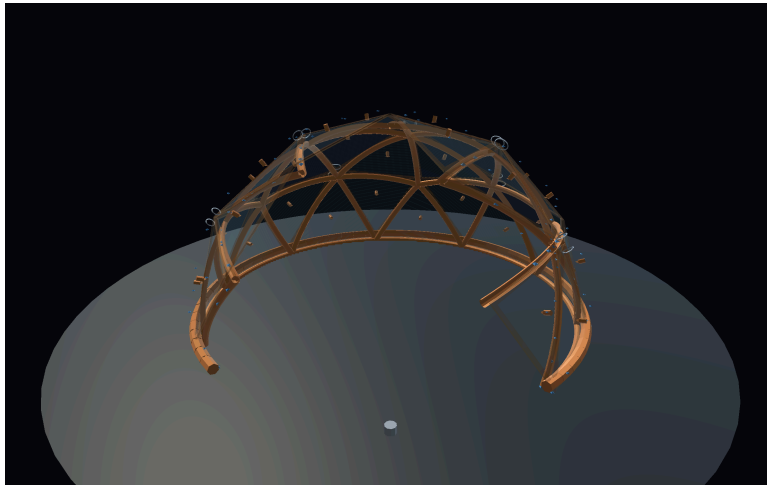


PLATE 14.1 Every layer at once

The water dome with its panels made translucent, so the veins, collector ring, downpipe and cistern show through them.

Rendered by Dome Forge with its interface hidden. The panel assemblies at a third opacity, with the cutaway open.

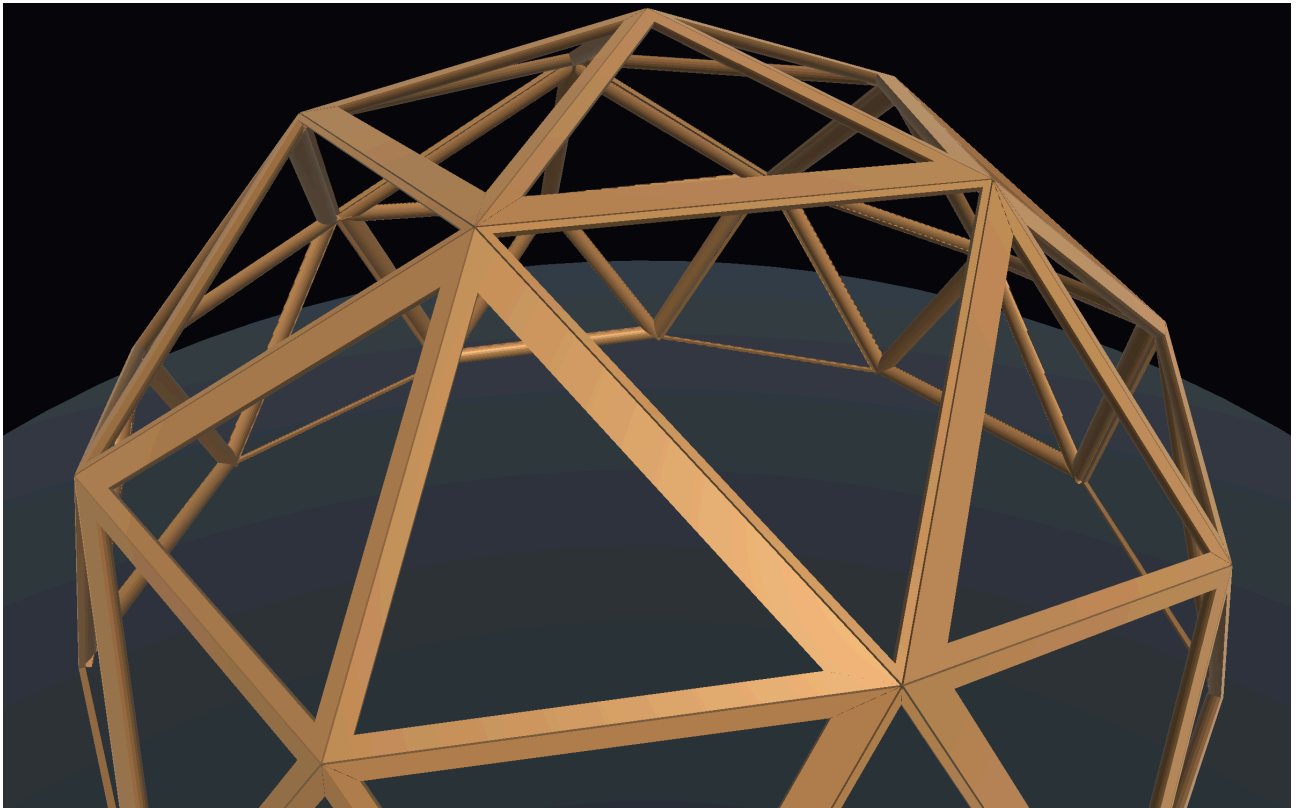


PLATE 14.2 Three kinds of stock in one frame

Dome Forge's split-log stack: half-round logs on the long seams, quarter-round on the short, and a lighter 2x2 on the equilateral caps, with the fills turned off so the frame shows.

Rendered by Dome Forge with its interface hidden. The split-log stack, fills off, moved in close so the different stock reads.

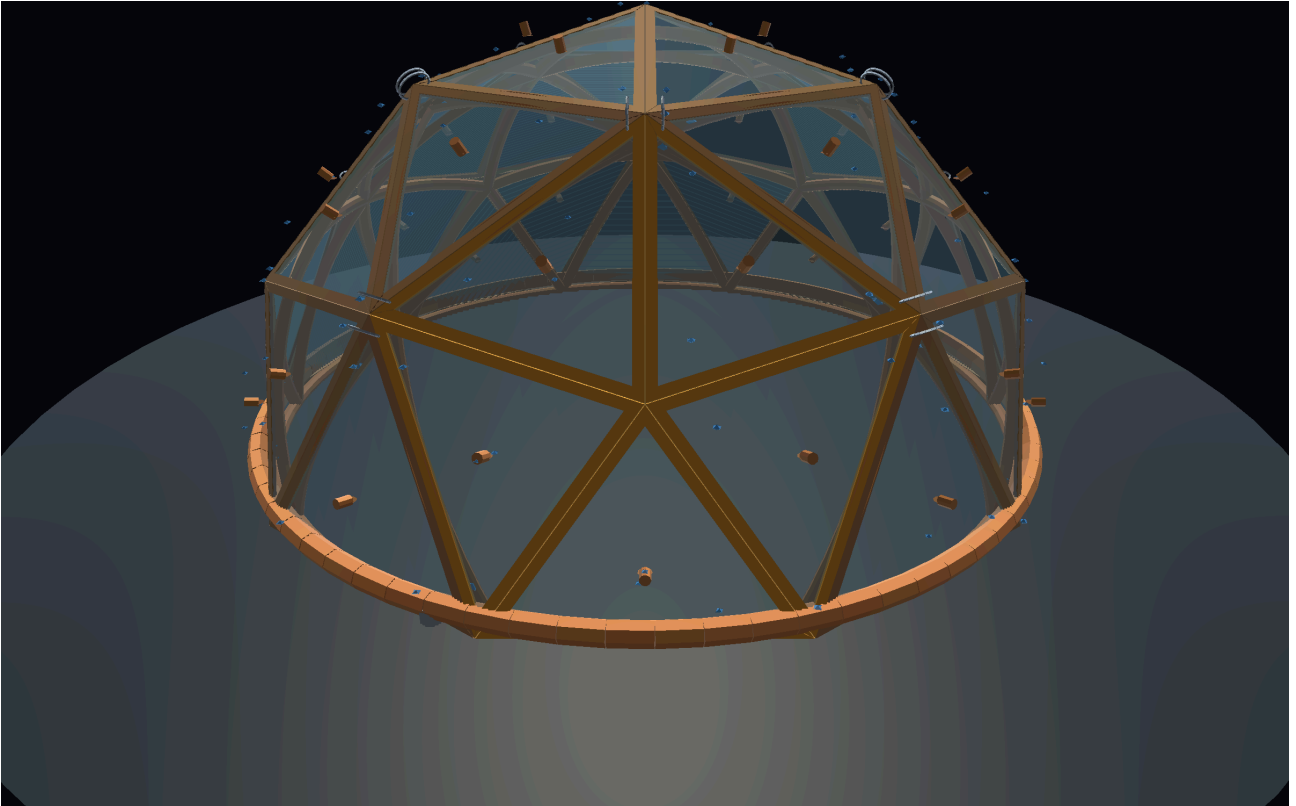
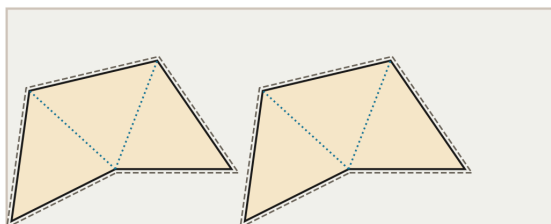


PLATE 14.3 A pentagon group

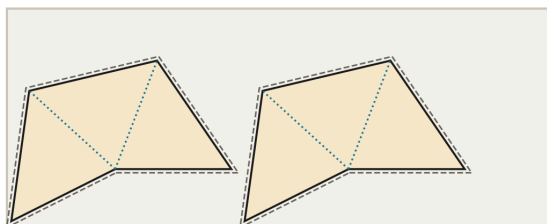
Five isosceles triangles around a five-way hub, edited as one group.

Rendered by Dome Forge with its interface hidden. The group editor with the first pentagon selected.

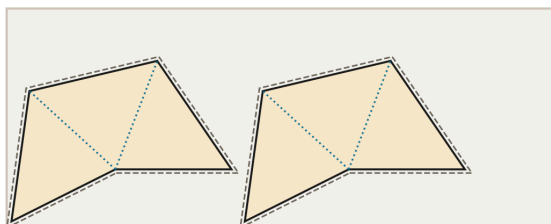
sheet 1: 35% used



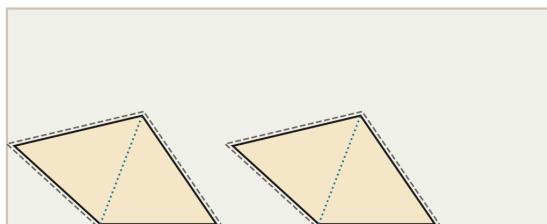
sheet 2: 35% used



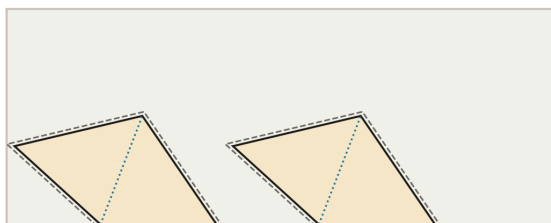
sheet 3: 35% used



sheet 4: 24% used



sheet 5: 24% used



sheet 6: 24% used

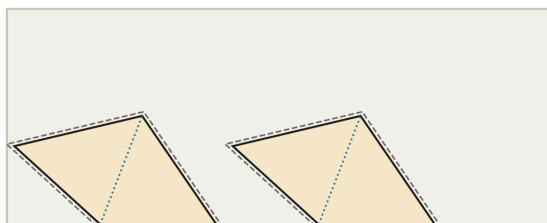


PLATE 14.4 Covers nested on the sheet

Pentagon covers split three-plus-two so each piece fits the sheet, packed by Dome Forge's own nesting routine.

Drawn when the book is built. From `dome_forge.patterns.auto_nest`.

15

The Frankendome

Strut-agnostic, asymmetric, joined from the top down: it stood, and it taught.

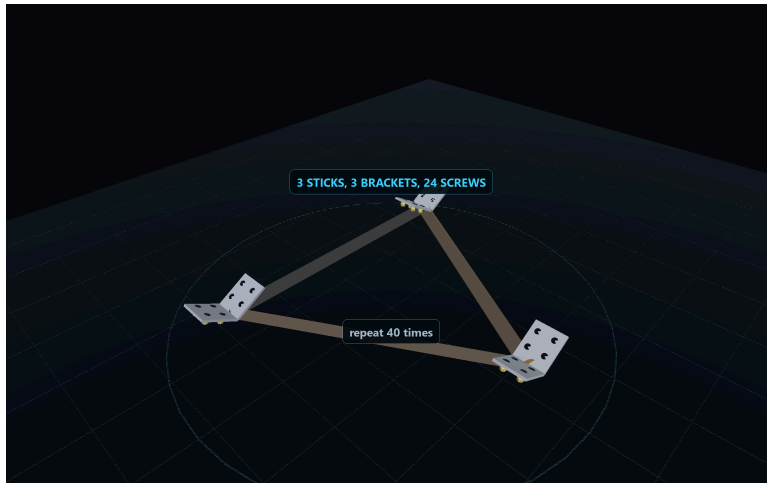


PLATE 15.1 Three different sticks, one rigid shape

A Frankendome triangle made from three unlike members.

Rendered by the DomeSim film engine from "The Franken-Dome", chapter "triangle", 80% through. The Frankendome film's triangle scene.



PLATE 15.2 A V-bracket between unlike members

The connector that let any stick meet any other.

Rendered by the DomeSim film engine from "The Franken-Dome", chapter "bracket_fitted", 80% through. The Frankendome film's fitted bracket.

**PLATE 15.3** Slack and settling

When every member is a little different, every joint takes up a little error and the frame settles.

Rendered by the DomeSim film engine from "The Franken-Dome", chapter "slack", 60% through. The Frankendome film's slack scene.

16

The Cut That Changed It

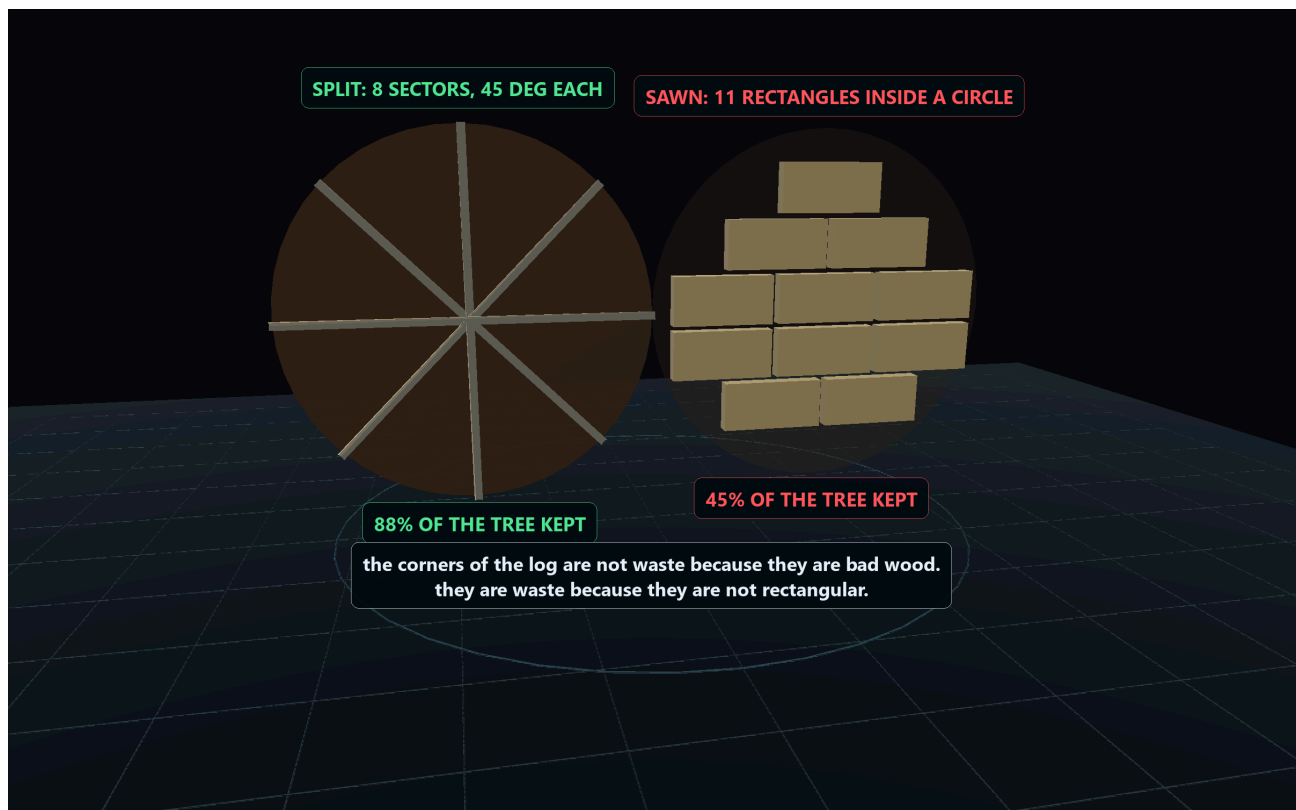
One chainsaw, eight sectors, and most of the tree kept.



PLATE 16.1 Split, not sawn

One section of trunk split radially into 8 sectors of 45 degrees each.

Rendered by the DomeSim film engine from "Eight Cuts to a House: A Dome Straight From the Tree", chapter "split", 80% through. The wedge film's split scene.

**PLATE 16.2** The same round, two ways

Split, this log keeps 88.4 percent before trimming; packed with two-by-fours it keeps 45.

Rendered by the DomeSim film engine from "Why Wedges: A Dome That Does Not Need a Sawmill", chapter "round", 90% through. The wedge argument's round-section scene, moved in so its labels separate.

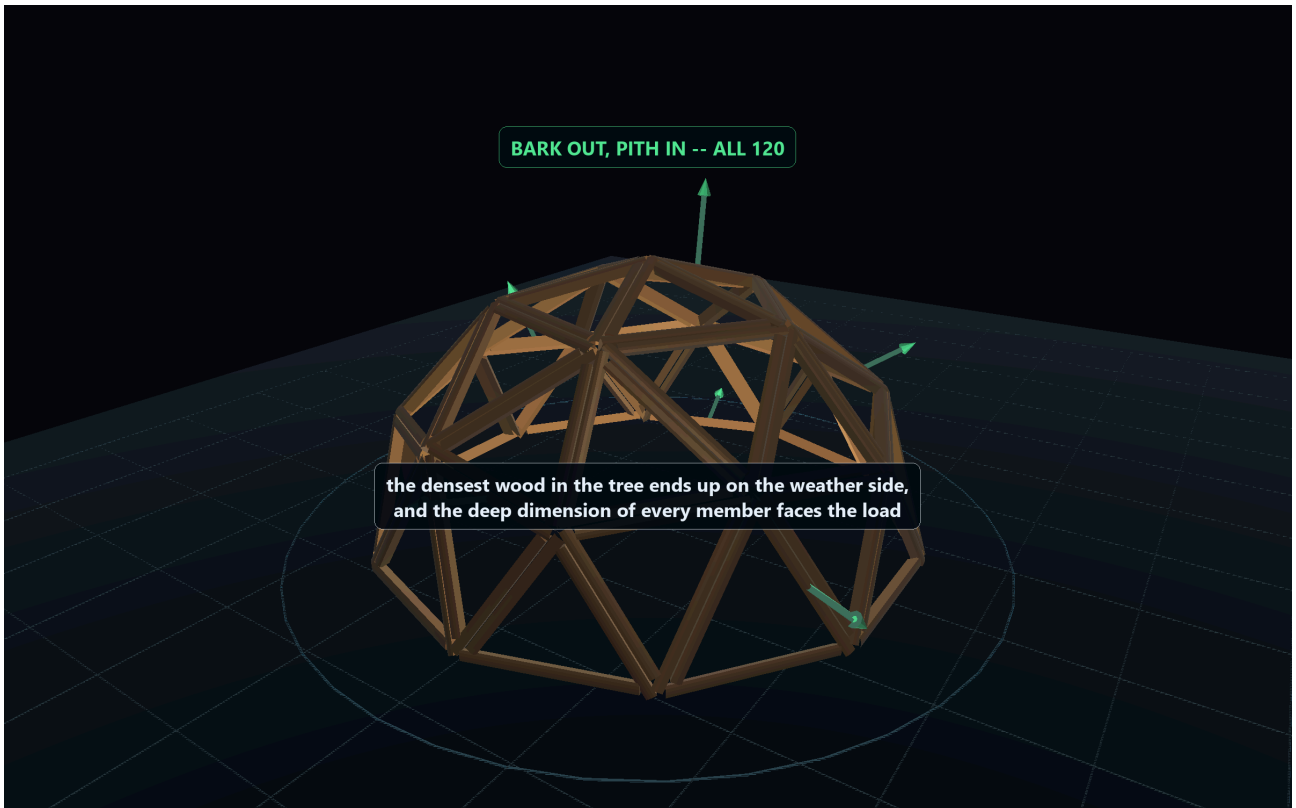


PLATE 16.3 Four ways to turn a wedge

The same wedge turned four ways in a panel makes four different buildings.

Rendered by the DomeSim film engine from "Eight Cuts to a House: A Dome Straight From the Tree", chapter "orient", 80% through. The wedge film's orientation scene.

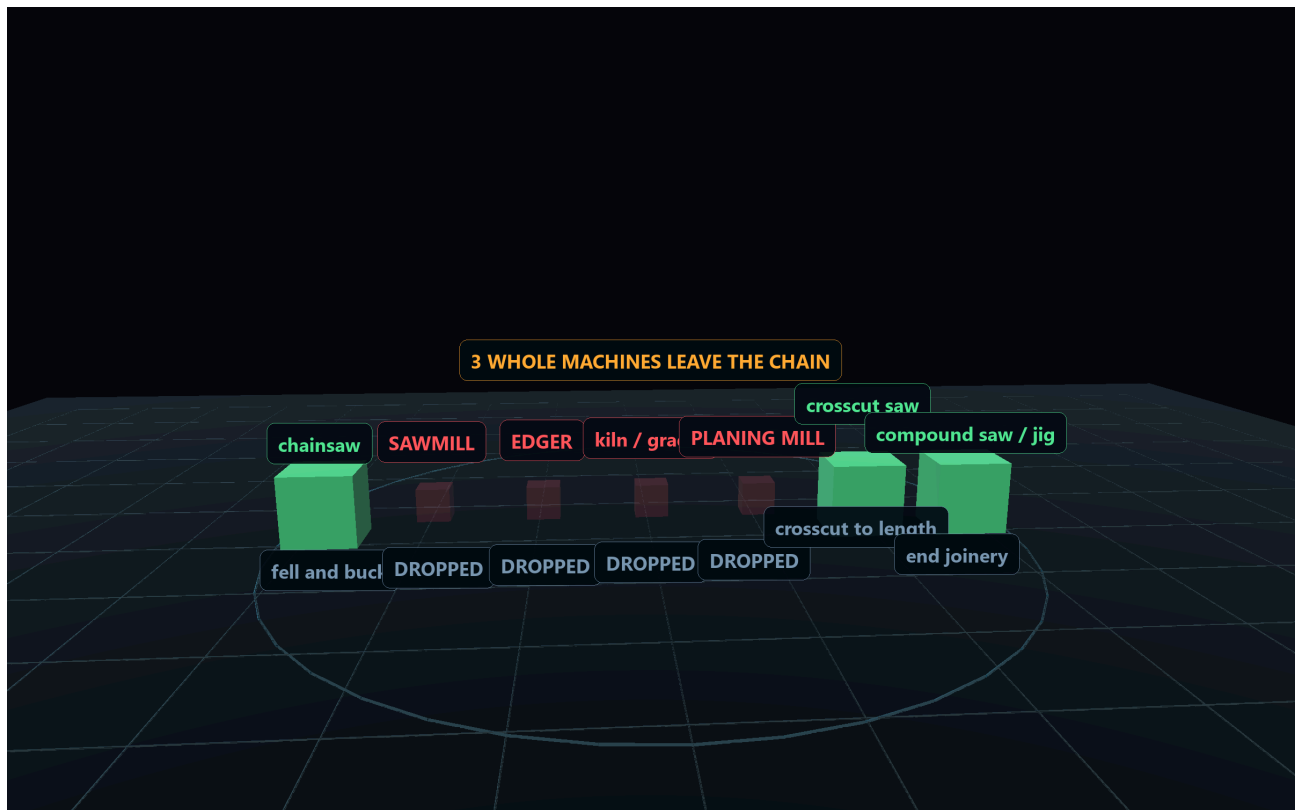


PLATE 16.4 The machines that drop out

Tree to frame, with the machines the wedge path never needs faded out.

Rendered by the DomeSim film engine from "Why Wedges: A Dome That Does Not Need a Sawmill", chapter "chain", 90% through. The wedge argument's processing-chain scene, moved in.

WORKED MATH

What a tree keeps

one trunk: 15.0 in at the butt, 5.5 in at the top, 60 ft usable

bucked into 5 sections of 12 ft

solid wood in those sections = 442.1 board feet

SPLIT RADially:

8 sectors per section, 45 deg each = 40 full-length sectors from one tree

the saw passes remove 51.2 bf of kerf

wood kept = $442.1 - 51.2 = 390.9$ bf = 88.4 %

measured across whole trees on site: 88 %

SAWN INTO TRUE TWO-BY-FOURS, same sections:

rectangles that grid-pack inside the round = 25 pieces

25×8 bf = 200.0 bf = 45.2 %

square milling in practice recovers 50-65 %, losing wood to slabs, edgings, trim, drying degrade and planer shavings

crosscut once, one tree gives 80 members; the dome needs 120 and 2 trees give 160

The same two trees yield 1.95 times as much usable structure split as sawn, because a wedge is not trying to be a rectangle and so throws none of the round away.

Computed by two_v_demo.wedge_why_facts.steps_yield. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

The machines you never buy

what has to happen to a log before it can hold a roof up:

BOTH fell and buck chainsaw
 identical in both, so it cancels
 SAWN primary breakdown SAWMILL
 wedge path splits radially with the same chainsaw
 SAWN edging EDGER
 there is no waney edge to remove from a split face
 SAWN drying and grading kiln / grader
 wedges are used green; the shell is not a graded-lumber assembly
 SAWN surfacing to dimension PLANING MILL
 the sawn faces come off the split already flat and already exact
 BOTH crosscut to length crosscut saw
 one square cut, both paths
 BOTH end joinery compound saw / jig
 sawn path measures two compound ends; wedge path cuts one butt and flush-cuts the head in place

machines the wedge path never buys, feeds or houses: 3

SAWMILL, EDGER, PLANING MILL

what is left on the wedge side is one chainsaw and a flat board

The dome removes 3 whole machines from the chain, and it can only do that because a triangulated shell does not care whether its sticks are rectangular.

Computed by two_v_demo.wedge_why_facts.steps_mills. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

Fifteen hands between the tree and the rack

a stud on a rack has been sold and handled this many times:

1. timber broker	sells the standing timber
2. logging contractor	fells and skids it
3. log haulier	trucks it to the mill yard
4. log scaler	measures and grades the log
5. headrig sawyer	breaks the log down
6. edger operator	squares the waney sides
7. trimmer operator	squares the ends
8. kiln operator	dries it to a moisture spec
9. lumber grader	stamps a grade on it
10. planing mill	surfaces four sides to dimension
11. bundler	packs and wraps a unit
12. mill sales	sells the unit
13. freight broker	books the flatbed
14. distributor	holds stock and breaks bulk
15. retail yard	sells you one stick

then sales tax: \$8.88 on the rack becomes \$10.00 paid

every one of the 15 took a margin, and every margin is in that price

the wedge path is: fell the tree, rip the trunk, use the wood

Cutting your own wedges removes 15 separate parties from between the tree and the wall, none of whom did anything to the wood that a geodesic shell needed done.

Computed by two_v_demo.wedge_why_facts.steps_middlemen. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

Four ways round

the wedge is not symmetric, so which way it faces is a decision.

turn it about its own long axis and the SHAPE stays the same
while the STIFFNESS does not, because bending in a shell is
radial: in and out along the dome's own radius.

DOVE-IN	I	5.32 in4	S	2.05 in3
a pair at one seam: both points inward, toward the centre of the dome				
PANEL-IN	I	2.50 in4	S	1.63 in3
a pair at one seam: points apart, each into the middle of its own triangle				
DOVE-OUT	I	5.32 in4	S	2.05 in3
a pair at one seam: both points outward, away from the centre, at the sky				
PANEL-OUT	I	2.50 in4	S	1.63 in3
a pair at one seam: points toward each other, across the seam				

area is 6.28 in2 in all four -- rotation moves no wood
but S runs 1.63 to 2.05 in3, a factor of 1.25

Pointing the wedge at the dome's centre puts 1.25 times the bending stiffness under the same wood, for free, by turning the stick before it goes down.

Computed by two_v_demo.wedge_why_facts.steps_orientation. Every line above is printed by that code when the book is built; none of it is typed.

17

Two Trees and a Fortnight

Size the dome to the tree, then give it fourteen days.

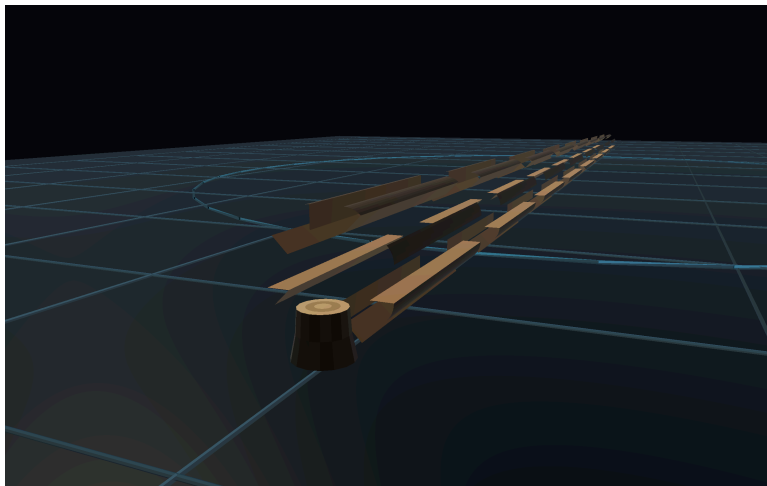


PLATE 17.1 A section opened into its wedges

One section of the bucked trunk, opened into the 8 wedges a radial split makes.

Rendered by the DomeSim film engine from "Two Trees, All at Once", chapter "why", 50% through. The harvest film's close look at one opened section.

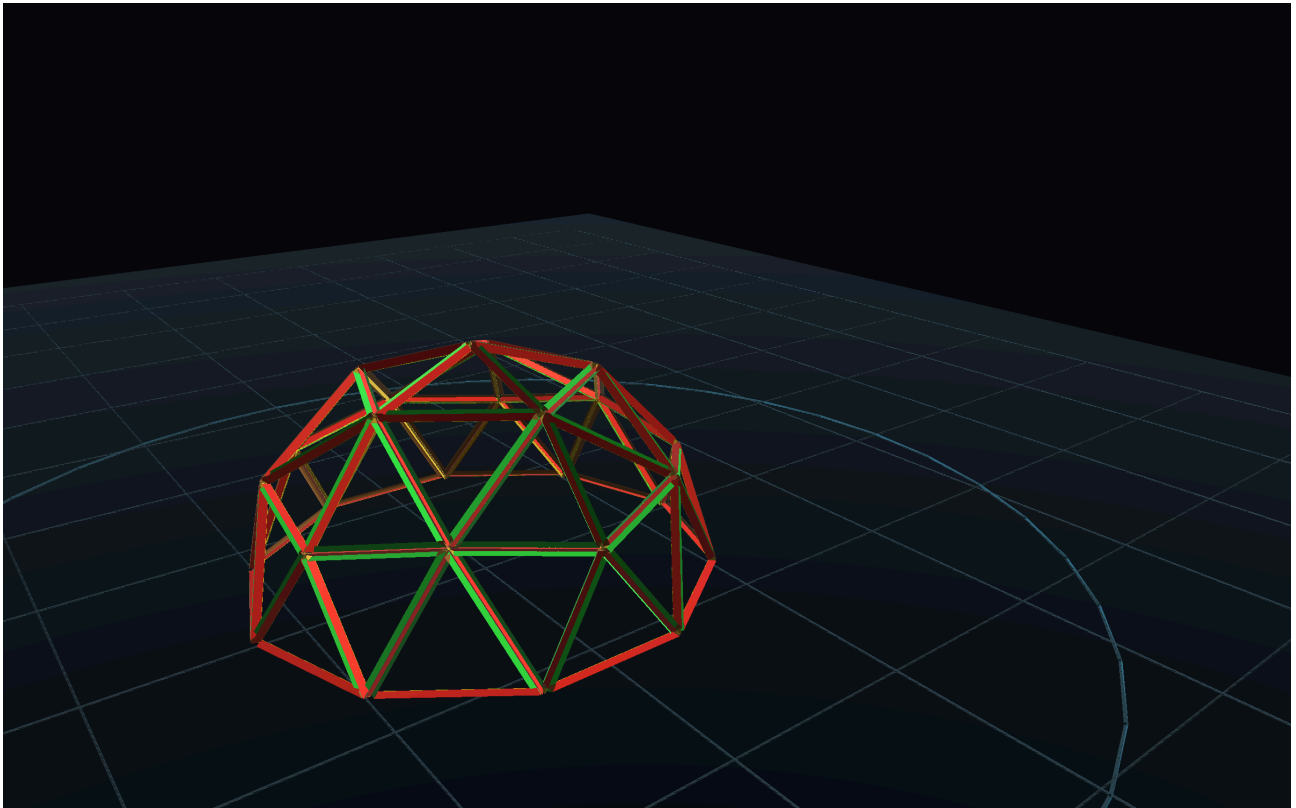


PLATE 17.2 The shell those wedges make

The raw-wedge simulator's solved dome, keys and all.

Rendered by the DomeSim film engine from "Two Trees, All at Once", chapter "worth", 50% through. The harvest film's dome scene.

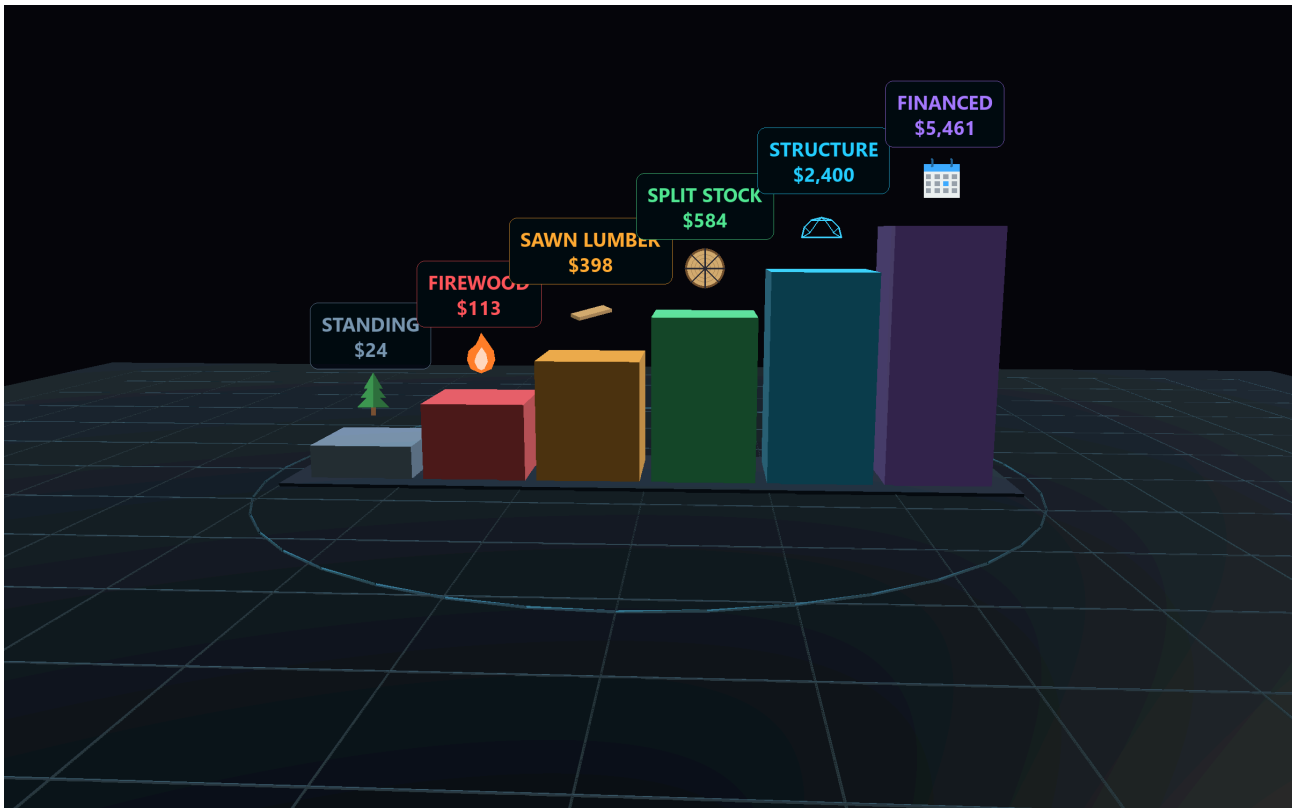


PLATE 17.3 One pine, every rung

Standing, burned, sawn, split, built and financed: one tree's values side by side.

Rendered by the DomeSim film engine from "The \$20 Pine", chapter "closing", 90% through. The \$20 Pine's closing ladder.

WORKED MATH

What was measured, and what was not

MEASURED fuel while ripping: 2 tanks an hour, one timed sitting
 MEASURED cutting rate: 30 struts in a 6-hour afternoon
 DERIVED ripping time: 120 struts at 5.0 an hour = 24 hours
 DECIDED the plan: felling days 1-2, bucking days 3-4, ripping days 5-8
 ESTIMATED the saw's tank: 0.28 to 0.37 L; the listings disagree
 NOT METERED fuel for felling and bucking: left out, not guessed

So the fuel is a range, and it is the ripping only

Computed by two_v_demo.lesson_harvest.assumption_lines. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

Two timed sessions

SESSION A, timed:
12 wedges in 2.0 h = 6.0 wedges/hour
\$10.00 of fuel and bar oil = \$5.00/hour, the only cash spent
SESSION B, timed separately:
6 passes at 7 ft = 42 ft of rip in 1 h, at 6 in+ of depth
2 tanks of fuel and 2 of bar oil, one sharpening
the two sessions measured different things, so divide them:
42 ft/h / 6.0 wedges/h = 7.00 ft of rip per wedge
and the geometry predicts the same number independently:
8 sectors takes 7 passes down a 7 ft section
7 x 7 / 8 = 6.12 ft per wedge
the pure cutting is 88 % of the measured time; the other 12 % is rolling the log, setting up and sharpening
planning uses 5.0 wedges/hour, 17 % below the measured rate

Two independent measurements and a geometric prediction agree to within 12 percent, which is why the rate below is worth anything at all.

Computed by two_v_demo.wedge_why_facts.steps_rate. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH

What the pine's ladder rests on

THE AUTHOR'S a representative pine: 60 ft of stem, 15 in to 5.5 in
THE AUTHOR'S 60% kept by a mill; 88% by splitting; 90 cu ft of wood a cord
THE AUTHOR'S sawing \$0.60/bd ft; a cord \$275 picked up, \$300 delivered
THE AUTHOR'S \$4,800 of framing for the shell; it uses 1.5 trees, 2 felled
PUBLISHED pine \$1.50/bd ft; stumpage \$23.23/ton; a cord 128 cu ft stacked
CHECK this log in the simulator: sawn to 2x4s 45%, split 88.4% before trimming

So the mill gets the benefit of the doubt, not the wedge

Computed by two_v_demo.pine_value_economics.source_lines. Every line above is printed by that code when the book is built; none of it is typed.

18

What I Got Wrong, and What Is Still Unproven

The claims that did not survive, and the tests still to run.

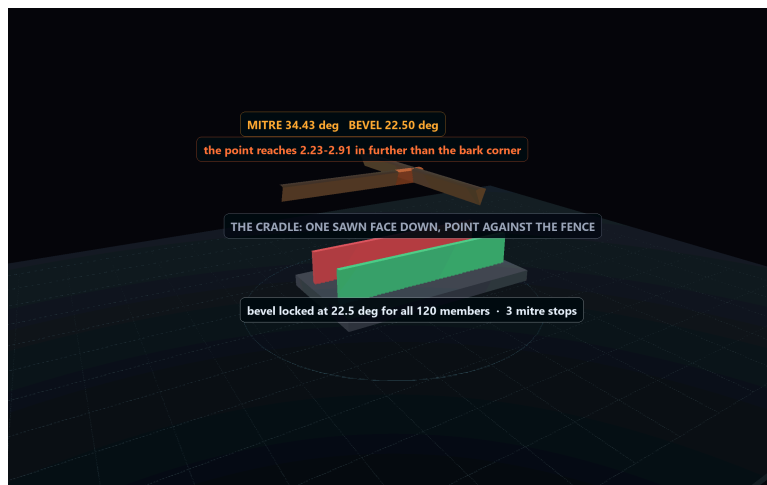
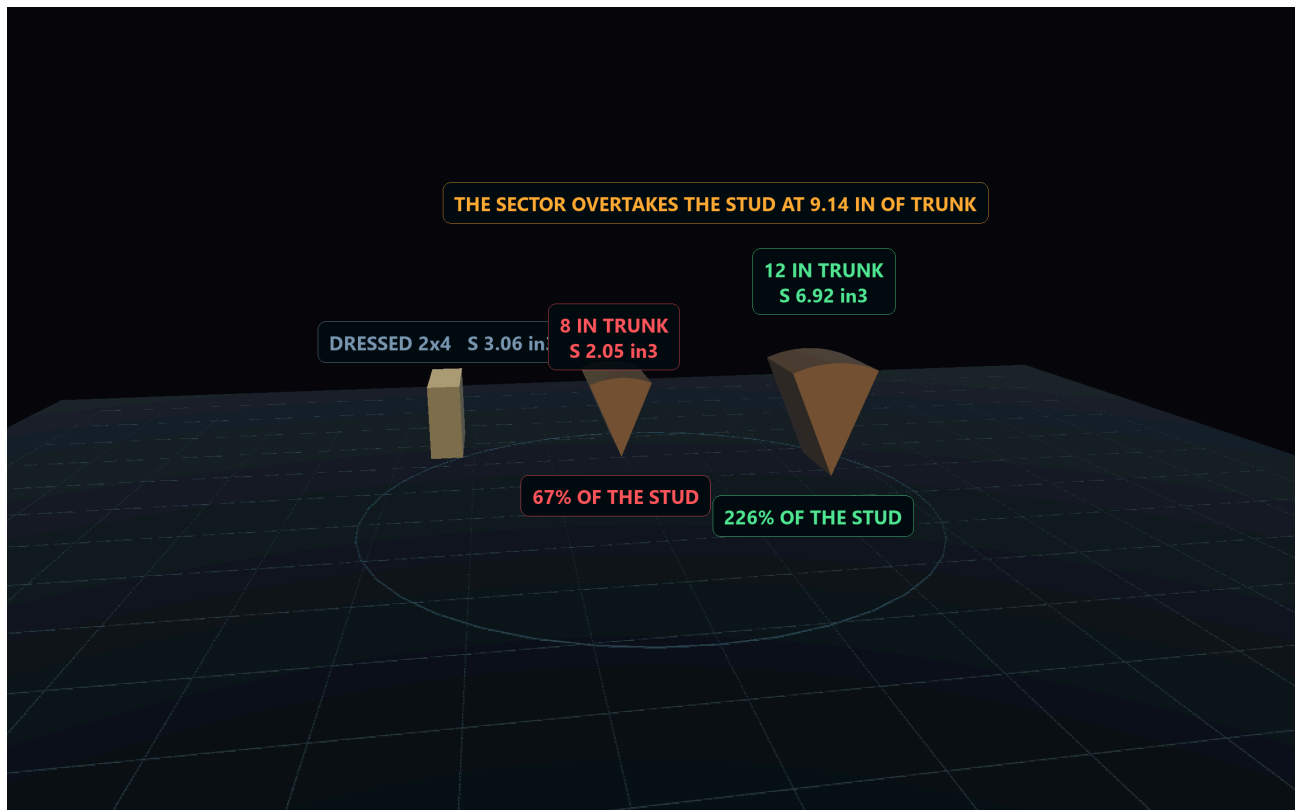


PLATE 18.1 The one compound cut

The butt end of each member: the one compound cut the wedge method keeps, made once, off the jig.

Rendered by the DomeSim film engine from "Why Wedges: A Dome That Does Not Need a Sawmill", chapter "buttcut", 80% through. The wedge argument's butt-cut scene.

**PLATE 18.2** A wedge against a board, in bending

One wedge and one dressed two-by-four in bending: the comparison that did not flatter the wedge, printed.

Rendered by the DomeSim film engine from "Why Wedges: A Dome That Does Not Need a Sawmill", chapter "stick", 85% through. The wedge argument's stick scene.

WORKED MATH**Strong enough?**

a raw sector is literally 1/8 of the tree, and how strong that is depends entirely on how big the tree was.

AT THE SIMULATOR'S 8 IN TRUNK:

area 6.28 in², S 2.05 in³

= 120 % of a dressed 2x4's area but only 67 % of its bending

AT THE 12 IN TRUNKS ACTUALLY BEING CUT:

area 14.14 in², I 26.98 in⁴, S 6.92 in³

= 269 % of a dressed 2x4's area and 226 % of its bending

and 177 % / 130 % against a FULL-dimension 2x4

section grows with the square of the radius and bending with the cube, so there is a diameter where the sector overtakes the stud:

more area than a dressed 2x4 above 7.31 in

stiffer than a dressed 2x4 above 9.14 in

stiffer than a FULL 2x4 above 11.00 in

and neighbouring panels never share a stick, so all 55 interior edges carry TWO sectors:

539 % of a stud's area, 452 % of its bending

Under 9.1 inches of trunk the wedge is the weaker member and this argument fails. The trees being cut are 12 to 15 inches, which is why it does not.

Computed by two_v_demo.wedge_why_facts.steps_structure. Every line above is printed by that code when the book is built; none of it is typed.

WORKED MATH**The one compound cut**

correction first. An earlier cut of this film said nothing in this frame is mitred. That is wrong, and it is the most important thing on this screen.

the PINWHEEL removes the shared vertex: no two ends ever meet each other, nothing is coped, nothing is shaved to a point.

but every butt end still lands on the SIDE of the next member, and that side is tilted two ways at once -- by the sector angle, and by the turn the member makes to close the triangle.

so the butt is a compound cut, and it is the hardest operation in the build. Here is what the solve says it costs:

BEVEL 22.500 deg on every one of 120 members

and 22.500 = half the sector angle (22.500), set by the SPLIT, not by the dome

MITRE takes 3 values in the whole shell:

34.431 deg x 60 members

21.138 deg x 30 members

30.000 deg x 30 members

largest is 34.431 deg, inside a common saw's 50 deg swing

the cut runs 2.23 to 2.91 in along the member, because it lands on a slope:

the sector's POINT reaches furthest and the bark corner least, so

the blank has to be long on the point side and the cut is walked from the short corner up

One bevel, locked at 22.5 degrees and never touched, and 3 mitre stops. That is what turns the hardest cut in the build from gauging by hand into a fixture.

Computed by two_v_demo.wedge_why_facts.steps_buttcut. Every line above is printed by that code when the book is built; none of it is typed.



PANEL 36 OF 40

BOOK THREE

The Build

each frame is finished on the ground and lifted whole

A guide to building a wedge dome the way it was built here, from choosing a tree to closing the shell.

- 19 Plan: Size, Site, and the Two Methods
- 20 The Tools
- 21 Harvest: Fell, Buck, Split
- 22 Build the Jig
- 23 Forty Panels
- 24 Seams, Keys, and the Fold
- 25 The Footing and the Ring
- 26 Raise the Shell
- 27 Skin, Openings, and Water
- 28 Inside a Round Room
- 29 Check, Correct, Record

19

Plan: Size, Site, and the Two Methods

Decide the dome from the floor you want, or from the tree you have.

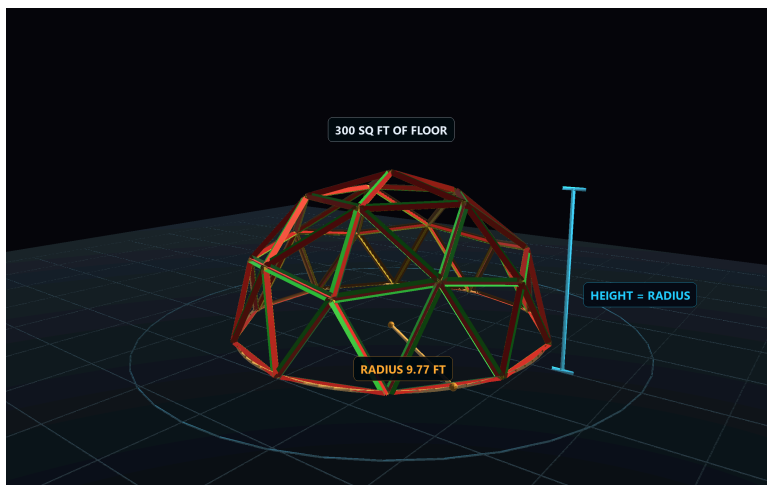


PLATE 19.1 A dome from the floor you want

A 300 square foot dome: radius 9.77 feet, longest member 64.5 inches.

Built for this book from the film engine's visual objects. The solved dome with its radius, height and floor marked.



PLATE 19.2 The floor it covers

The same dome from above: the floor circle and the ring of hubs it stands on.

Built for this book from the film engine's visual objects. The solved dome from overhead with the floor circle drawn.

20

The Tools

Every program that made this book, and how to run it yourself.

21

Harvest: Fell, Buck, Split

Turning two standing pines into a pile of struts.

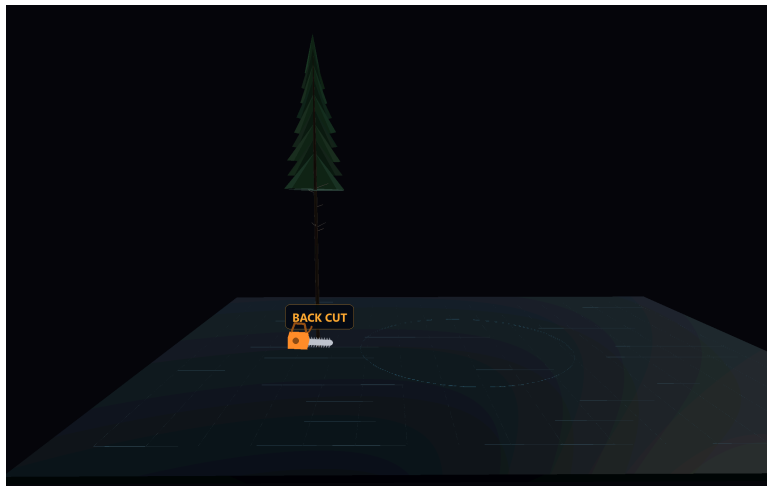


PLATE 21.1 The hinge steers the tree

The face notch and the back cut, and the hinge between them.

Rendered by the DomeSim film engine from "Two Trees, All at Once", chapter "fell", 42% through. The harvest film's felling scene at the back cut.

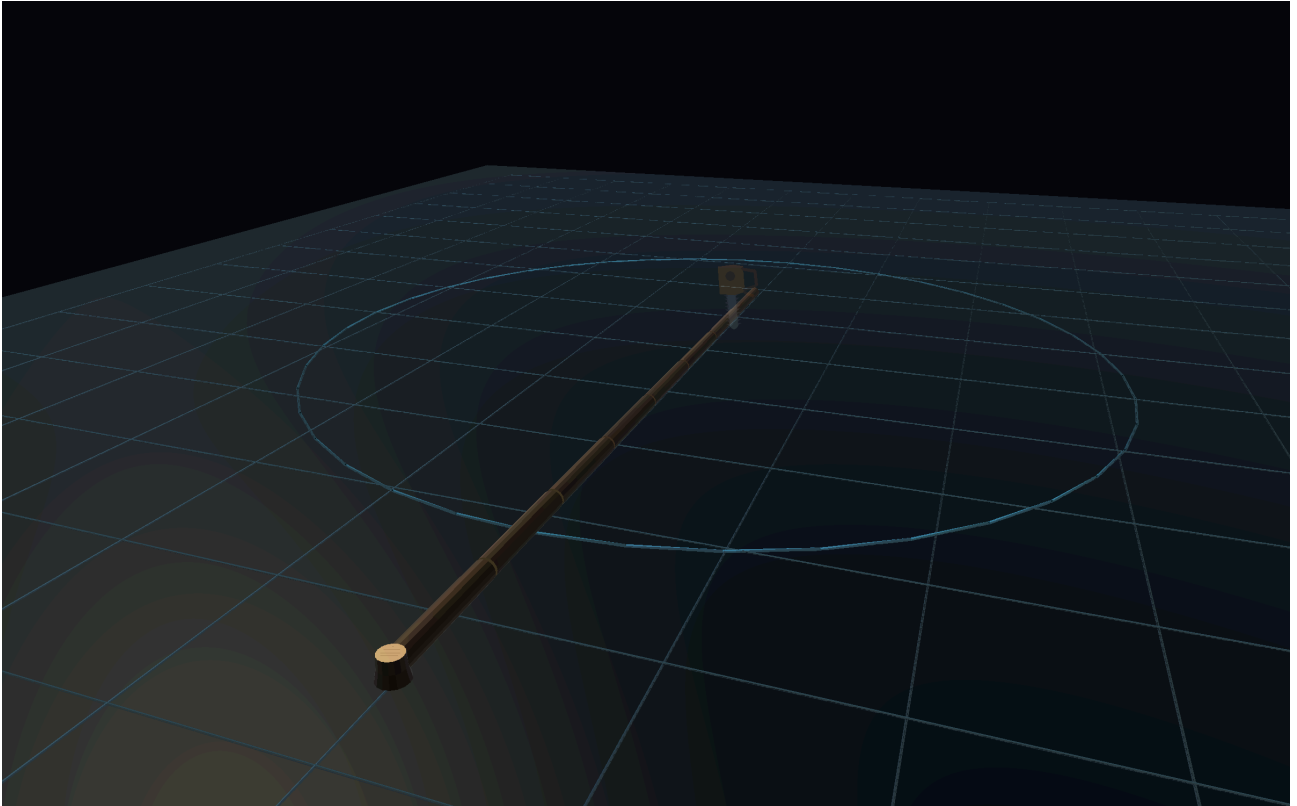


PLATE 21.2 Bucking to length

The stem cut into sections as long as the longest member.

Rendered by the DomeSim film engine from "Two Trees, All at Once", chapter "explode", 26% through. The harvest film as the bucking cuts are made, re-aimed along the stem.

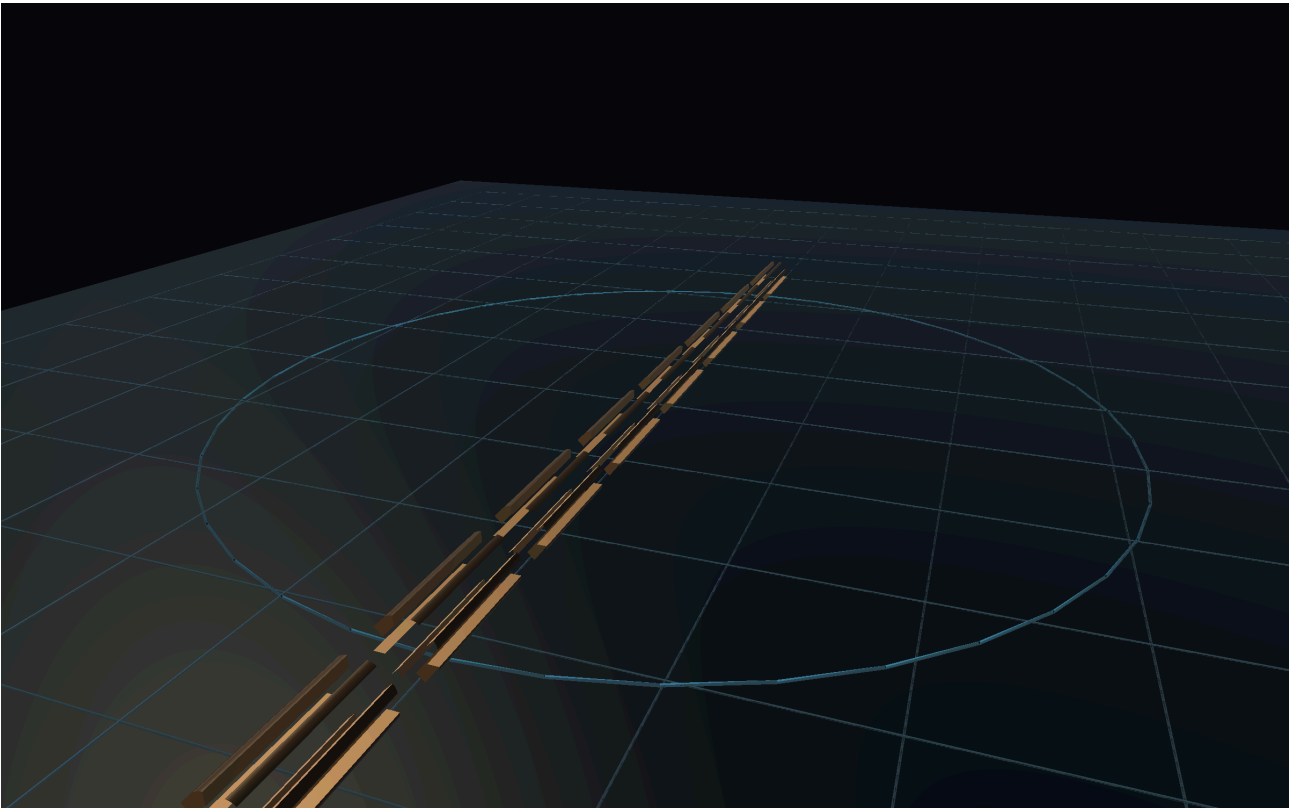


PLATE 21.3 Two trees, split
The harvest as a pile of wedges.

Rendered by the DomeSim film engine from "Two Trees, All at Once", chapter "cost", 50% through. The harvest film's pile, seen along the row from its near end.

22

Build the Jig

Solve the geometry once, in plywood, so you never solve it again.

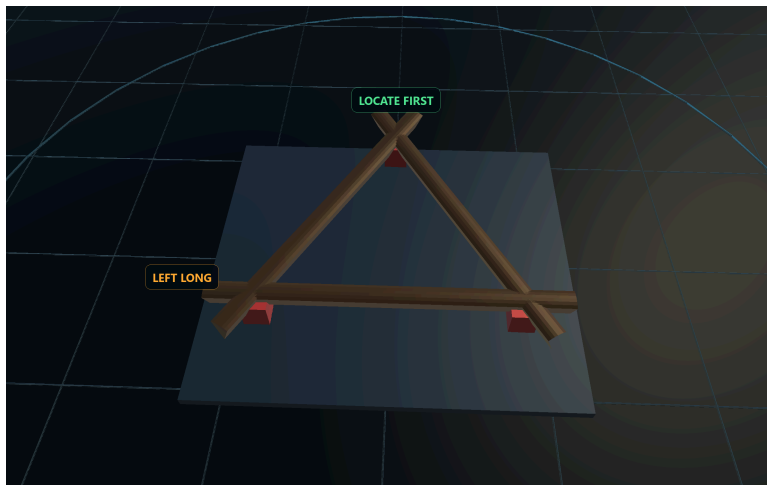
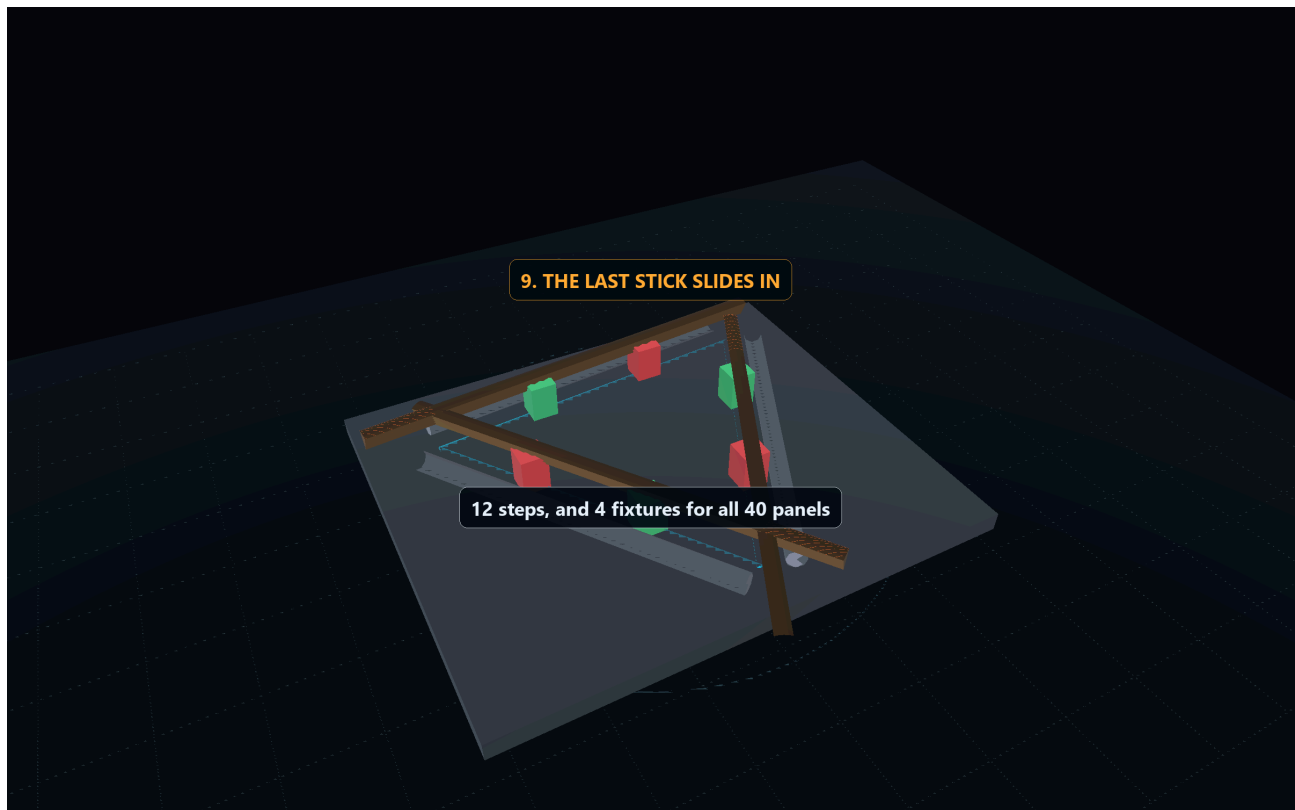


PLATE 22.1 Locate first, cut second

Members held in their true relationship before the excess is trimmed.

Rendered by the DomeSim film engine from “Why Build This Way”, chapter “jig”, 50% through. Why Build This Way’s jig scene as the members arrive.

**PLATE 22.2** The fabrication bench

One flat jig that turns sticks into panels.

Rendered by the DomeSim film engine from "Why Wedges: A Dome That Does Not Need a Sawmill", chapter "bench", 80% through. The wedge argument's bench scene.

WORKED MATH**One jig, forty panels**

40 triangles, 120 members, 120 corner joints -- all off one flat board

the fixture comes in 12 steps and holds three things:

a RAIL per member, set to the sawn point-axis, not the bark

a RED and a GREEN plate that make a wrongly-turned wedge physically refuse to seat

a FENCE at each head end, which is the plane the saw runs on

each member arrives with ONE cut already made:

the butt, cut to the angle its neighbour presents, batched

and one deliberate error: the head is left 6 in too long

three butts are pulled up, the loop closes, and only then is

each head sawn against its fence -- unmeasured, in place

the offcut is 60 ft per dome = 8.3 % of raw length

and 40 panels need only 4 physically different fixtures

Leaving 8.3 percent of the length as offcut is what buys the accuracy: error in stock length or butt angle leaves in the offcut instead of accumulating around the triangle.

Computed by two_v_demo.wedge_why_facts.steps_jig. Every line above is printed by that code when the book is built; none of it is typed.

23

Forty Panels

Butt cut, head overfit, flush cut: one panel, forty times.

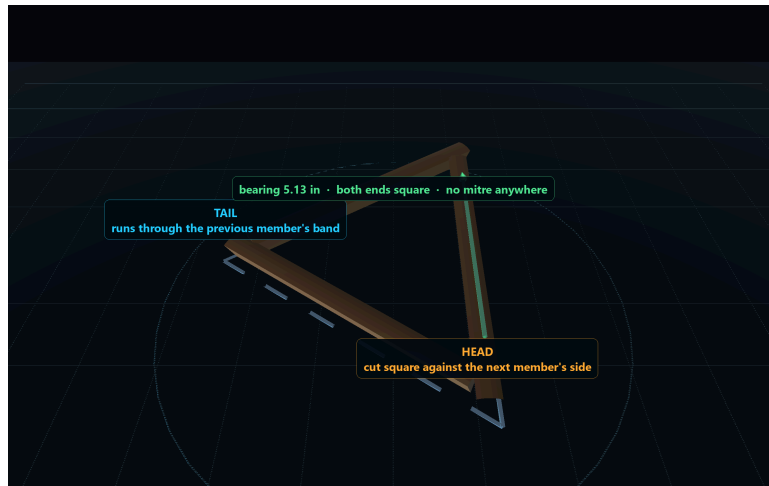
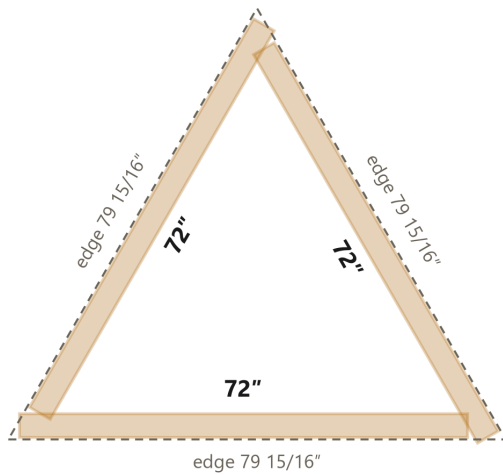
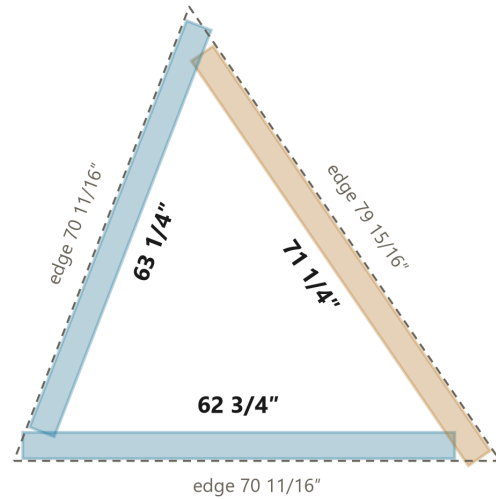


PLATE 23.1 The pinwheel panel

Each member's end butts the side of the next, around the triangle.

Rendered by the DomeSim film engine from "Eight Cuts to a House: A Dome Straight From the Tree", chapter "pinwheel", 85% through. The wedge film's pinwheel scene.

Equilateral panel: 3 long; 10 per dome**Isosceles panel: 1 long, 2 short; 30 per dome**

Build radius 10.8 ft; member width 3.8 in; gasket 0.75 in. Dashed: the true triangle. Every end stops 2.6 to 7.0 in short of its corner.

PLATE 23.2 The two panels, drawn to build

Both panel shapes at the book's build size, flattened: each stick's length, where its head stops against the next stick's side, and the corner left open. The ends are square because this drawing models a flat band; a real log sector needs the compound butt cut described in the text.

Drawn when the book is built. From two_v_demo.wedge_geometry.pinwheel_panels at the build plan's radius.



PLATE 23.3 Flush cutting in place

The overlong head trimmed against the panel after the jig has located it.

Rendered by the DomeSim film engine from "Why Wedges: A Dome That Does Not Need a Sawmill", chapter "flush", 85% through. The wedge argument's flush-cut scene.

WORKED MATH

The one compound cut

correction first. An earlier cut of this film said nothing in this frame is mitred. That is wrong, and it is the most important thing on this screen.

the PINWHEEL removes the shared vertex: no two ends ever meet each other, nothing is coped, nothing is shaved to a point.

but every butt end still lands on the SIDE of the next member, and that side is tilted two ways at once -- by the sector angle, and by the turn the member makes to close the triangle.

so the butt is a compound cut, and it is the hardest operation in the build. Here is what the solve says it costs:

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and 22.500 = half the sector angle (22.500), set by the SPLIT, not by the dome

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34.431 deg x 60 members

21.138 deg x 30 members

30.000 deg x 30 members

largest is 34.431 deg, inside a common saw's 50 deg swing

the cut runs 2.23 to 2.91 in along the member, because it lands on a slope:

the sector's POINT reaches furthest and the bark corner least, so

the blank has to be long on the point side and the cut is walked from the short corner up

One bevel, locked at 22.5 degrees and never touched, and 3 mitre stops. That is what turns the hardest cut in the build from gauging by hand into a fixture.

Computed by two_v_demo.wedge_why_facts.steps_buttcut. Every line above is printed by that code when the book is built; none of it is typed.

24

Seams, Keys, and the Fold

Where two panels meet, and why the key holds the angle.

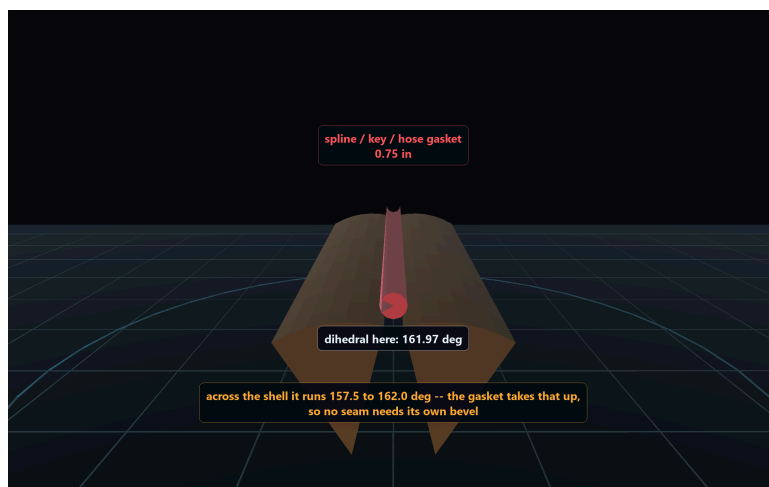
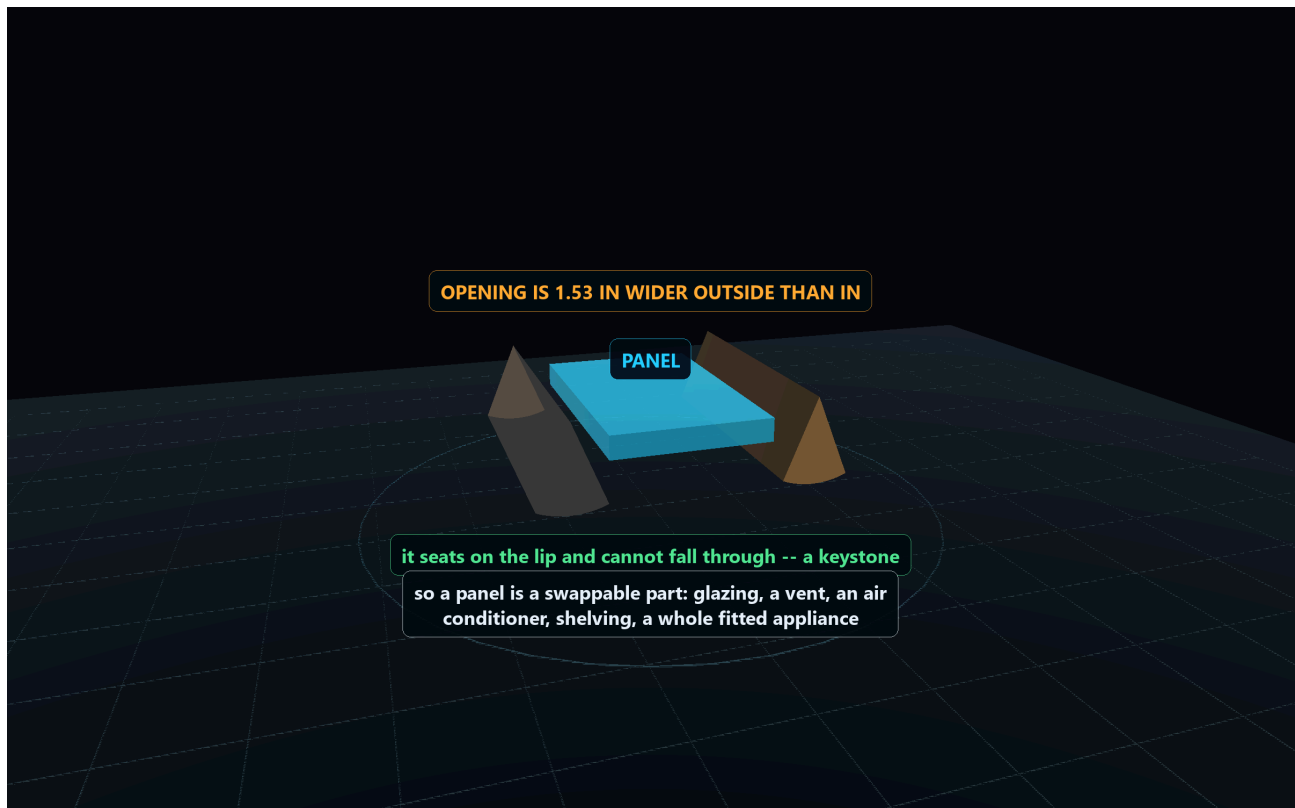


PLATE 24.1 The seam and its gasket

Where two panels meet, with the gasket between them.

Rendered by the DomeSim film engine from "Eight Cuts to a House: A Dome Straight From the Tree", chapter "gasket", 85% through. The wedge film's gasket scene.

**PLATE 24.2** The key in the seam

A shaped key holds the fold angle between two panels.

Rendered by the DomeSim film engine from "Why Wedges: A Dome That Does Not Need a Sawmill", chapter "keystone", 80% through. The wedge argument's keystone scene.

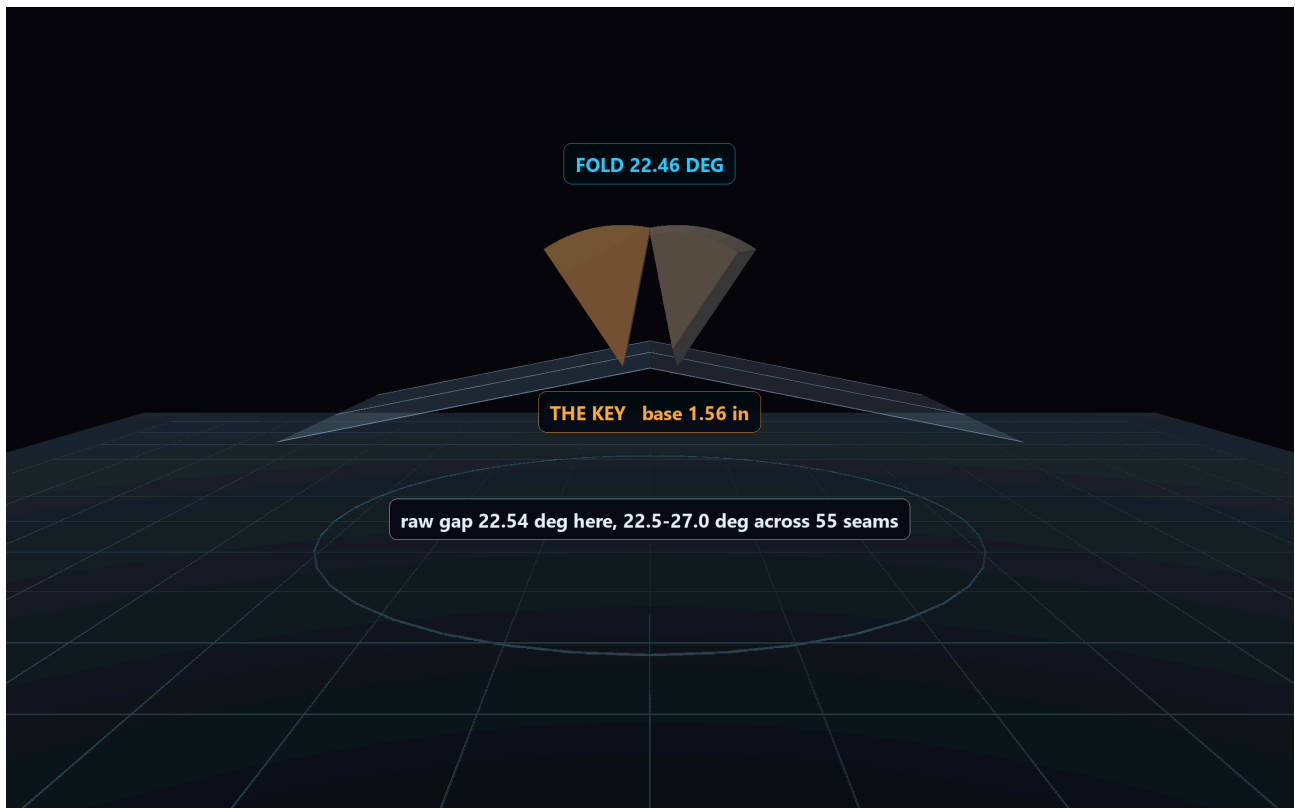


PLATE 24.3 Two fold angles

The dome's panels fold at only 2 angles: 18.03 and 22.46 degrees.

Rendered by the DomeSim film engine from "Why Wedges: A Dome That Does Not Need a Sawmill", chapter "fold", 80% through. The wedge argument's fold scene.

WORKED MATH

Two fold angles

two panels meeting along one edge are not in the same plane.
 the fold between them runs 18.03 to 22.46 deg
 a spread of 4.43 deg across 55 interior seams
 each panel brings its own member, so a seam has two raw sawn
 faces 45 deg apart before anything folds
 after folding, the gap left between them runs 22.54 to 26.97 deg
 which is exactly 45 deg minus the fold
 there are only two ways to close a gap that is never the same:
 shave the wood to suit each seam -- a different bevel, 55 times
 or leave the wood alone and fit a key that varies instead
 the key's base runs 1.56 to 1.87 in
 one part, cut on a taper, doing what would otherwise be a setup change per seam

The key exists so the 4.43 degrees of dihedral variation live in a small replaceable part instead of in every structural member of the building.

Computed by two_v_demo.wedge_why_facts.steps_dihedral. Every line above is printed by that code when the book is built; none of it is typed.

25

The Footing and the Ring

A level ring is most of a dome's foundation.

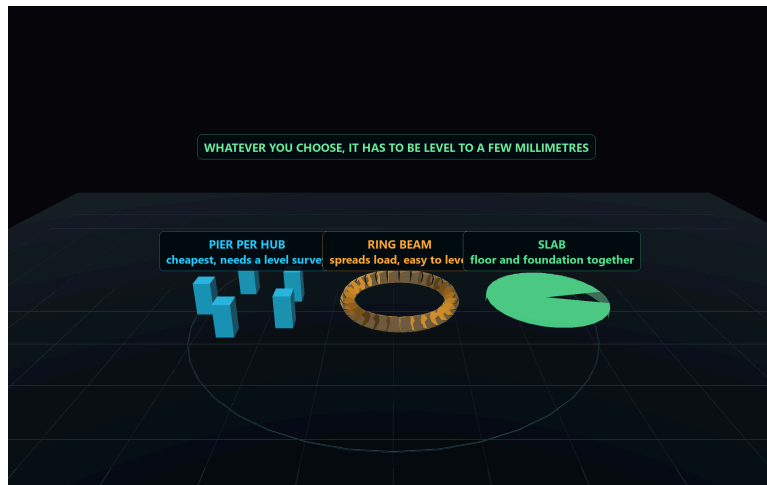


PLATE 25.1 A level ring

Footings and a level ring: most of a dome's foundation.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "foundation", 85% through. The construction masterclass's foundation scene.

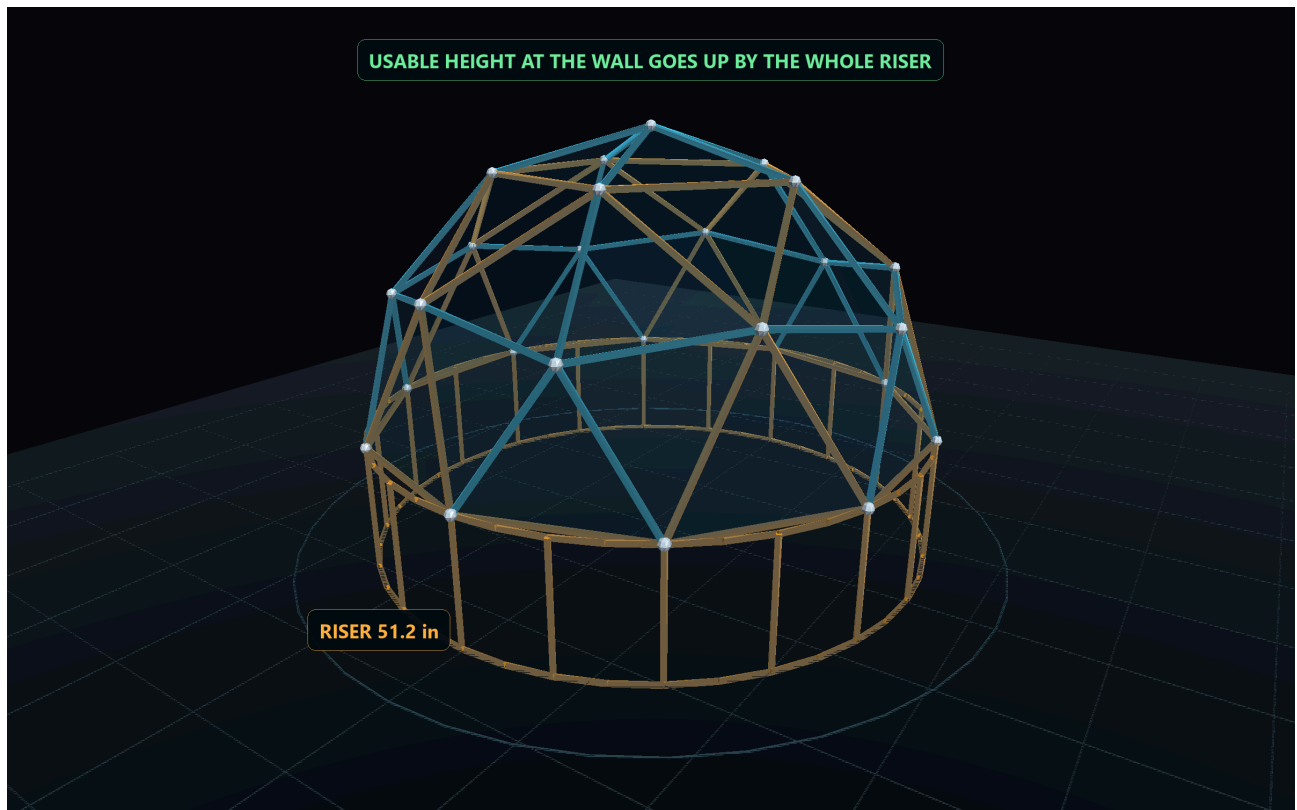


PLATE 25.2 A riser wall

A short wall under the dome gains headroom at the edge.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "riser", 85% through. The construction masterclass's riser scene.

26

Raise the Shell

Ring by ring from the ground up, stable at every step.

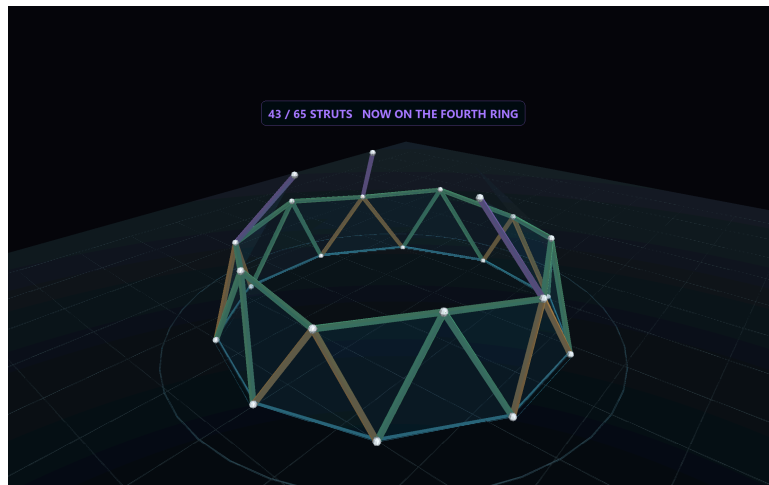


PLATE 26.1 Ring by ring

Raising the shell from the ground up.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "raise", 60% through. The construction masterclass's raising scene.



PLATE 26.2 The top

The last pieces at the apex.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "apex", 85% through. The construction masterclass's apex scene.

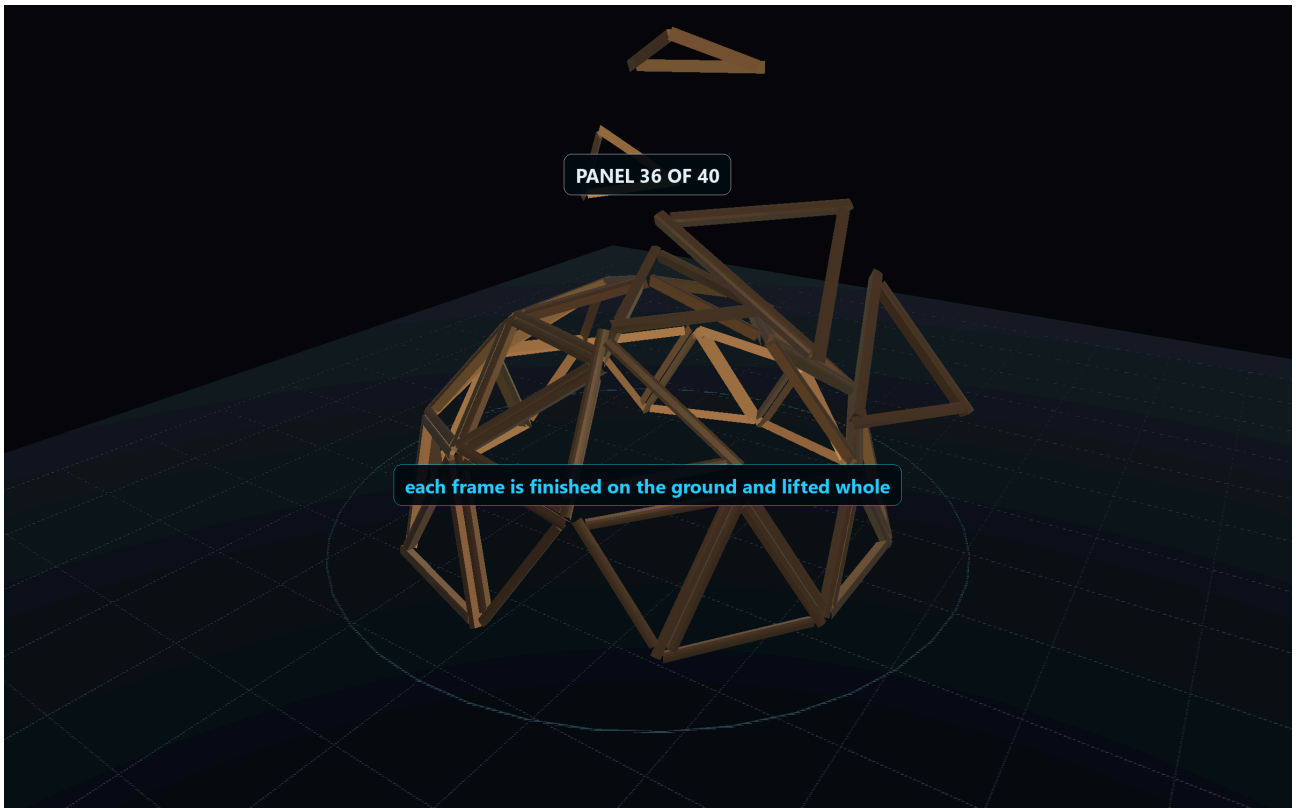


PLATE 26.3 Forty panels into a shell
The pinwheel panels assembled into the dome.

Rendered by the DomeSim film engine from "Eight Cuts to a House: A Dome Straight From the Tree", chapter "assemble", 70% through. The wedge film's assembly scene.

27

Skin, Openings, and Water

Close the shell against the weather, and let the seams drink the rain.

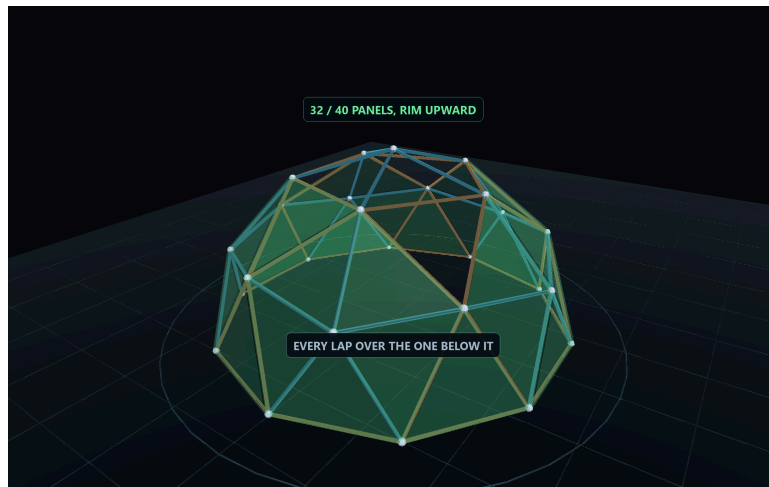


PLATE 27.1 Skinning the shell

Closing the triangles against the weather.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "skin", 70% through. The construction masterclass's skin scene.

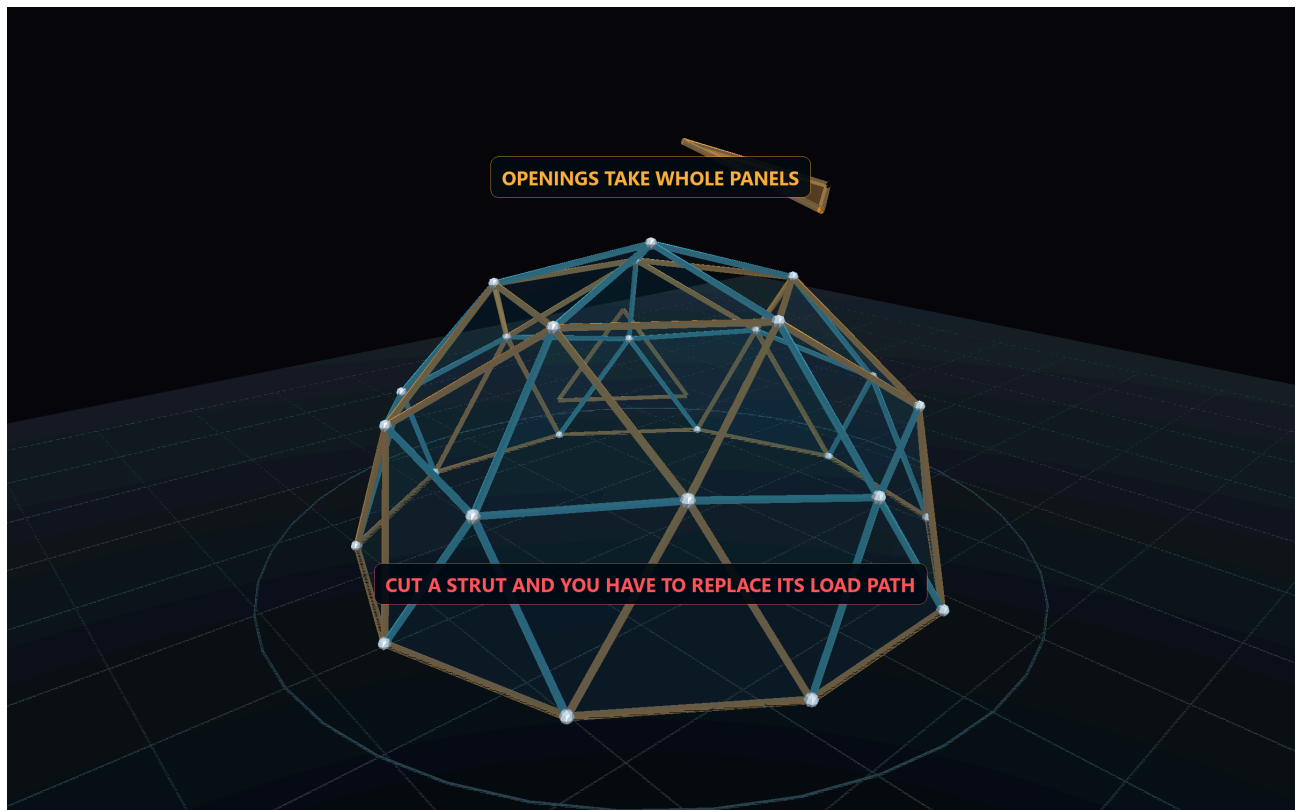


PLATE 27.2 An opening in a triangulated shell
Framing a door or window without breaking the triangles.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "openings", 80% through. The construction masterclass's openings scene.

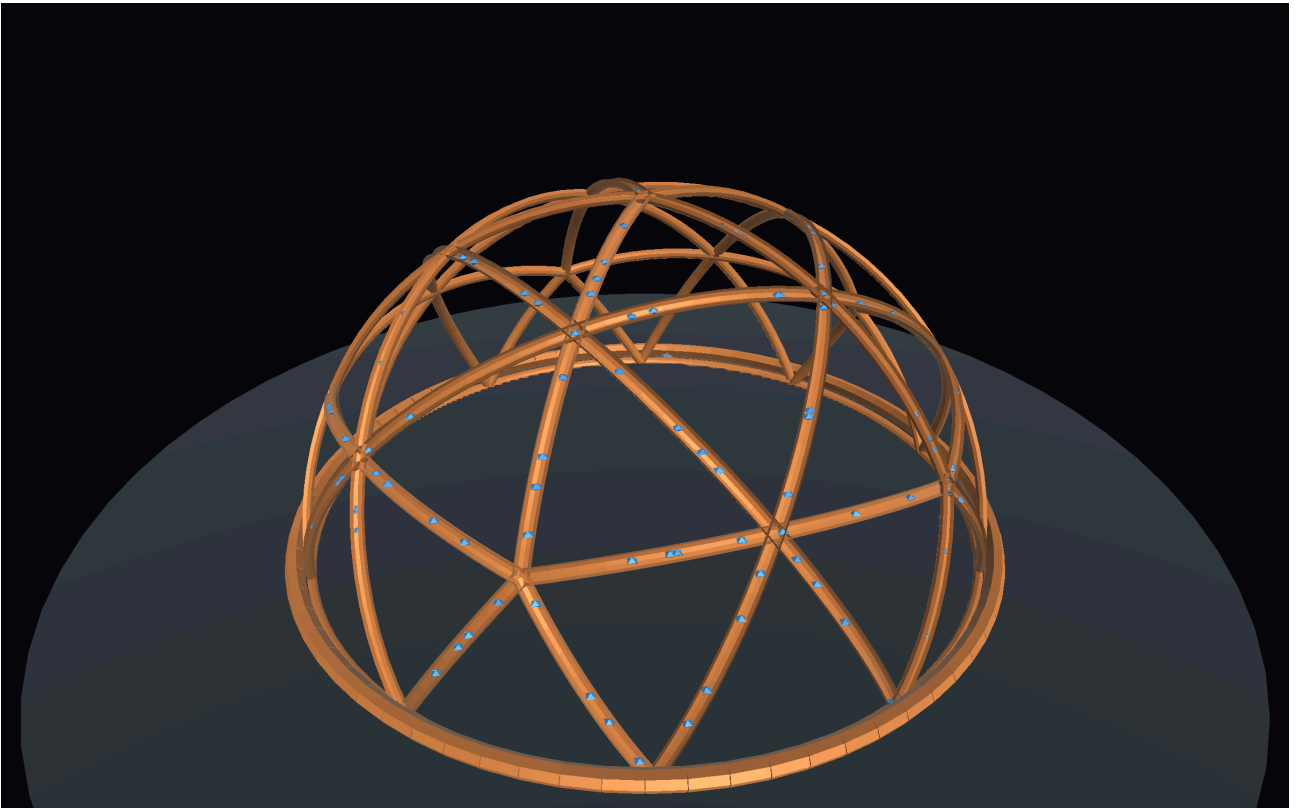


PLATE 27.3 The seam veins alone

A channel beneath every seam, standing off the skin, draining to the collector ring.

Rendered by Dome Forge with its interface hidden. Only the veins, the water in them, the collector, downpipe and cistern.

28

Inside a Round Room

Light, power, heat and furniture in a room with no corners.

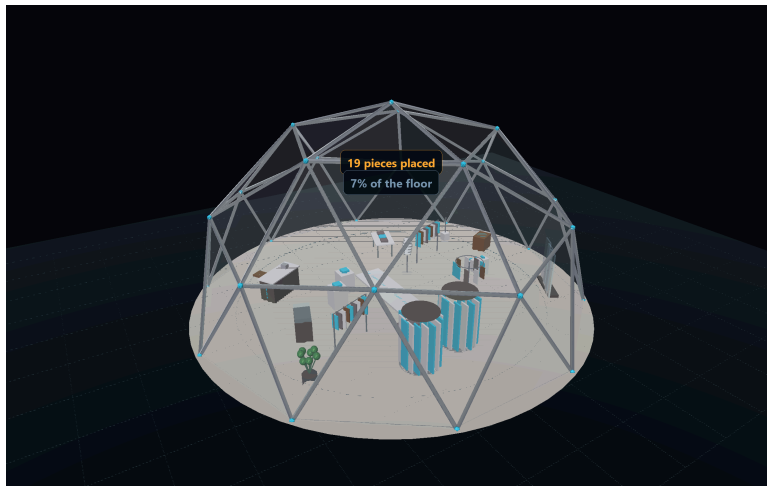


PLATE 28.1 A furnished round room

A dome furnished as a home.

Rendered by the DomeSim film engine from "The Dome House Lookbook", chapter "furnished", 60% through. The lookbook's furnished dome.

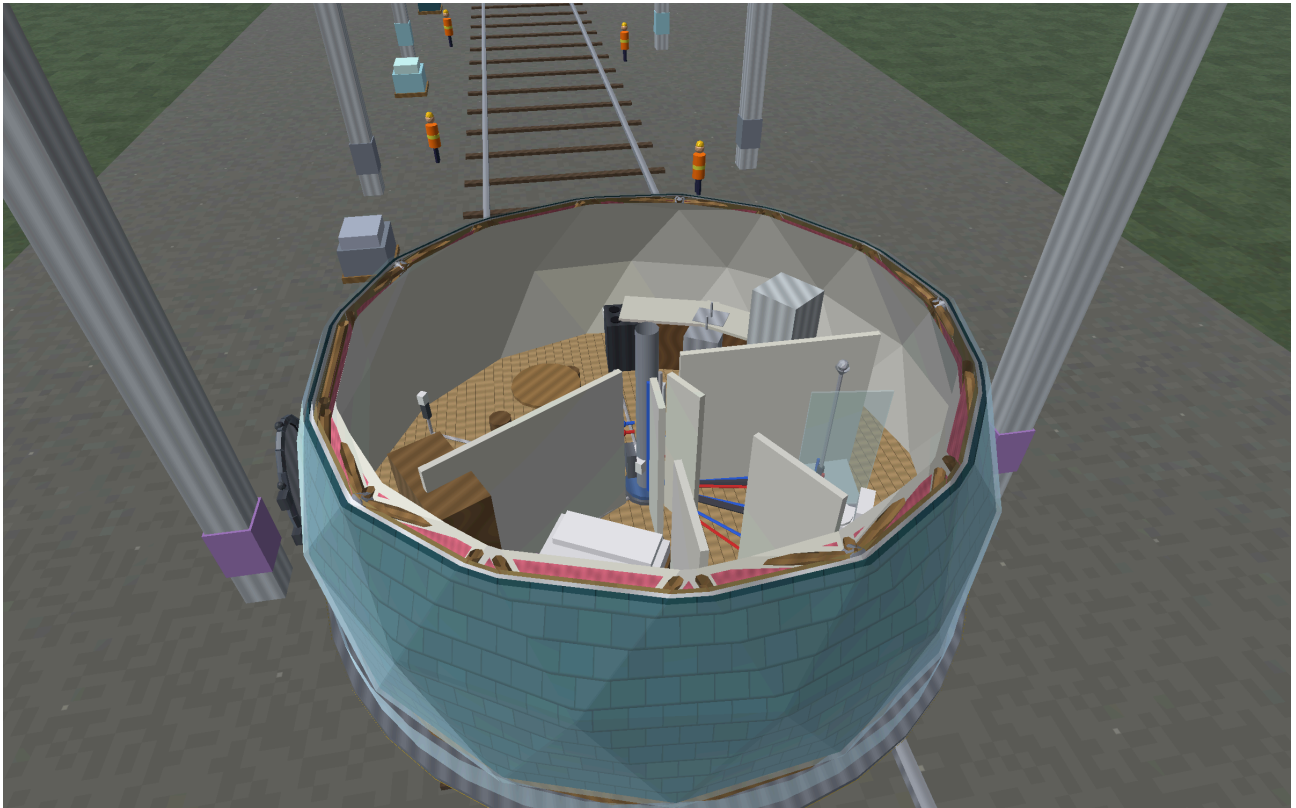


PLATE 28.2 The room around the column

The interior station seen closer and from the side: the service column at the centre of the room.

Rendered by the Dome Home Assembly Line. The interior station from a lower, closer angle.

29

Check, Correct, Record

Where the error goes, how to measure the finished dome, and what to write down.

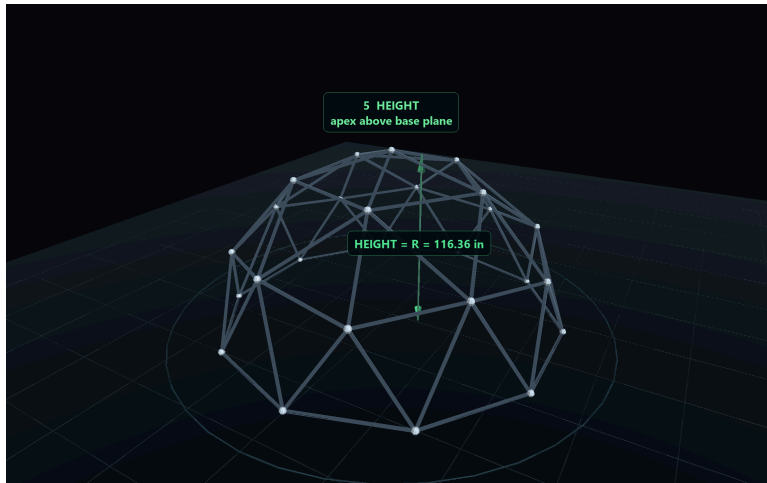
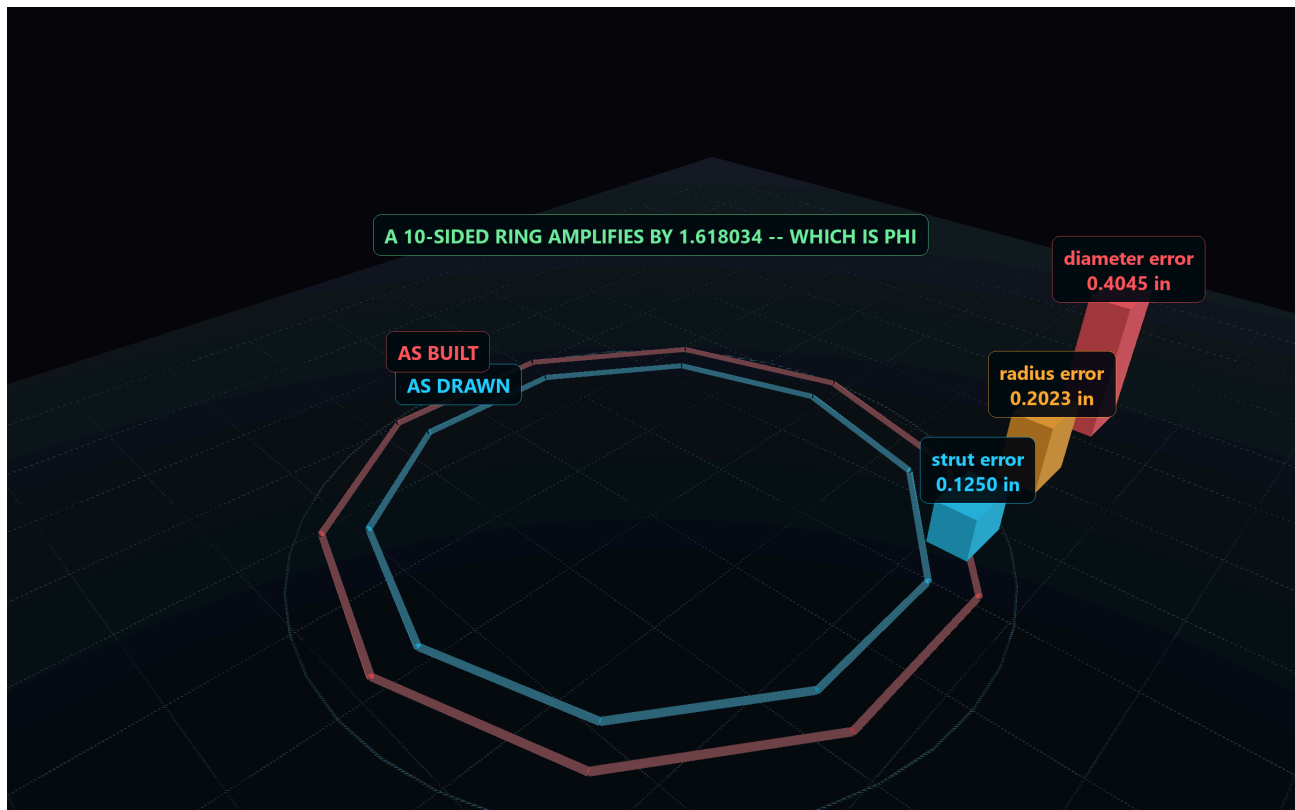


PLATE 29.1 Checking the shell
Checking the finished shell against the drawing.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "check", 80% through. The construction masterclass's check scene.

**PLATE 29.2** Where error goes

Small errors at every joint add up around a ring.

Rendered by the DomeSim film engine from "2V Dome Construction Masterclass", chapter "error", 80% through. The construction masterclass's error scene.

WORKED MATH**Bends, knots and small units**

a knot or a bend does not ruin wood. It ruins whatever length has
to be thrown away around it -- and that length is the piece you
were trying to make.

this dome's members run 5.13 to 5.90 ft, cut from 6 ft stock

a 16 ft stick from the same trunk is 2.7 times longer

one defect costs, out of 60 usable ft:

short members 6 ft = 10.0 % of the tree

long sticks 16 ft = 26.7 % of the tree

and the trunk gives 10 short pieces against 3 long ones, so there are more places for a
defect to fall harmlessly between

a real bend is cut out at its middle: go six feet clear either side

and the sections above and below it are both still good

in square milling the same bend follows the board down its whole

length, because the rectangle has to be straight before it exists

Short modular members make a defect 2.7 times cheaper, which is what lets a crooked farm tree be structure instead of firewood.

Computed by two_v_demo.wedge_why_facts.steps_defects. Every line above is printed by that code when the book is built; none of it is typed.

Worksheets

Blank forms for the tree register, the member log and the time and money.

Glossary

The words beside the cuts.

Every word below is used somewhere in this book. The definitions are the project's own, from the lexicon the films are checked against, and any number in them is live, like the rest of the book's.

B

Beat — One chapter, or a claim with the screen that proves it, rendered to its own file so that fixing it costs one short render.

F

Frequency — How many pieces each edge of the starting icosahedron is split into. 2V means two, which leaves just two strut lengths.

G

Geodesic dome — A roughly spherical shell built from triangles. The book's is half a sphere of 40 triangles, 21.6 feet across and 10.8 tall at the centre.

J

Jig — A fixture that holds parts in the same place every time, so that forty panels come out identical. Here it is one flat bench.

L

Load — The weight and push a structure has to carry: its own weight, snow, wind and people.

M

Member — Any structural piece of the frame. In the wedge dome every member is one raw wedge, and there are 120 of them.

S

Seam — The line where two neighbouring panels meet. The hemisphere has 55, and they close at only 2 different fold angles.

Seam key — The separate strip that sits in every seam between two panels' members and takes up the angle between them, so the wood

never has to be cut to it. It can be a rigid spline, a rubber hose or a gasket, 0.75 inches thick here.

Skin — What covers the frame against weather: sheathing, membrane, cladding. A frame is not a shelter until it has one.

Strut — A straight member of a dome frame. An ordinary 2V hub dome has 65; the panelised wedge dome has 120, because every panel keeps its own.

T

Tree — A standing pine: the raw material of the whole method. The book's tree has 48 feet of straight trunk, 12 inches across at the butt and 8 at the top.

Triangle — The only shape that cannot change without changing a side's length, which is why every dome panel is one.

The Numbers Behind This Book

Every live figure the book quotes, its value, and what computes it.

Every figure this book quotes is a live value. The prose holds a reference to it, not the number: the value was computed by the project's code when this copy was built. This is the full list -- each figure's name in the code, the value it had at build time, what it measures, and the chapters that print it. Change the code and rebuild, and every page that quotes a figure changes with it. Nothing here was typed.

Geometry, comparisons and the catalogue

The tree

The dome two trees make

The frame

The seams and the fold

The jig

Method A, worked

The twenty-dollar pine

Figure	Value	What it is	Chapters
dm.box_floor	640	the comparison house's floor, square feet	1
dm.box_skin	1,559	the box house's walls and roof, square feet	1
dm.cheapest_sqft	7	lowest cost per square foot in the catalogue	10
dm.dearest_sqft	68	highest cost per square foot in the catalogue	10
dm.deficit5	15.69	angle missing at a 5-way hub, degrees	6
dm.descartes	720	total missing angle of any closed cage	6
dm.equi_count	10	equilateral triangles in the hemisphere	5
dm.f1_classes	1	strut lengths in a 1V dome	4
dm.f1_struts	25	struts in a 1V dome	4
dm.f2_classes	2	strut lengths in a 2V dome	4
dm.f2_struts	65	struts in a 2V dome	4
dm.f3_classes	5	strut lengths in a 3V dome	4
dm.f3_struts	165	struts in a 3V dome	4
dm.f4_classes	6	strut lengths in a 4V dome	4
dm.f4_struts	250	struts in a 4V dome	4
dm.hemi_edges	65	struts in a hubbed 2V hemisphere	7
dm.hemi_faces	40	triangles in the 2V hemisphere	5, 7
dm.hemi_hubs	26	hubs in the 2V hemisphere	7
dm.home_cut_pct	3	the finished dome's saving	1
dm.hubless_past_saw	6	setups past a typical saw's stop	7
dm.hubless_setups	6	distinct compound saw setups	7
dm.ico_faces	20	faces of an icosahedron	3, 4
dm.iso_count	30	isosceles triangles in the hemisphere	5
dm.long_count	35	long struts in a hubbed hemisphere	2, 5

Figure	Value	What it is	Chapters
dm.panel_skin	1,188	2V panels over the same floor, square feet	1
dm.pentagons	12	pentagons in any closed pentagon-hexagon cage	9
dm.phi	1.618034	the golden ratio, phi	3
dm.preset	12	designs in the Dome Creator catalogue	1
dm.ratio	1.1308	long strut over short strut	5
dm.shed_cut_pct	24	the dome shed's saving	1
dm.short_count	30	short struts in a hubbed hemisphere	2, 5
dm.skin_saving_pct	24	less envelope than the box house, same floor	1
dm.sphere_faces	80	triangles of the full 2V sphere	4
dm.sphere_saving_pct	19.4	less skin a sphere needs than a cube of the same volume	1
dm.zome_faces	30	rhombic faces of the golden zonohedron	9

Figure	Value	What it is	Chapters
tree.butt_diameter_in	12	trunk diameter at the butt	Glossary
tree.sector_angle_deg	45	angle of one sector	16
tree.sectors	8	sectors a section is split into	16, 17
tree.top_diameter_in	8	trunk diameter at the top	Glossary
tree.usable_length_ft	48	usable straight trunk	Glossary

Figure	Value	What it is	Chapters
dome.diameter_ft	21.6	dome diameter, feet	Glossary
dome.height_ft	10.8	standing height at the centre	Glossary

Figure	Value	What it is	Chapters
frame.edges	65	unique edges in the frame	Glossary
frame.members	120	members in the frame	Glossary
frame.panels	40	triangular panels in the frame	Glossary
frame.seams	55	seams between two panels	Glossary

Figure	Value	What it is	Chapters
seam.distinct_angles	2	how many fold angles the dome has	24, Glossary
seam.fold_a_deg	18.03	the tighter of the two fold angles	24
seam.fold_b_deg	22.46	the wider of the two fold angles	24

Figure	Value	What it is	Chapters
jig.gasket_in	0.75	thickness of the seam key	Glossary

Figure	Value	What it is	Chapters
method_a.floor_sqft	300	the worked example's floor area	19
method_a.longest_in	64.5	its longest member	19
method_a.radius_ft	9.77	the same radius in feet	19

Figure	Value	What it is	Chapters
pine.kerf_only_pct	88.4	the simulator's split recovery for this log	16
pine.sawn_model_pct	45	the simulator's sawn recovery for this log	16

The Plates

Every illustration, the tool that rendered it, and what was changed to make it.

No illustration in this book is a stock picture or a generic view. Each is a scene from one of the project's tools, set up to show one idea -- a moment chosen, a layer hidden, a triangle lifted out, a label added -- and this list says which tool drew it and what was changed. The three book dividers reuse renders from the same tools as backgrounds.

Plate	Title, and what was changed	Made with	Page
i	The shell. <i>Why Build This Way's dome, centred and rendered as a full page.</i>	Film engine	ii
1.1	Twelve domes, one geometry engine. <i>The lineup is frozen after the last design has risen, with its names kept.</i>	Film engine	3
1.2	Less skin for the same floor. <i>From domology.science, using al_build's comparison house.</i>	Chart drawn from the code	4
1.3	Where the saving stops. <i>From al_build.building_comparisons, with the fit-out drawn as its own band.</i>	Chart drawn from the code	5
2.1	A triangle does not lean. <i>Stopped at the instant the square leans furthest.</i>	Film engine	7
2.2	Compression and tension in one triangle. <i>Built for the book: three members, their forces, and the supports.</i>	Film engine, scene built for this book	8
2.3	Forty triangles, two shapes. <i>The classes scene once every strut has taken its class colour.</i>	Film engine	9
3.1	The five Platonic solids. <i>Stopped once every edge has drawn, with the icosahedron highlighted.</i>	Film engine	11
3.2	Twelve corners from one number. <i>The coordinates scene with its axes and one vertex labelled.</i>	Film engine	12
4.1	Push the new points out. <i>Near the end of the projection, with the points on the sphere.</i>	Film engine	15
4.2	Split every edge. <i>Stopped after the midpoints appear, before they are pushed out.</i>	Film engine	16
4.3	What frequency costs. <i>From world_facts.frequency_ladder, which builds each dome with the simulator.</i>	Chart drawn from the code	17
5.1	Two lengths of board. <i>The masterclass opening, once the dome has drawn, beside its two boards.</i>	Film engine	21
5.2	Two jigs make every triangle. <i>Dome Forge's jig shop, both jigs loaded and seen from directly above so their true shapes compare, each named.</i>	Dome Forge, two views	22
5.3	The whole cut list. <i>The masterclass cut-list scene at full reveal.</i>	Film engine	23
6.1	Why a flat fan rises. <i>The jig shop's last stage, which assembles a fan around one hub.</i>	Dome Forge	25

Plate	Title, and what was changed	Made with	Page
6.2	A pentagon, laid flat. <i>From Dome Forge's cover-pattern geometry, dome_forge.patterns.</i>	Chart drawn from the code	26
6.3	Curvature is bought with missing angle. <i>The hex masterclass's deficit scene.</i>	Film engine	27
7.1	Forty separate triangles. <i>Only the panel assemblies shown, fills off; every face selected and popped out.</i>	Dome Forge	29
7.2	A dome on hubs. <i>The construction masterclass's hub scene near its end.</i>	Film engine	30
7.3	Two angles at once. <i>The compound-cut film's opening scene, moved in so its labels separate.</i>	Film engine	31
7.4	An hourglass and its waist. <i>The group editor on the first hourglass, moved in close on its braced waist.</i>	Dome Forge	32
8.1	The water dome in section. <i>The default water stack with the cutaway open and the flow mid-run.</i>	Dome Forge	35
8.2	Wind around a round body. <i>The campaign film's wind scene.</i>	Film engine	36
8.3	In the rain. <i>Rain and panel runoff on, from low on the ground.</i>	Dome Forge	37
9.1	A soccer-ball dome. <i>The hex masterclass's soccer-ball scene.</i>	Film engine	39
9.2	A zome. <i>The zome masterclass's opening zome, fully drawn.</i>	Film engine	40
9.3	Three ways to frame. <i>The world film's framing scene.</i>	Film engine	41
10.1	The shell being priced. <i>Why Build This Way's dome scene.</i>	Film engine	43
10.2	The same shape, clad twelve ways. <i>The world film's cladding scene.</i>	Film engine	44
10.3	Where a new house's price goes. <i>From house_economics, NAHB's 2024 cost survey.</i>	Chart drawn from the code	45
11.1	The Split-Log Homestead. <i>The world film's Split-Log Homestead chapter.</i>	Film engine	49

Plate	Title, and what was changed	Made with	Page
11.2	The Whole Trunk Lodge. <i>The world film's Whole Trunk Lodge chapter.</i>	Film engine	50
12.1	The shell on the line. <i>The line stopped as the framing station finishes.</i>	Assembly Line	51
12.2	Services before skin. <i>The line at the power station, seen from above.</i>	Assembly Line	52
12.3	Fit-out through a cutaway. <i>The interior station with the cutaway forced open.</i>	Assembly Line	53
13.1	From geometry to a pixel. <i>The from-scratch film's pipeline scene, moved in so each stage reads.</i>	Film engine	55
13.2	A worksheet on screen. <i>Why Build This Way's sources chapter with its worksheet kept.</i>	Film engine	56
14.1	Every layer at once. <i>The panel assemblies at a third opacity, with the cutaway open.</i>	Dome Forge	57
14.2	Three kinds of stock in one frame. <i>The split-log stack, fills off, moved in close so the different stock reads.</i>	Dome Forge	58
14.3	A pentagon group. <i>The group editor with the first pentagon selected.</i>	Dome Forge	59
14.4	Covers nested on the sheet. <i>From dome_forge.patterns.auto_nest.</i>	Chart drawn from the code	60
15.1	Three different sticks, one rigid shape. <i>The Frankendome film's triangle scene.</i>	Film engine	61
15.2	A V-bracket between unlike members. <i>The Frankendome film's fitted bracket.</i>	Film engine	62
15.3	Slack and settling. <i>The Frankendome film's slack scene.</i>	Film engine	63
16.1	Split, not sawn. <i>The wedge film's split scene.</i>	Film engine	65
16.2	The same round, two ways. <i>The wedge argument's round-section scene, moved in so its labels separate.</i>	Film engine	66
16.3	Four ways to turn a wedge. <i>The wedge film's orientation scene.</i>	Film engine	67

Plate	Title, and what was changed	Made with	Page
16.4	The machines that drop out. <i>The wedge argument's processing-chain scene, moved in.</i>	Film engine	68
17.1	A section opened into its wedges. <i>The harvest film's close look at one opened section.</i>	Film engine	71
17.2	The shell those wedges make. <i>The harvest film's dome scene.</i>	Film engine	72
17.3	One pine, every rung. <i>The \$20 Pine's closing ladder.</i>	Film engine	73
18.1	The one compound cut. <i>The wedge argument's butt-cut scene.</i>	Film engine	75
18.2	A wedge against a board, in bending. <i>The wedge argument's stick scene.</i>	Film engine	76
19.1	A dome from the floor you want. <i>The solved dome with its radius, height and floor marked.</i>	Film engine, scene built for this book	81
19.2	The floor it covers. <i>The solved dome from overhead with the floor circle drawn.</i>	Film engine, scene built for this book	82
21.1	The hinge steers the tree. <i>The harvest film's felling scene at the back cut.</i>	Film engine	85
21.2	Bucking to length. <i>The harvest film as the bucking cuts are made, re-aimed along the stem.</i>	Film engine	86
21.3	Two trees, split. <i>The harvest film's pile, seen along the row from its near end.</i>	Film engine	87
22.1	Locate first, cut second. <i>Why Build This Way's jig scene as the members arrive.</i>	Film engine	89
22.2	The fabrication bench. <i>The wedge argument's bench scene.</i>	Film engine	90
23.1	The pinwheel panel. <i>The wedge film's pinwheel scene.</i>	Film engine	91
23.2	The two panels, drawn to build. <i>From two_v_demo.wedge_geometry.pinwheel_panels at the build plan's radius.</i>	Chart drawn from the code	92
23.3	Flush cutting in place. <i>The wedge argument's flush-cut scene.</i>	Film engine	93

Plate	Title, and what was changed	Made with	Page
24.1	The seam and its gasket. <i>The wedge film's gasket scene.</i>	Film engine	95
24.2	The key in the seam. <i>The wedge argument's keystone scene.</i>	Film engine	96
24.3	Two fold angles. <i>The wedge argument's fold scene.</i>	Film engine	97
25.1	A level ring. <i>The construction masterclass's foundation scene.</i>	Film engine	99
25.2	A riser wall. <i>The construction masterclass's riser scene.</i>	Film engine	100
26.1	Ring by ring. <i>The construction masterclass's raising scene.</i>	Film engine	101
26.2	The top. <i>The construction masterclass's apex scene.</i>	Film engine	102
26.3	Forty panels into a shell. <i>The wedge film's assembly scene.</i>	Film engine	103
27.1	Skinning the shell. <i>The construction masterclass's skin scene.</i>	Film engine	105
27.2	An opening in a triangulated shell. <i>The construction masterclass's openings scene.</i>	Film engine	106
27.3	The seam veins alone. <i>Only the veins, the water in them, the collector, downpipe and cistern.</i>	Dome Forge	107
28.1	A furnished round room. <i>The lookbook's furnished dome.</i>	Film engine	109
28.2	The room around the column. <i>The interior station from a lower, closer angle.</i>	Assembly Line	110
29.1	Checking the shell. <i>The construction masterclass's check scene.</i>	Film engine	111
29.2	Where error goes. <i>The construction masterclass's error scene.</i>	Film engine	112

Sources, Assumptions, and Corrections

What each figure rests on, and what has been corrected.

The Tools

Every program that made this book, and how to run it yourself.

About the Author

What Comes Next

More domes, more films, more of the story.
